

37. PTP Module for the Ethernet Controller (EPTPCb)

37.1 Overview

This MCU has an on-chip Precision Time Protocol (PTP) module for the Ethernet controller (EPTPC). The module applies the PTP defined in version 2 of the IEEE 1588-2008 standard to handle timing and synchronization between devices. The EPTPC is composed of synchronization frame processing units (SYNFP0 and SYNFP1), a packet relation controller unit (PRC-TC), and a statistical time correction algorithm unit (STCA).

Use the EPTPC in combination with the on-chip Ethernet controller (ETHERC) and the DMA controller for the PTP Ethernet controller (PTPEDMAC).

Table 37.1 lists the EPTPC specifications, and Figure 37.1 shows the EPTPC configuration.

Table 37.1 EPTPC Specifications

Item	Description
Protocol	Compliant with the Precision Time Protocol (PTP) defined in IEEE 1588
Synchronization frame processing units (SYNFP0 and SYNFP1)	- Transmits and receives PTP messages as a master or slave. - The following four clock devices are supported: - Ordinary clock (OC) - Boundary clock (BC) - End-to-end transparent clock (E2E TC) - Peer-to-peer transparent clock (P2P TC) - Calculates the meanPathDelay and offsetFromMaster values defined in IEEE 1588. - Capable of generating a master clock. - Hardware filtering of received multicast packets with a MAC address - Capable of hardware filtering in accordance with the type of PTP message - Supports PTP message frames in layer 4 (IPv4 and UDP) and layer 2 (Ethernet frames). - Can be used as a normal Ethernet port when time synchronization is not in use.
Packet relation controller unit (PRC-TC)	- Relaying of received data between Ethernet ports 0 and 1 - Setting the same MAC address for Ethernet ports 0 and 1 allows transmission of data from the two ports or from only one of them. - Store-and-forward method or cut-through method selectable for the relaying of packets
Statistical time correction algorithm unit (STCA)	- Frequency of the clock signal supplied to the statistical time correction algorithm unit is selectable as 20, 25, 50, or 100 MHz. - In slave operation, the synchronized state can be indicated from the offsetFromMaster value staying below a previously specified threshold. Additionally, the threshold can be calculated statistically from collected positive and negative gradient values (worst-10 acquisition). - The local clock counter holds corrected time information obtained from a master clock. - The STCA clock can be used as the clock source for generating pulse signals from pulse output timer m (m = 0 to 5). - Peripheral modules such as MTU3 and GPT can be started or stopped on the edge of pulses synchronized with the master clock in response to interrupt requests by the pulse output timer or the output of event signals to the ELC. - The pulses generated by the pulse output timers 0 and 1 can be output from the EPLSOUT0 and EPLSOUT1 pins, respectively.
Interrupt sources	MINT interrupt - Requested when the state of the individual modules is changed. - Requested on rising edges of the pulse signal generated by the pulse output timer. IPLS interrupt - Requested on rising or falling edges of the pulse signal generated by the previously selected pulse output timer group. - Can be requested on every edge or only once.
Event linking	- An event signal is output to the ELC on a rising or falling edge of the pulse signal generated by the pulse output timer. - An event signal can be output on every edge or only once.

When using the functions of EPTPC, set the ETHERCn.ECMR.PRM bit to 1 (promiscuous mode) and the SYBYPSSR.BYPASSn bit to 0 (SYNFPn module is used).

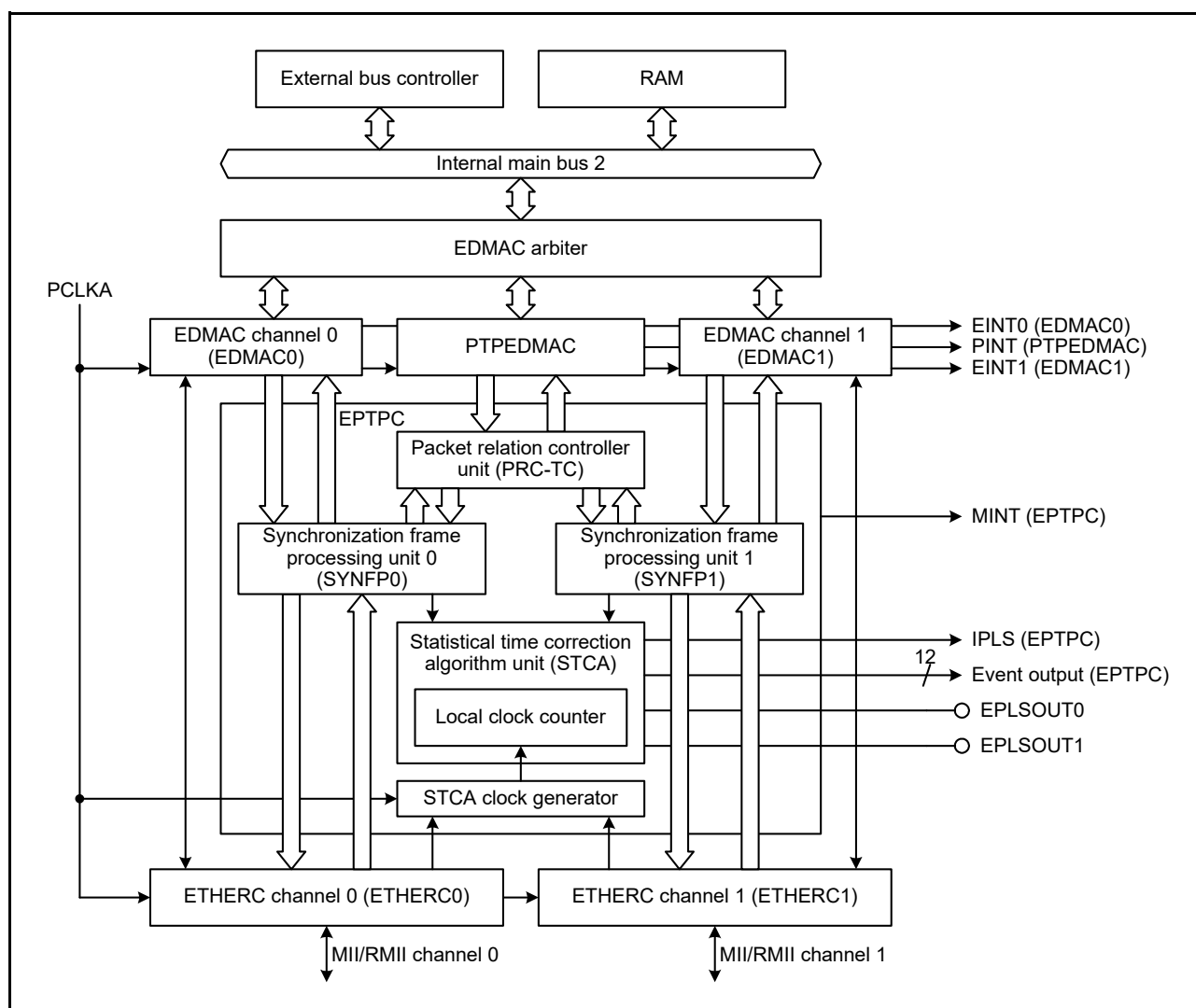


Figure 37.1 EPTPC Configuration

Table 37.2 lists I/O pins used for the EPTPC.

Table 37.2 EPTPC I/O Pins

Pin Name	I/O	Function
EPLSOUT0	Output	Pulse output pin for the pulse output timer 0
EPLSOUT1	Output	Pulse output pin for the pulse output timer 1

In this section, channels may not be mentioned in overall descriptions of modules that have multiple channels. Examples of the notation are listed in Table 37.3.

Table 37.3 Examples of the Notation

Module Name	Channel	Meaning
SYNFPn module	n = 0, 1	Synchronization processing units 0 and 1 (SYNFP0 and SYNFP1)
Pulse output timer m	m = 0 to 5	Pulse output timer channels 0 to 5

37.1.1 Combination of Clock Device and Ethernet Port

The EPTPC supports operation as three types of clock device: an ordinary clock (OC), boundary clock (BC), or transparent clock (TC). Furthermore, it supports both end-to-end (E2E) and peer-to-peer (P2P) operation for all three clock devices.

Available combinations for usage of Ethernet ports 0 and 1 are listed in Table 37.4.

Table 37.4 Combination of Clock Device and Ethernet Port

Clock Device	Ethernet Port 0		Ethernet Port 1	
No control by EPTPC	PTP packets are not handled.			
Ordinary Clock (OC) Only Ethernet port 0 is used for handling PTP packets	Master	End-to-end (E2E)	PTP packets are not handled.	
		Peer-to-peer (P2P)		
	Slave	E2E		
		P2P		
Ordinary Clock (OC) Only Ethernet port 1 is used for handling PTP packets	PTP packets are not handled.		Master	E2E
				P2P
			Slave	E2E
				P2P
Boundary clock (OC)	Master	E2E	Master	E2E
		E2E		P2P
		P2P		E2E
		P2P		P2P
	Master	E2E	Slave	E2E
		E2E		P2P
		P2P		E2E
		P2P		P2P
	Slave	E2E	Master	E2E
		E2E		P2P
		P2P		E2E
		P2P		P2P
Transparent clock (TC)	E2E TC			
	P2P TC			

37.1.2 Frame Format of PTP Messages

The frame format of PTP messages can be selected from the four types by setting the FORM0 and FORM1 bits in the SYNFP frame format setting register (SYFORMR).

Figure 37.2 shows the PTP message formats for transmission and reception by the EPTPC.

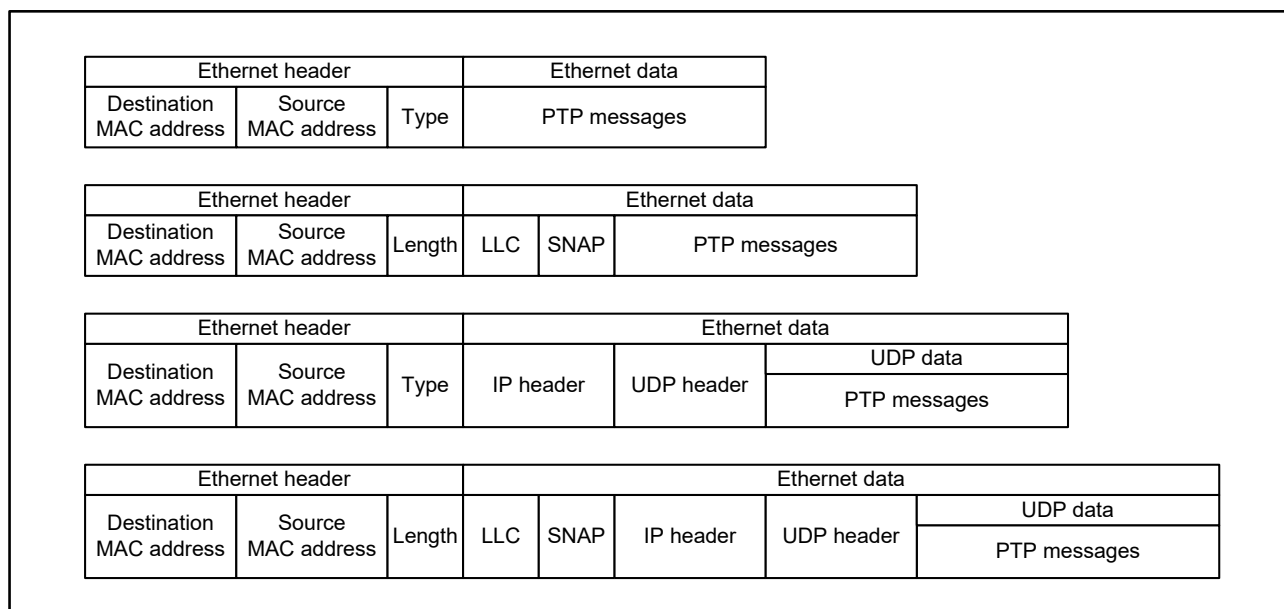


Figure 37.2 Frame Format of PTP Messages

The EPTPC module is capable of transmitting PTP messages. When it sends a PTP message, multicast addresses defined by IEEE 1588 are normally specified as the destination MAC address and IP address, depending on the type of the PTP message to be sent.

Furthermore, the port number when a PTP message is encapsulated for use with the UDP must also be specified in accordance with the type of message as stipulated by IEEE 1588.

Table 37.5 gives a summary of the information to specify the Ethernet frame format for PTP messages.

Table 37.5 Type of PTP Message for Multicast and Information to Specify the Ethernet Frame Format

Type of PTP Message			IEEE802.3 Frame Format (SYFORMR.FORM0 Bit = 1)		Ethernet II Frame Format (SYFORMR.FORM0 Bit = 0)		UDP Port No. *1	
			MAC Address	IP Address	MAC Address	Ethertype		
PTP-primary	Event messages	Sync	01-00-5E-00-01-81	224.0.1.129	01-1B-19-00-00-00	88F7h	319	
		Delay_Req						
PTP-pdelay		Pdelay_Req	01-00-5E-00-00-6B	224.0.0.107	01-80-C2-00-00-0E			320
		Pdelay_Resp						
PTP-primary	General messages	Pdelay_Resp_Follow_Up	01-00-5E-00-01-81	224.0.1.129	01-1B-19-00-00-00			
		Announce						
		Follow_Up						
		Delay_Resp						
		Signaling						
		Management						

Note 1. The port number must be specified only when a PTP message is encapsulated for use with the UDP (i.e. the SYFORMR.FORM1 bit = 1).

37.1.3 Type of PTP Message and Details of Processing

Table 37.6 and Table 37.7 respectively give details of processing by the EPTPC for receiving and transmitting PTP messages.

Table 37.6 List of PTP Messages for Reception by the EPTPC

Message Type	Message	Details of Processing
Event	Sync	Calculates the value of offsetFromMaster if twoStepFlag in flagField is FALSE.
	Delay_Req	Responds to Delay_Resp.
	Pdelay_Req	Responds to Pdelay_Resp.
	Pdelay_Resp	Calculates the value of meanPathDelay if twoStepFlag in flagField is FALSE.
General	Announce	—
	Follow_Up	Calculates the value of offsetFromMaster if twoStepFlag in flagField of the most recently received Sync message was TRUE and the value of meanPathDelay is fixed.
	Delay_Resp	Calculates the value of meanPathDelay.
	Pdelay_Resp_Follow_Up	Calculates the value of meanPathDelay if twoStepFlag in flagField of the most recently received Pdelay_Resp message was TRUE.
	Management	—
	Signaling	—

Table 37.7 List of PTP Messages for Transmission by the EPTPC

Message Type	Message	Details of Processing
Event	Sync	Sync messages are transmitted at the fixed interval specified by the SYTLIR.SYNC[7:0] bits.
	Delay_Req	Transmission proceeds with an interval from 0 to twice the interval set by the value of the SYTLIR.DREQ[7:0] bits and determined by a random number.
	Pdelay_Req	Pdelay_Req messages are transmitted at the fixed interval specified by the SYTLIR.DREQ[7:0] bits.
	Pdelay_Resp	Transmission of responses to Pdelay_Req.
General	Announce	Announce messages are transmitted at the fixed interval specified by the SYTLIR.ANCE[7:0] bits.
	Follow_Up	—
	Delay_Resp	Transmission of responses to Delay_Req
	Pdelay_Resp_Follow_Up	—
	Management	—
	Signaling	—

37.1.4 Operations for Non-PTP Messages

Table 37.8 lists the operations on receiving non-PTP messages with or without the settings of promiscuous mode and extended promiscuous mode.

Table 37.8 Operations on Receiving Non-PTP Messages

ETHERCn.ECMR.PRM Bit	FFLTR.EXTPRM Bit	Frame Type	Operations
0 (non-promiscuous mode)	0 (normal operation)	unicast frame	Only messages for this station are transferred to EDMACn.
		multicast frame	Messages are transferred in accordance with the settings of frame reception filter.
		broadcast frame	Messages are transferred to EDMACn and the PRC-TC module.
	1 (extended promiscuous mode)	unicast frame	Only messages for this station are transferred to EDMACn.
		multicast frame	All messages are transferred to EDMACn.
		broadcast frame	All messages are transferred to EDMACn.
1 (promiscuous mode)	0 (normal operation)	unicast frame	Messages for this station are transferred to EDMACn and messages for other station are transferred to the PRC-TC module.
		multicast frame	Messages are transferred in accordance with the settings of frame reception filter.
		broadcast frame	Messages are transferred to EDMACn and the PRC-TC module.
	1 (extended promiscuous mode)	unicast frame	All messages are transferred to EDMACn regardless the destination address.
		multicast frame	All messages are transferred to EDMACn.
		broadcast frame	All messages are transferred to EDMACn.

When the ETHERCn.ECMR.PRM bit is 0, ETHERCn receives messages only for this station. When it is 1, ETHERCn receives all messages.

The SYNFPn module in EPTPC can only handle messages received by ETHERCn.

For details of operations of the SYNFPn module at this time, refer to section 37.3.1, Transmission and Reception and Relaying of Non-PTP Messages.

37.2 Register Descriptions

37.2.1 MINT Interrupt Source Status Register (MIESR)

Address(es): EPTPC.MIESR 000C 4000h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	CYC5	CYC4	CYC3	CYC2	CYC1	CYC0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	—	PRC	SY1	SY0	ST
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	ST	STCA Status Flag	0: No change in the state of the STCA module 1: A change in the state of the STCA module	R
b1	SY0	SYNFP0 Status Flag	0: No change in the state of the SYNFP0 module 1: A change in the state of the SYNFP0 module	R
b2	SY1	SYNFP1 Status Flag	0: No change in the state of the SYNFP1 module 1: A change in the state of the SYNFP1 module	R
b3	PRC	PRC-TC Status Flag	0: No change in the state of the PRC-TC module 1: A change in the state of the PRC-TC module	R
b15 to b4	—	Reserved	These bits are read as 0. The write value should be 0.	R
b16	CYC0	Pulse Output Timer 0 Rising Edge Detection Flag	0: A rising edge in the periodic pulse signal from pulse output timer 0 is not detected. 1: A rising edge in the periodic pulse signal from pulse output timer 0 is detected.	R/W*1
b17	CYC1	Pulse Output Timer 1 Rising Edge Detection Flag	0: A rising edge in the periodic pulse signal from pulse output timer 1 is not detected. 1: A rising edge in the periodic pulse signal from pulse output timer 1 is detected.	R/W*1
b18	CYC2	Pulse Output Timer 2 Rising Edge Detection Flag	0: A rising edge in the periodic pulse signal from pulse output timer 2 is not detected. 1: A rising edge in the periodic pulse signal from pulse output timer 2 is detected.	R/W*1
b19	CYC3	Pulse Output Timer 3 Rising Edge Detection Flag	0: A rising edge in the periodic pulse signal from pulse output timer 3 is not detected. 1: A rising edge in the periodic pulse signal from pulse output timer 3 is detected.	R/W*1
b20	CYC4	Pulse Output Timer 4 Rising Edge Detection Flag	0: A rising edge in the periodic pulse signal from pulse output timer 4 is not detected. 1: A rising edge in the periodic pulse signal from pulse output timer 4 is detected.	R/W*1
b21	CYC5	Pulse Output Timer 5 Rising Edge Detection Flag	0: A rising edge in the periodic pulse signal from pulse output timer 5 is not detected. 1: A rising edge in the periodic pulse signal from pulse output timer 5 is detected.	R/W*1
b31 to b22	—	Reserved	These bits are read as 0. The write value should be 0.	R

Note 1. Writing 1 clears the flag. Writing 0 does not affect the flag value.

The MIESR register indicates changes in the states of the STCA, SYNFPn, and PRC-TC modules which act as sources of MINT interrupts and the detection of rising edges of pulse output timers m (m = 0 to 5).

For the MINT interrupt, see section 37.4, Interrupts.

ST Flag (STCA Status Flag)

This flag indicates changes in the state of the STCA module.

[Setting condition]

- A change in the state of a flag of the STSR register for which the setting in the STIPR register enables notification

[Clearing conditions]

When any of the following conditions is met:

- The flags in the STSR register are all 0.
- The bits in the STIPR register are all 0.
- There is a bit set to 1 in the STIPR register, but the corresponding flag in the STSR register is 0.

SYn Flag (SYNFPn Status Flag)

This flag indicates changes in the state of the SYNFPn module.

[Setting condition]

- A change in the state of a flag of the SYSR register for which the setting in the SYIPR register enables notification

[Clearing conditions]

When any of the following conditions is met:

- The flags in the SYSR register are all 0.
- The bits in the SYIPR register are all 0.
- There is a bit set to 1 in the SYIPR register, but the corresponding flag in the SYSR register is 0.

PRC Flag (PRC-TC Status Flag)

This flag indicates changes in the state of the PRC-TC module.

[Setting condition]

- A change in the state of a flag of the PRSR register for which the setting in the PRIPR register enables notification

[Clearing conditions]

When any of the following conditions is met:

- The flags in the PRSR register are all 0.
- The bits in the PRIPR register are all 0.
- There is a bit set to 1 in the PRIPR register, but the corresponding bit in the PRSR register is 0.

CYCm Flag (Pulse Output Timer m Rising Edge Detection Flag)

This flag indicates detection of a rising edge in the periodic pulse signal produced by the corresponding pulse output timer m (m = 0 to 5).

[Setting condition]

- Detection of a rising edge in the periodic pulse signal produced by a pulse output timer for which the setting in the MITSELR register enables notification

[Clearing condition]

- 1 being written to this flag.

After the flag is cleared to 0, it is set to 1 again on detection of a rising edge in the periodic pulse signal from the corresponding pulse output timer.

37.2.2 MINT Interrupt Request Enable Register (MIEIPR)

Address(es): EPTPC.MIEIPR 000C 4004h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	CYC5	CYC4	CYC3	CYC2	CYC1	CYC0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	—	PR	SY1	SY0	ST
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	ST	STCA Status Interrupt Request Enable	0: Disables the generation of MINT interrupt requests by the STCA status flag. 1: Enables the generation of MINT interrupt requests by the STCA status flag.	R/W
b1	SY0	SYNFP0 Status Interrupt Request Enable	0: Disables the generation of MINT interrupt requests by the SYNFP0 status flag. 1: Enables the generation of MINT interrupt requests by the SYNFP0 status flag.	R/W
b2	SY1	SYNFP1 Status Interrupt Request Enable	0: Disables the generation of MINT interrupt requests by the SYNFP1 status flag. 1: Enables the generation of MINT interrupt requests by the SYNFP1 status flag.	R/W
b3	PR	PRC-TC Status Interrupt Request Enable	0: Disables the generation of MINT interrupt requests by the PRC-TC status flag. 1: Enables the generation of MINT interrupt requests by the PRC-TC status flag.	R/W
b15 to b4	—	Reserved	These bits are read as 0. The write value should be 0.	R
b16	CYC0	Pulse Output Timer 0 Rising Edge Detection Interrupt Request Enable	0: Disables the generation of MINT interrupt requests in response to detection of a rising edge of pulse output timer 0. 1: Enables the generation of MINT interrupt requests in response to detection of a rising edge of pulse output timer 0.	R/W
b17	CYC1	Pulse Output Timer 1 Rising Edge Detection Interrupt Request Enable	0: Disables the generation of MINT interrupt requests in response to detection of a rising edge of pulse output timer 1. 1: Enables the generation of MINT interrupt requests in response to detection of a rising edge of pulse output timer 1.	R/W
b18	CYC2	Pulse Output Timer 2 Rising Edge Detection Interrupt Request Enable	0: Disables the generation of MINT interrupt requests in response to detection of a rising edge of pulse output timer 2. 1: Enables the generation of MINT interrupt requests in response to detection of a rising edge of pulse output timer 2.	R/W
b19	CYC3	Pulse Output Timer 3 Rising Edge Detection Interrupt Request Enable	0: Disables the generation of MINT interrupt requests in response to detection of a rising edge of pulse output timer 3. 1: Enables the generation of MINT interrupt requests in response to detection of a rising edge of pulse output timer 3.	R/W
b20	CYC4	Pulse Output Timer 4 Rising Edge Detection Interrupt Request Enable	0: Disables the generation of MINT interrupt requests in response to detection of a rising edge of pulse output timer 4. 1: Enables the generation of MINT interrupt requests in response to detection of a rising edge of pulse output timer 4.	R/W
b21	CYC5	Pulse Output Timer 5 Rising Edge Detection Interrupt Request Enable	0: Disables the generation of MINT interrupt requests in response to detection of a rising edge of pulse output timer 5. 1: Enables the generation of MINT interrupt requests in response to detection of a rising edge of pulse output timer 5.	R/W
b31 to b22	—	Reserved	These bits are read as 0. The write value should be 0.	R

The MIEIPR register is used to enable or disable the generation of MINT interrupt requests when MINT interrupt source conditions are satisfied.

37.2.3 ELC Output/IPLS Interrupt Request Enable Register (ELIPPR)

Address(es): EPTPC.ELIPPR 000C 4010h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	PLSN	—	—	—	—	—	—	—	PLSP
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	CYCN5	CYCN4	CYCN3	CYCN2	CYCN1	CYCN0	—	—	CYCP5	CYCP4	CYCP3	CYCP2	CYCP1	CYCP0
Value after reset:	0	0	1	1	1	1	1	1	0	0	1	1	1	1	1	1

Bit	Symbol	Bit Name	Description	R/W
b0	CYCP0	Pulse Output Timer 0 Rising Edge Detection Event Output Enable	0: Rising edges of the signal from pulse output timer 0 are not conveyed to the ELC as event signals. 1: Rising edges of the signal from pulse output timer 0 are conveyed to the ELC as event signals.	R/W
b1	CYCP1	Pulse Output Timer 1 Rising Edge Detection Event Output Enable	0: Rising edges of the signal from pulse output timer 1 are not conveyed to the ELC as event signals. 1: Rising edges of the signal from pulse output timer 1 are conveyed to the ELC as event signals.	R/W
b2	CYCP2	Pulse Output Timer 2 Rising Edge Detection Event Output Enable	0: Rising edges of the signal from pulse output timer 2 are not conveyed to the ELC as event signals. 1: Rising edges of the signal from pulse output timer 2 are conveyed to the ELC as event signals.	R/W
b3	CYCP3	Pulse Output Timer 3 Rising Edge Detection Event Output Enable	0: Rising edges of the signal from pulse output timer 3 are not conveyed to the ELC as event signals. 1: Rising edges of the signal from pulse output timer 3 are conveyed to the ELC as event signals.	R/W
b4	CYCP4	Pulse Output Timer 4 Rising Edge Detection Event Output Enable	0: Rising edges of the signal from pulse output timer 4 are not conveyed to the ELC as event signals. 1: Rising edges of the signal from pulse output timer 4 are conveyed to the ELC as event signals.	R/W
b5	CYCP5	Pulse Output Timer 5 Rising Edge Detection Event Output Enable	0: Rising edges of the signal from pulse output timer 5 are not conveyed to the ELC as event signals. 1: Rising edges of the signal from pulse output timer 5 are conveyed to the ELC as event signals.	R/W
b7, b6	—	Reserved	These bits are read as 0. The write value should be 0.	R
b8	CYCN0	Pulse Output Timer 0 Falling Edge Detection Event Output Enable	0: Falling edges of the signal from pulse output timer 0 are not conveyed to the ELC as event signals. 1: Falling edges of the signal from pulse output timer 0 are conveyed to the ELC as event signals.	R/W
b9	CYCN1	Pulse Output Timer 1 Falling Edge Detection Event Output Enable	0: Falling edges of the signal from pulse output timer 1 are not conveyed to the ELC as event signals. 1: Falling edges of the signal from pulse output timer 1 are conveyed to the ELC as event signals.	R/W
b10	CYCN2	Pulse Output Timer 2 Falling Edge Detection Event Output Enable	0: Falling edges of the signal from pulse output timer 2 are not conveyed to the ELC as event signals. 1: Falling edges of the signal from pulse output timer 2 are conveyed to the ELC as event signals.	R/W

Bit	Symbol	Bit Name	Description	R/W
b11	CYCN3	Pulse Output Timer 3 Falling Edge Detection Event Output Enable	0: Falling edges of the signal from pulse output timer 3 are not conveyed to the ELC as event signals. 1: Falling edges of the signal from pulse output timer 3 are conveyed to the ELC as event signals.	R/W
b12	CYCN4	Pulse Output Timer 4 Falling Edge Detection Event Output Enable	0: Falling edges of the signal from pulse output timer 4 are not conveyed to the ELC as event signals. 1: Falling edges of the signal from pulse output timer 4 are conveyed to the ELC as event signals.	R/W
b13	CYCN5	Pulse Output Timer 5 Falling Edge Detection Event Output Enable	0: Falling edges of the signal from pulse output timer 5 are not conveyed to the ELC as event signals. 1: Falling edges of the signal from pulse output timer 5 are conveyed to the ELC as event signals.	R/W
b15, b14	—	Reserved	These bits are read as 0. The write value should be 0.	R
b16	PLSP	Pulse Output Timer Rising Edge Detection IPLS Interrupt Request Enable	0: Disables IPLS interrupt requests due to rising edges of signals from the selected pulse output timer. 1: Enables IPLS interrupt requests due to rising edges of signals from the selected pulse output timer.	R/W
b23 to b17	—	Reserved	These bits are read as 0. The write value should be 0.	R
b24	PLSN	Pulse Output Timer Falling Edge Detection IPLS Interrupt Request Enable	0: Disables IPLS interrupt requests due to falling edges of signals from the selected pulse output timer. 1: Enables IPLS interrupt requests due to falling edges of signals from the selected pulse output timer.	R/W
b31 to b25	—	Reserved	These bits are read as 0. The write value should be 0.	R

The ELIPPR register determines whether or not rising and falling edges of the periodic pulse signals produced by pulse output timers *m* are output to the ELC. The register also enables and disables IPLS interrupts due to rising or falling edges of signals from the pulse output timer selected in the IPTSELR register.

Peripheral modules such as the MTU3 or GPT can be controlled with the clock synchronized by the PTP by using the ELC linking function to set a periodic pulse generated by pulse output timer *m* as a trigger for operations of the peripheral module.

The ELIPACR register can be used to set up the one-time-only output of event signals to the ELC or of IPLS interrupt requests.

For the IPLS interrupt, see [section 37.4, Interrupts](#).

37.2.4 ELC Output/IPLS Interrupt Enable Automatic Clearing Register (ELIPACR)

Address(es): EPTPC.ELIPACR 000C 4014h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	PLSN	—	—	—	—	—	—	—	PLSP
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	CYCN5	CYCN4	CYCN3	CYCN2	CYCN1	CYCN0	—	—	CYCP5	CYCP4	CYCP3	CYCP2	CYCP1	CYCP0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	CYCP0	ELIPPR.CYCP0 Bit Automatic Clearing	0: Disables automatic clearing of the enable bit for the output of rising edges of pulse output timer 0. 1: Enables automatic clearing of the enable bit for the output of rising edges of pulse output timer 0.	R/W
b1	CYCP1	ELIPPR.CYCP1 Bit Automatic Clearing	0: Disables automatic clearing of the enable bit for the output of rising edges of pulse output timer 1. 1: Enables automatic clearing of the enable bit for the output of rising edges of pulse output timer 1.	R/W
b2	CYCP2	ELIPPR.CYCP2 Bit Automatic Clearing	0: Disables automatic clearing of the enable bit for the output of rising edges of pulse output timer 2. 1: Enables automatic clearing of the enable bit for the output of rising edges of pulse output timer 2.	R/W
b3	CYCP3	ELIPPR.CYCP3 Bit Automatic Clearing	0: Disables automatic clearing of the enable bit for the output of rising edges of pulse output timer 3. 1: Enables automatic clearing of the enable bit for the output of rising edges of pulse output timer 3.	R/W
b4	CYCP4	ELIPPR.CYCP4 Bit Automatic Clearing	0: Disables automatic clearing of the enable bit for the output of rising edges of pulse output timer 4. 1: Enables automatic clearing of the enable bit for the output of rising edges of pulse output timer 4.	R/W
b5	CYCP5	ELIPPR.CYCP5 Bit Automatic Clearing	0: Disables automatic clearing of the enable bit for the output of rising edges of pulse output timer 5. 1: Enables automatic clearing of the enable bit for the output of rising edges of pulse output timer 5.	R/W
b7, b6	—	Reserved	These bits are read as 0. The write value should be 0.	R
b8	CYCN0	ELIPPR.CYCN0 Bit Automatic Clearing	0: Disables automatic clearing of the enable bit for the output of falling edges of pulse output timer 0. 1: Enables automatic clearing of the enable bit for the output of falling edges of pulse output timer 0.	R/W
b9	CYCN1	ELIPPR.CYCN1 Bit Automatic Clearing	0: Disables automatic clearing of the enable bit for the output of falling edges of pulse output timer 1. 1: Enables automatic clearing of the enable bit for the output of falling edges of pulse output timer 1.	R/W
b10	CYCN2	ELIPPR.CYCN2 Bit Automatic Clearing	0: Disables automatic clearing of the enable bit for the output of falling edges of pulse output timer 2. 1: Enables automatic clearing of the enable bit for the output of falling edges of pulse output timer 2.	R/W
b11	CYCN3	ELIPPR.CYCN3 Bit Automatic Clearing	0: Disables automatic clearing of the enable bit for the output of falling edges of pulse output timer 3. 1: Enables automatic clearing of the enable bit for the output of falling edges of pulse output timer 3.	R/W

Bit	Symbol	Bit Name	Description	R/W
b12	CYCN4	ELIPPR.CYCN4 Bit Automatic Clearing	0: Disables automatic clearing of the enable bit for the output of falling edges of pulse output timer 4. 1: Enables automatic clearing of the enable bit for the output of falling edges of pulse output timer 4.	R/W
b13	CYCN5	ELIPPR.CYCN5 Bit Automatic Clearing	0: Disables automatic clearing of the enable bit for the output of falling edges of pulse output timer 5. 1: Enables automatic clearing of the enable bit for the output of falling edges of pulse output timer 5.	R/W
b15, b14	—	Reserved	These bits are read as 0. The write value should be 0.	R
b16	PLSP	ELIPPR.PLSP Bit Automatic Clearing	0: Disables automatic clearing of the enable bit for IPLS interrupt requests in response to detection of rising edges of the pulse output timer. 1: Enables automatic clearing of the enable bit for IPLS interrupt requests in response to detection of rising edges of the pulse output timer.	R/W
b23 to b17	—	Reserved	These bits are read as 0. The write value should be 0.	R
b24	PLSN	ELIPPR.PLSN Bit Automatic Clearing	0: Disables automatic clearing of the enable bit for IPLS interrupt requests in response to detection of falling edges of the pulse output timer. 1: Enables automatic clearing of the enable bit for IPLS interrupt requests in response to detection of falling edges of the pulse output timer.	R/W
b31 to b25	—	Reserved	These bits are read as 0. The write value should be 0.	R

The ELIPACR register is used to enable one-time output of each event to the ELC or each IPLS interrupt request triggered by detecting edges of the periodic pulses of pulse output timer m.

Normally, an event is output to the ELC or an IPLS interrupt request is generated at each edge of the periodic pulses of pulse output timer m while the corresponding bit in the ELIPPR register is 1 (enabled),

When a bit in the ELIPPR register is 1 while the corresponding bit in the ELIPACR register is also 1, the bit in the ELIPPR register automatically becomes 0 when the event signal for the ELC or IPLS interrupt request is generated.

For the IPLS interrupt, see section 37.4, Interrupts.

37.2.5 STCA Status Register (STSR)

Address(es): EPTPC.STSR 000C 4040h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	W10D	SYNTO UT	—	SYNCO UT	SYNC
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	SYNC	Synchronized State Detection Flag	0: Synchronization is not detected. 1: Synchronization is detected.	R/W*1
b1	SYNCOUT	Synchronization Loss Detection Flag	0: A loss of synchronization is not detected. 1: A loss of synchronization is detected.	R/W*1
b2	—	Reserved	This bit is read as 0. The write value should be 0.	R
b3	SYNTOUIT	Sync Message Reception Timeout Detection Flag	0: A Sync message reception timeout is not detected. 1: A Sync message reception timeout is detected.	R/W*1
b4	W10D	Worst 10 Acquisition Completion Flag	0: The worst 10 values have not been acquired yet. 1: The worst 10 values have been acquired.	R/W*1
b31 to b5	—	Reserved	These bits are read as 0. The write value should be 0.	R

Note: When the SYNSTARTR.STR bit is 0, the value of the corresponding flag remains.

Note 1. Writing 1 clears the flag. Writing 0 does not affect the flag value.

The STSR register indicates the state of the STCA module.

SYNC Flag (Synchronized State Detection Flag)

This flag indicates that synchronization has occurred more than the number of times specified by the STMR.SYTH[3:0] bits in succession when the STMR.ALEN0 bit is 1. When the STMR.ALEN0 bit is 0, the SYNC flag is not set to 1 even if synchronization has occurred more than the specified number of times in succession.

SYNCOUT Flag (Synchronization Loss Detection Flag)

This flag indicates that loss of synchronization has occurred more than the number of times specified by the STMR.DVTH[3:0] bits in succession when the STMR.ALEN0 bit is 1.

Since the time is not synchronized immediately after time synchronization is started (SYNSTARTR.STR bit is set to 1), the SYNCOUT becomes 1 regardless of the STMR.ALEN0 bit setting. When using the SYNTOUIT flag, set the SYNTOUIT flag to 0 immediately after starting time synchronization.

When the STMR.ALEN0 bit is 0, the SYNCOUT flag does not become 1 even if loss of synchronization has occurred more than the specified number of times in succession after time synchronization is started and the SYNTOUIT flag is immediately set to 0.

SYNTOUIT Flag (Sync Message Reception Timeout Detection Flag)

This flag indicates that a Sync message has not been received during the period specified by the SYNTOR register when the STMR.ALEN1 bit is 1. The SYNTOUIT flag becomes 1 immediately if time synchronization is started (SYNSTARTR.STR bit is set to 1) when no Sync message has been received after the EPTPC starts. When using the SYNTOUIT flag, set the SYNTOUIT flag to 0 immediately after starting time synchronization.

W10D Flag (Worst 10 Acquisition Completion Flag)

This flag indicates that acquisition of the worst 10 has been completed.

37.2.6 STCA Status Notification Enable Register (STIPR)

Address(es): EPTPC.STIPR 000C 4044h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	W10D	SYNTO UT	—	SYNCO UT	SYNC
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	SYNC	SYNC Status Notification Enable	0: Disables notification of the state of STSR.SYNC. 1: Enables notification of the state of STSR.SYNC.	R/W
b1	SYNCOUT	SYNCOUT Status Notification Enable	0: Disables notification of the state of STSR.SYNCOUT. 1: Enables notification of the state of STSR.SYNCOUT.	R/W
b2	—	Reserved	This bit is read as 0. The write value should be 0.	R
b3	SYNTOUT	SYNTOUT Status Notification Enable	0: Disables notification of the state of STSR.SYNTOUT. 1: Enables notification of the state of STSR.SYNTOUT.	R/W
b4	W10D	W10D Status Notification Enable	0: Disables notification of the state of STSR.W10D. 1: Enables notification of the state of STSR.W10D.	R/W
b31 to b5	—	Reserved	These bits are read as 0. The write value should be 0.	R

The STIPR register specifies whether the MIESR.ST flag does or does not reflect changes in the state of the STCA module.

37.2.7 STCA Clock Frequency Setting Register (STCFR)

Address(es): EPTPC.STCFR 000C 4050h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	STCF[1:0]	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b1, b0	STCF[1:0]	STCA Clock Frequency	b1 b0 0 0: 20 MHz 0 1: 25 MHz 1 0: 50 MHz 1 1: 100 MHz	R/W
b31 to b2	—	Reserved	These bits are read as 0. The write value should be 0.	R

Note: Set a value in this register before starting the EDMAC, ETHERC, and PTPEDMAC. Do not change the settings during operations.

The STCFR register is used to select the frequency of the clock source for the STCA module (STCA clock).

The setting in the STCFR register must be for the same frequency as that selected by the setting of the STCSELR register.

STCF[1:0] Bits (STCA Clock Frequency)

These bits select the frequency of the STCA clock.

To enable synchronous control in compliance with IEEE 1588, the STCA clock frequency must be selected from 20 MHz, 25 MHz, 50 MHz, or 100 MHz. If the frequency selected by these bits differs from the clock frequency actually input to the STCA module, operation is not guaranteed.

37.2.8 STCA Operating Mode Register (STMR)

Address(es): EPTPC.STMR 000C 4054h

b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
—	—	ALEN1	ALEN0	—	—	—	—	DVTH[3:0]			SYTH[3:0]				
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
W10S	—	CMOD	—	—	—	—	—	WINT[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b7 to b0	WINT[7:0]	Worst 10 Acquisition Time	00h: The worst 10 values are not acquired. 01h: Sync message reception: 1 time : Fh: Sync message reception: 255 times	R/W
b12 to b8	—	Reserved	These bits are read as 0. The write value should be 0.	R
b13	CMOD	Time Synchronization Correction Mode	0: Mode 1 1: Mode 2	R/W
b14	—	Reserved	This bit is read as 0. The write value should be 0.	R
b15	W10S	Worst 10 Acquisition Control Select	0: Measurement is started by hardware and the value acquired in the PW10VR or MW10R register is used as the limit for filtering. 1: Measurement is started by the GETW10R.GW10 bit. Also, the value set in the PLIMITR or MLIMITR register is used as the limit for filtering.	R/W
b19 to b16	SYTH[3:0]	Synchronized State Detection Threshold Setting	0h: None *1 1h: 1 time : Fh: 15 times	R/W
b23 to b20	DVTH[3:0]	Synchronization Loss Detection Threshold Setting	0h: None *2 1h: 1 time : Fh: 15 times	R/W
b27 to b24	—	Reserved	These bits are read as 0. The write value should be 0.	R
b28	ALEN0	Alarm Detection Enable 0	0: The STSR.SYNC or SYNCOUT flag is not set to 1 on detection of synchronization or loss of synchronization. 1: The STSR.SYNC or SYNCOUT flag is set to 1 on detection of synchronization or loss of synchronization.	R/W
b29	ALEN1	Alarm Detection Enable 1	0: The STSR.SYNTOUT flag is not set to 1 on detection of the Sync message reception timeout interrupt. 1: The STSR.SYNTOUT flag is not set to 1 on detection of the Sync message reception timeout interrupt.	R/W
b31, b30	—	Reserved	These bits are read as 0. The write value should be 0.	R

Note 1. The STSR.SYNC flag does not become 1 regardless of the ALEN0 bit setting.

Note 2. The STSR.SYNCOUT flag does not become 1 regardless of the ALEN0 bit setting.

The STMR register is used to set the operating mode of the STCA module.

WINT[7:0] Bits (Worst 10 Acquisition Time)

These bits set the time for acquiring the worst 10 gradients (the number of times Sync messages are received). Usually, it is recommended to set the number of times of Sync message receptions to 32 or more.

CMOD Bit (Time Synchronization Correction Mode)

Mode 1 or mode 2 can be selected to correct the local time information when the EPTPC operates as a slave clock. Select the appropriate mode in consideration of system configuration, etc.

Table 37.9 is a summary of the two individual correction modes.

Table 37.9 Types and Features of Correction Mode

Correction Mode	Function	Features	Notes
Mode 1	Mode for correcting the counter every Sync message reception by using the current offsetFromMaster (operation is in mode 1 after the start of correction, and then shifts to the specified mode).	The time information of the master clock is set as the local time information at a time.	Synchronization cannot be guaranteed if calculating offsetFromMaster is not possible (e.g. packets are temporarily being discarded due to a failure of communications).
Mode 2	Mode where the gradient value calculated from offsetFromMaster (worst-10 control) is retained and used in correcting the local time information so that it approximates the time information of the master clock.	Even if calculating offsetFromMaster is not possible, a certain level of synchronization can be guaranteed in this mode (the counter is still corrected from the gradient information).	Establishing synchronization takes longer.

W10S Bit (Worst 10 Acquisition Control Select)

This bit selects the value used for measurement and filtering of worst 10 gradients. When this bit is set to 0, the values acquired in registers PW10VRU, PW10VRM, and PW10VRL and registers MW10RU, MW10RM, and MW10RL are used as the limit for the filter. When the bit is set to 1, the values set in registers PLIMITRU, PLIMITRM, and PLIMITRL and registers MLIMITRU, MLIMITRM, and MLIMITRL are used as the limit for the filter.

SYTH[3:0] Bits (Synchronized State Detection Threshold Setting)

These bits specify a value for the number of consecutive times being within the thresholds will be judged as synchronization. Specifically, the value falling within the thresholds set in registers SYNTDBRU and SYNTDBRL as many times as the value of these bits indicates the synchronized state. When the ALEN0 bit is 1, the STSR.SYNCOUT flag becomes 1.

DVTH[3:0] Bits (Synchronization Loss Detection Threshold Setting)

These bits specify a value for the number of consecutive times exceeding a threshold will be judged as a loss of synchronization. Specifically, synchronization is judged to have been lost when the value in registers SYNTDARU and SYNTDARL is consecutively exceeded as many times as the value of these bits. When the ALEN0 bit is 1, the STSR.SYNCOUT flag becomes 1.

ALEN0 Bit (Alarm Detection Enable 0)

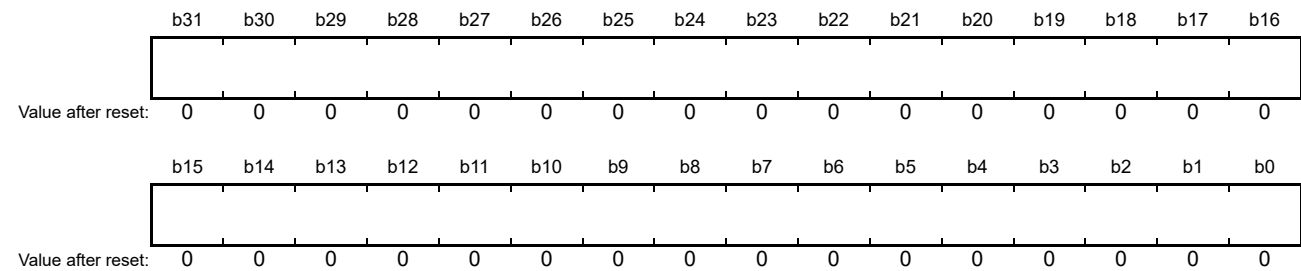
When this bit is 1, the STSR.SYNC or SYNCOUT flag is set to 1 on detection of synchronization or loss of synchronization. When this bit is 0, the SYNC or SYNCOUT flag is not set to 1 even if synchronization or loss of synchronization is detected.

ALEN1 Bit (Alarm Detection Enable 1)

When this bit is 1, the STSR.SYNTOUT flag is set to 1 if a Sync message is not received within the time specified by the SYNTOR register. When this bit is 0, the SYNTOUT flag is not set to 1 even if the reception timeout has occurred.

37.2.9 Sync Message Reception Timeout Register (SYNTOR)

Address(es): EPTPC.SYNTOR 000C 4058h



Bit	Symbol	Bit Name	Description	R/W
b31 to b0	—	—	A Sync message not being received within 1024 × n (ns), where n is the setting, leads to a timeout for reception of Sync messages, leading to the STSR.SYNTOUT flag being set to 1.	R/W

The SYNTOR register is used to specify the timeout period for reception of Sync messages. The timeout period is 1024 times the setting of these bits, in units of nanoseconds.

A Sync message not being received within the period specified by these bits is judged to represent a timeout.

When the SYNTOR register is 0, the STSR.SYNTOUT flag does not become 1.

37.2.10 IPLS Interrupt Request Timer Select Register (IPTSELR)

Address(es): EPTPC.IPTSELR 000C 4060h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	IPTSEL 5	IPTSEL 4	IPTSEL 3	IPTSEL 2	IPTSEL 1	IPTSEL 0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	IPTSEL0	Pulse Output Timer 0 Select	0: Pulse output timer 0 is not selected as a source of IPLS interrupt requests. 1: Pulse output timer 0 is selected as a source of IPLS interrupt requests.	R/W
b1	IPTSEL1	Pulse Output Timer 1 Select	0: Pulse output timer 1 is not selected as a source of IPLS interrupt requests. 1: Pulse output timer 1 is selected as a source of IPLS interrupt requests.	R/W
b2	IPTSEL2	Pulse Output Timer 2 Select	0: Pulse output timer 2 is not selected as a source of IPLS interrupt requests. 1: Pulse output timer 2 is selected as a source of IPLS interrupt requests.	R/W
b3	IPTSEL3	Pulse Output Timer 3 Select	0: Pulse output timer 3 is not selected as a source of IPLS interrupt requests. 1: Pulse output timer 3 is selected as a source of IPLS interrupt requests.	R/W
b4	IPTSEL4	Pulse Output Timer 4 Select	0: Pulse output timer 4 is not selected as a source of IPLS interrupt requests. 1: Pulse output timer 4 is selected as a source of IPLS interrupt requests.	R/W
b5	IPTSEL5	Pulse Output Timer 5 Select	0: Pulse output timer 5 is not selected as a source of IPLS interrupt requests. 1: Pulse output timer 5 is selected as a source of IPLS interrupt requests.	R/W
b31 to b6	—	Reserved	These bits are read as 0. The write value should be 0.	R

The IPTSELR register is used to select the pulse output timer that generates IPLS interrupt requests.

Each pulse output timer m ($m = 0$ to 5) takes the clock signal from the STCA as its clock source and produces pulses with a specified period and duty cycle. An IPLS interrupt is requested on rising edges if the ELIPPR.PLSP bit is set to 1 and on falling edges if the PLSN bit in the same register is set to 1. The interrupt request signal becomes the logical OR of the interrupt requests from the given channels when bits for multiple channels in register IPTSELR are set to 1. For the IPLS interrupt, see section 37.4, Interrupts.

37.2.11 MINT Interrupt Request Timer Select Register (MITSELR)

Address(es): EPTPC.MITSELR 000C 4064h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	MINTEN5	MINTEN4	MINTEN3	MINTEN2	MINTEN1	MINTEN0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	MINTEN0	Pulse Output Timer 0 MINT Interrupt Output Enable	0: Output of rising edges by pulse output timer 0 is not reflected by the MIESR.CYC0 flag as a MINT interrupt source. 1: Output of rising edges by pulse output timer 0 is reflected by the MIESR.CYC0 flag as a MINT interrupt source.	R/W
b1	MINTEN1	Pulse Output Timer 1 MINT Interrupt Output Enable	0: Output of rising edges by pulse output timer 1 is not reflected by the MIESR.CYC1 flag as a MINT interrupt source. 1: Output of rising edges by pulse output timer 1 is reflected by the MIESR.CYC1 flag as a MINT interrupt source.	R/W
b2	MINTEN2	Pulse Output Timer 2 MINT Interrupt Output Enable	0: Output of rising edges by pulse output timer 2 is not reflected by the MIESR.CYC2 flag as a MINT interrupt source. 1: Output of rising edges by pulse output timer 2 is reflected by the MIESR.CYC2 flag as a MINT interrupt source.	R/W
b3	MINTEN3	Pulse Output Timer 3 MINT Interrupt Output Enable	0: Output of rising edges by pulse output timer 3 is not reflected by the MIESR.CYC3 flag as a MINT interrupt source. 1: Output of rising edges by pulse output timer 3 is reflected by the MIESR.CYC3 flag as a MINT interrupt source.	R/W
b4	MINTEN4	Pulse Output Timer 4 MINT Interrupt Output Enable	0: Output of rising edges by pulse output timer 4 is not reflected by the MIESR.CYC4 flag as a MINT interrupt source. 1: Output of rising edges by pulse output timer 4 is reflected by the MIESR.CYC4 flag as a MINT interrupt source.	R/W
b5	MINTEN5	Pulse Output Timer 5 MINT Interrupt Output Enable	0: Output of rising edges by pulse output timer 5 is not reflected by the MIESR.CYC5 flag as a MINT interrupt source. 1: Output of rising edges by pulse output timer 5 is reflected by the MIESR.CYC5 flag as a MINT interrupt source.	R/W
b31 to b6	—	Reserved	These bits are read as 0. The write value should be 0.	R

The MITSELR register is used to select pulse output timer m that generates MINT interrupt requests.

Each pulse output timer m (m = 0 to 5) takes the clock signal from the STCA as its clock source and produces pulses with a specified period and duty cycle. A MINT interrupt is requested on rising edges of the pulse signal from the corresponding pulse output timer m if the setting of the MIEIPR.CYCM bit is 1.

For the MINT interrupt, see section 37.4, Interrupts.

37.2.12 ELC Output Timer Select Register (ELTSELR)

Address(es): EPTPC.ELTSELR 000C 4068h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	ELTDIS 5	ELTDIS 4	ELTDIS 3	ELTDIS 2	ELTDIS 1	ELTDIS 0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	ELTDIS0	Pulse Output Timer 0 Event Generation Disable	0: Pulse output timer 0 is used for the generation of event signals for the ELC. 1: Pulse output timer 0 is not used for the generation of event signals for the ELC.	R/W
b1	ELTDIS1	Pulse Output Timer 1 Event Generation Disable	0: Pulse output timer 1 is used for the generation of event signals for the ELC. 1: Pulse output timer 1 is not used for the generation of event signals for the ELC.	R/W
b2	ELTDIS2	Pulse Output Timer 2 Event Generation Disable	0: Pulse output timer 2 is used for the generation of event signals for the ELC. 1: Pulse output timer 2 is not used for the generation of event signals for the ELC.	R/W
b3	ELTDIS3	Pulse Output Timer 3 Event Generation Disable	0: Pulse output timer 3 is used for the generation of event signals for the ELC. 1: Pulse output timer 3 is not used for the generation of event signals for the ELC.	R/W
b4	ELTDIS4	Pulse Output Timer 4 Event Generation Disable	0: Pulse output timer 4 is used for the generation of event signals for the ELC. 1: Pulse output timer 4 is not used for the generation of event signals for the ELC.	R/W
b5	ELTDIS5	Pulse Output Timer 5 Event Generation Disable	0: Pulse output timer 5 is used for the generation of event signals for the ELC. 1: Pulse output timer 5 is not used for the generation of event signals for the ELC.	R/W
b31 to b6	—	Reserved	These bits are read as 0. The write value should be 0.	R

The ELTSELR register is used to select pulse output timer m that outputs event signals to the ELC.

Each pulse output timer m (m = 0 to 5) takes the clock signal from the STCA as its clock source and produces pulses with a specified period and duty cycle. An event signal is output to the ELC on rising edges if the ELIPPR.CYCPm bit is set to 1 and on falling edges if the CYCNm bit in the same register is set to 1.

For the output of event signals to the ELC, see [section 37.4, Interrupts](#).

37.2.13 Time Synchronization Channel Select Register (STCHSELR)

Address(es): EPTPC.STCHSELR 000C 406Ch

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	SYSEL
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	SYSEL	Timer Information Input Select	0: Time information from the SYNFP0 module is used. 1: Time information from the SYNFP1 module is used.	R/W*1
b31 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R

Note 1. Do not change the value of this bit while the SYNSTARTR.STR bit is 1.

The STCHSELR register selects the time information input to the STCA module.

37.2.14 Slave Time Synchronization Start Register (SYNSTARTR)

Address(es): EPTPC.SYNSTARTR 000C 4080h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	STR
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	STR	Slave Time Synchronization Control	0: Slave time synchronization is stopped. 1: Slave time synchronization is started.	R/W
b31 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R

The SYNSTARTR register is used to start or stop time synchronization. This register is used when the EPTPC is operating as a slave device.

37.2.15 Local Clock Counter Initial Value Load Directive Register (LCIVLDR)

Address(es): EPTPC.LCIVLDR 000C 4084h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	LOAD
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

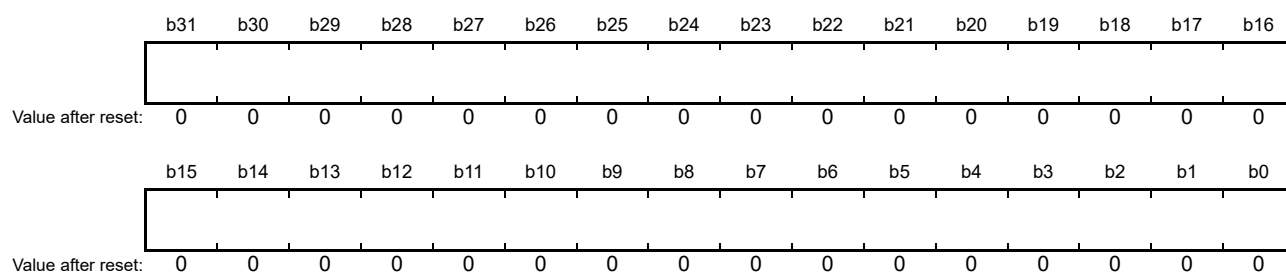
Bit	Symbol	Bit Name	Description	R/W
b0	LOAD	Local Clock Counter Initial Value Load Directive	0: The initial value is not loaded into the local clock counter. 1: The initial value is loaded into the local clock counter.	W*1
b31 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R

Note 1. Do not change the value of this bit while the SYNSTARTR.STR bit is 1.

The LCIVLDR register is used to set the value of registers LCIVRU, LCIVRM, and LCIVRL as the initial value of the local clock counter.

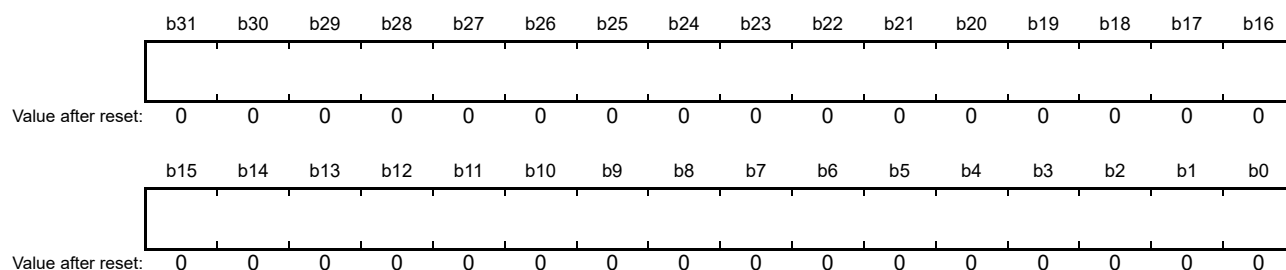
37.2.16 Synchronization Loss Detection Threshold Registers (SYNTDARU, SYNTDARL)

Address(es): EPTPC.SYNTDARU 000C 4090h



Bit	Symbol	Bit Name	Description	R/W
b31 to b0	—	—	These bits hold the setting for the higher-order 32 bits of the threshold for detection of loss of synchronization.	R/W

Address(es): EPTPC.SYNTDARL 000C 4094h



Bit	Symbol	Bit Name	Description	R/W
b31 to b0	—	—	These bits hold the setting for the lower-order 32 bits of the threshold for detection of loss of synchronization.	R/W

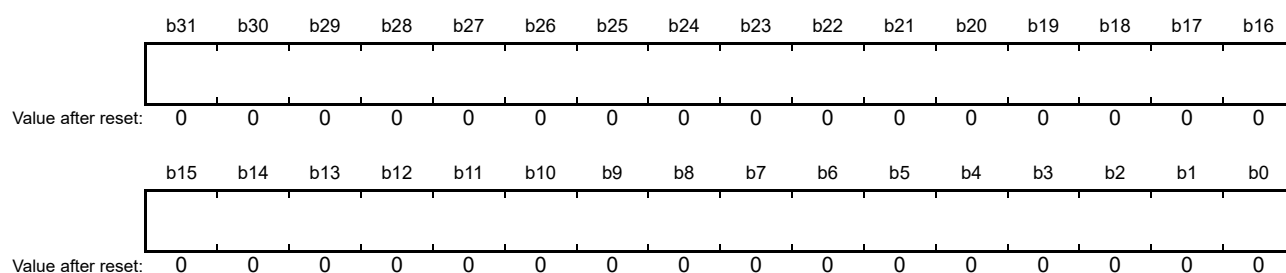
The settings of the SYNTDARU and SYNTDARL registers specify the threshold value for offsetFromMaster to be used in judging loss of synchronization. When setting a threshold value, write the higher-order 32 bits to SYNTDARU and the lower-order 32 bits to SYNTDARL, in that order and in consecutive operations.

If the offsetFromMaster value exceeds the value specified in SYNTDARU and SYNTDARL, a loss of synchronization is detected. Set a value in SYNTDARU and SYNTDARL in units of nanoseconds.

SYNTDARU and SYNTDARL are not used when the device is operating as a master clock.

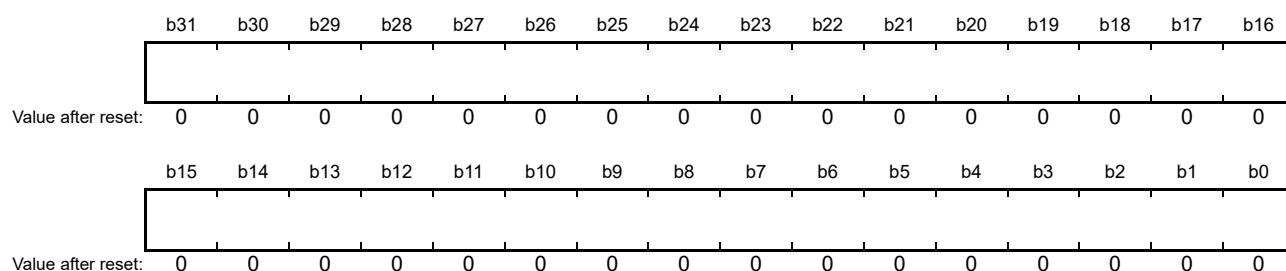
37.2.17 Synchronization Detection Threshold Registers (SYNTDBRU, SYNTDBRL)

Address(es): EPTPC.SYNTDBRU 000C 4098h



Bit	Symbol	Bit Name	Description	R/W
b31 to b0	—	—	These bits hold the setting for the higher-order 32 bits of the threshold for detection of synchronization.	R/W

Address(es): EPTPC.SYNTDBRL 000C 409Ch



Bit	Symbol	Bit Name	Description	R/W
b31 to b0	—	—	These bits hold the setting for the lower-order 32 bits of the threshold for detection of synchronization.	R/W

The settings of the SYNTDBRU and SYNTDBRL registers specify the threshold value for offsetFromMaster to be used in judging synchronization. When setting a threshold value, write the higher-order 32 bits to SYNTDBRU and the lower-order 32 bits to SYNTDBRL, in that order and in consecutive operations.

If the offsetFromMaster value is less than the value specified in SYNTDBRU and SYNTDBRL, synchronization is detected. Set a value in SYNTDBRU and SYNTDBRL in units of nanoseconds.

SYNTDBRU and SYNTDBRL are not used when the device is operating as a master clock.

37.2.18 Local Clock Counter Initial Value Registers (LCIVRU, LCIVRM, LCIVRL)

LCIVRU

Address(es): EPTPC.LCIVRU 000C 40B0h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b15 to b0	—	—	These bits hold the setting for the higher-order 16 bits of the integer portion of the initial value for the local clock counter.	R/W
b31 to b16	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Address(es): EPTPC.LCIVRM 000C 40B4h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b31 to b0	—	—	These bits hold the setting for the lower-order 32 bits of the integer portion of the initial value for the clock counter.	R/W

Address(es): EPTPC.LCIVRL 000C 40B8h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b31 to b0	—	—	These bits hold the setting for the fractional portion of the initial value of the clock counter in nanoseconds.	R/W

Registers LCIVRU, LCIVRM, and LCIVRL specify the initial value in seconds of the clock counter. When setting an initial value, write the higher-order 16 bits of the integer portion to LCIVRU, the lower-order 32 bits of the integer portion to LCIVRM, and the fractional portion in nanoseconds to LCIVRL, in that order and in consecutive operations. The value of registers LCIVRU, LCIVRM, and LCIVRL can be used as the initial value of the local clock counter. When setting the value of registers LCIVRU, LCIVRM, and LCIVRL in the local clock counter, set the LCIVLDR.LOAD bit to 1.

Example) When 2.000000025s is to be set as the initial value, write the following values to each of the registers.

LCIVRU: 0000 0000h

LCIVRM: 0000 0002h

LCIVRL: 0000 0019h

37.2.19 Worst 10 Acquisition Directive Register (GETW10R)

Address(es): EPTPC.GETW10R 000C 4124h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	GW10
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	GW10	Worst 10 Acquisition Directive	0: The worst-10 values are not acquired. 1: Starts acquisition of the worst-10 values.	R/W*1
b31 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R

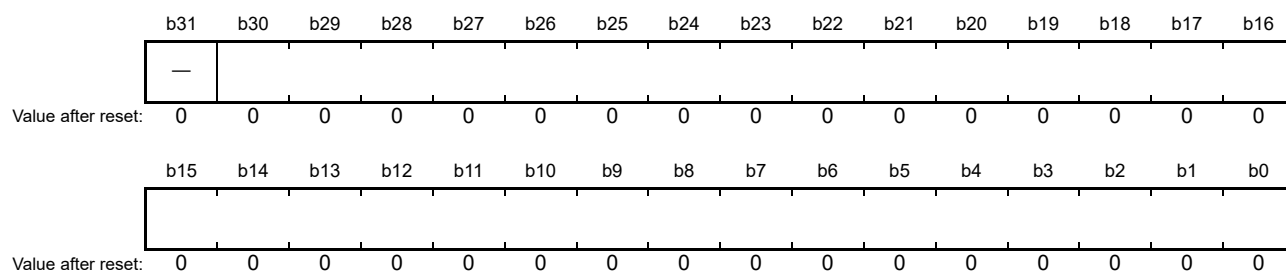
Note 1. Do not set this bit to 1 while the STMR.W10S bit is 0.

The GETW10R register is used in software control to start calculation of gradient values for use in selecting the worst-10 values.

A gradient value is the amount by which the timer counter of a slave is incremented when a given interval elapses. Setting the GW10 bit to 1 while the value of the STMR.W10S bit is 1 selects calculation of a gradient value by the EPTPC each time it receives a Sync message. Gradient values are calculated the number of times specified by the STMR.WINT[7:0] bits. The GW10 bit is cleared to 0 on completion of this number of calculations. The GETW10R register is not used in operation as a master clock.

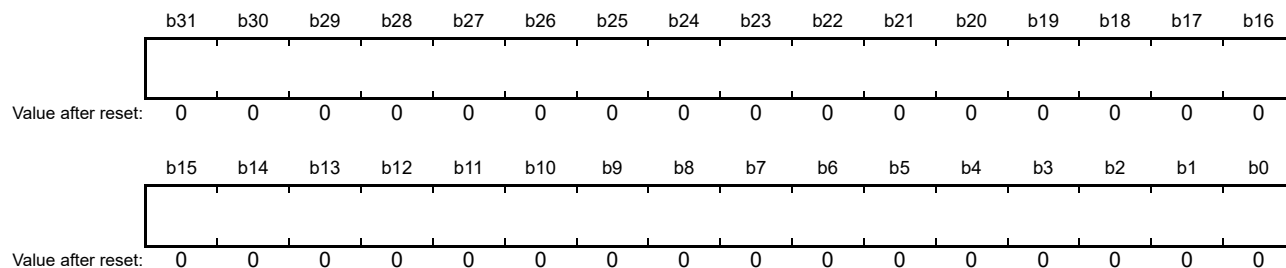
37.2.20 Positive Gradient Limit Registers (PLIMITRU, PLIMITRM, PLIMITRL)

Address(es): EPTPC.PLIMITRU 000C 4128h



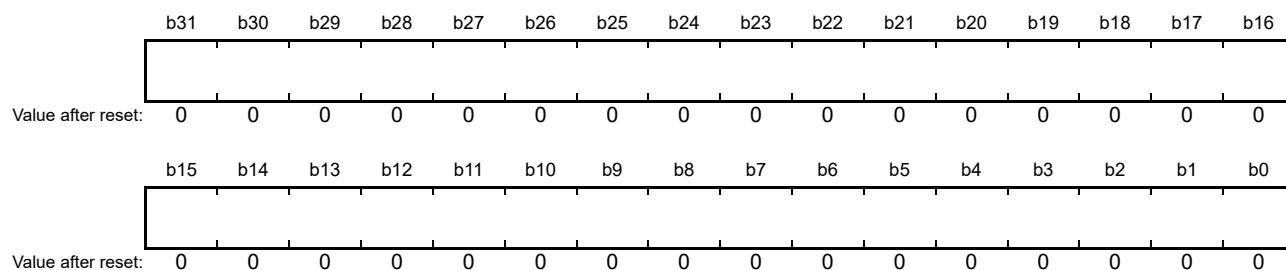
Bit	Symbol	Bit Name	Description	R/W
b30 to b0	—	—	These bits hold the setting for the higher-order 31 bits of the limit for the positive gradient.	R/W
b31	—	Reserved	This bit is read as 0. The write value should be 0.	R

Address(es): EPTPC.PLIMITRM 000C 412Ch



Bit	Symbol	Bit Name	Description	R/W
b31 to b0	—	—	These bits hold the setting for the middle-order 32 bits of the limit for the positive gradient.	R/W

Address(es): EPTPC.PLIMITRL 000C 4130h



Bit	Symbol	Bit Name	Description	R/W
b31 to b0	—	—	These bits hold the setting for the lower-order 32 bits of the limit for the positive gradient.	R/W

Registers PLIMITRU, PLIMITRM, and PLIMITRL are used to specify an upper limit on gradient (= positive gradient) for use in time synchronization. When setting an upper limit, write values consecutively to PLIMITRU, PLIMITRM, and PLIMITRL in that order.

The gradients that exceed the value specified in registers PLIMITRU, PLIMITRM, and PLIMITRL are not used in time synchronization. Registers PLIMITRU, PLIMITRM, PLIMITRL are not used when the device is operating as a master clock. Registers PLIMITRU, PLIMITRM, and PLIMITRL are effective while the STMR.CMOD and W10S bits are 1.

The gradient value to be set in the register is calculated by using the following expression.

$$\text{PLIMITRU, PLIMITRM, and PLIMITRL register values} = A(s)/T(s) \times 2^{32}$$

A: The time (s) by which the slave clock advances during the interval between received Sync messages

T: The actual time (s) between received Sync messages

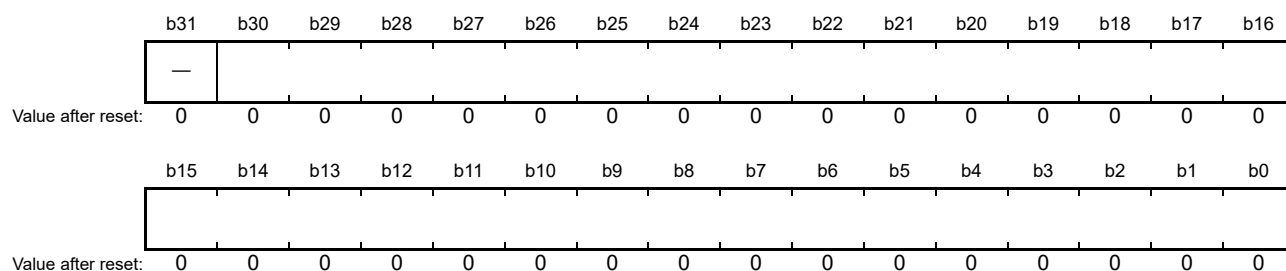
For example, if the interval between Sync messages is 0.5 seconds and the slave clock advances by 0.7 seconds during that time, and this is to be set as the limit, then the setting for PLIMITR = $0.7/0.5 \times 2^{32} = 6\,012\,954\,214 = 1\,6666\,6666h$, and the settings for the individual registers are as follows: PLIMITRU = 0000 0000h, PLIMITRM = 0000 0001h, and PLIMITRL = 6666 6666h.

The minimum setting depends on the STCA clock frequency as the clock source for counting by the local clock counter. For example, if the STCA clock frequency is 50 MHz, then the minimum allowable setting for registers PLIMITRU, PLIMITRM, and PLIMITRL = $(1/50 \text{ (MHz)}) (s)/0.5 (s) \times 2^{32} = 172 = ACh$, and the settings for the individual registers are as follows: PLIMITRU = 0000 0000h, PLIMITRM = 0000 0000h, and PLIMITRL = 0000 00ACh.

The gradient limit values to be set are valid when time synchronization correction mode is mode 2 (the STMR.CMOD bit is 1) and the gradient is controlled by software (the STMR.W10S bit is 1).

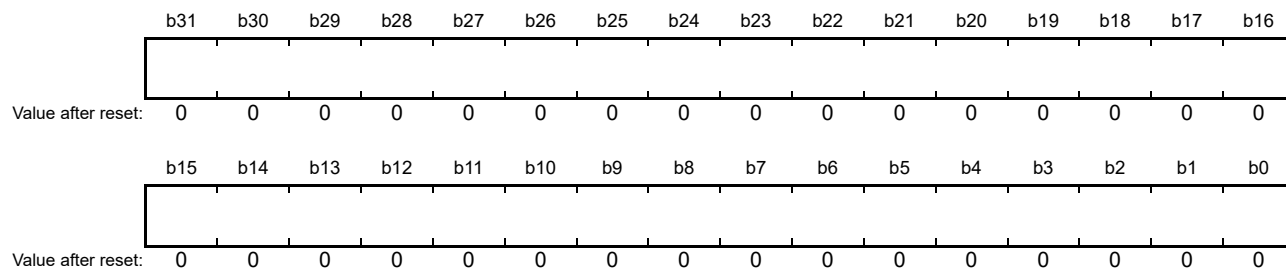
37.2.21 Negative Gradient Limit Registers (MLIMITRU, MLIMITRM, MLIMITRL)

Address(es): EPTPC.MLIMITRU 000C 4134h



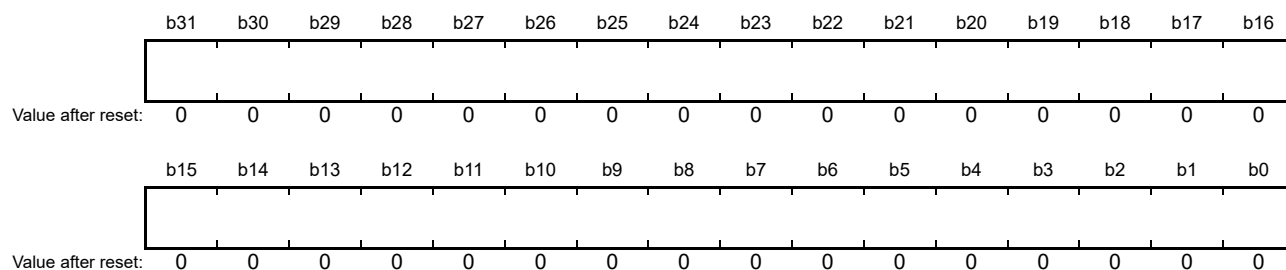
Bit	Symbol	Bit Name	Description	R/W
b30 to b0	—	—	These bits hold the setting for the higher-order 31 bits of the limit for the negative gradient.	R/W
b31	—	Reserved	This bit is read as 0. The write value should be 0.	R

Address(es): EPTPC.MLIMITRM 000C 4138h



Bit	Symbol	Bit Name	Description	R/W
b31 to b0	—	—	These bits hold the setting for the middle-order 32 bits of the limit for the negative gradient.	R/W

Address(es): EPTPC.MLIMITRL 000C 413Ch



Bit	Symbol	Bit Name	Description	R/W
b31 to b0	—	—	These bits hold the setting for the lower-order 32 bits of the limit for the negative gradient.	R/W

Registers MLIMITRU, MLIMITRM, and MLIMITRL are used to specify a lower limit for the gradient (= negative gradient) for use in time synchronization. Use two's-complement representation to set a lower limit. When setting a lower limit, write values consecutively to MLIMITRU, MLIMITRM, and MLIMITRL in that order.

The gradients that are less than the value specified in registers MLIMITRU, MLIMITRM, and MLIMITRL are not used in time synchronization. Registers MLIMITRU, MLIMITRM, and MLIMITRL are not used when the device is operating as a master clock. Registers MLIMITRU, MLIMITRM, and MLIMITRL are effective while the STMR.CMOD and W10S bits are 1.

The procedure for setting a value and the minimum value that can be set are the same as for registers PLIMITRU, PLIMITRM, and PLIMITRL.

37.2.22 Statistical Information Retention Control Register (GETINFOR)

Address(es): EPTPC.GETINFOR 000C 4140h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	INFO
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	INFO	Information Retention Control	[When written] 0: Has no effects. 1: Information is retained. [When read] 0: Information retention is completed. 1: Processing for information retention is in progress. Once information fetching is directed, values of various statistical information read before completion of information fetching are not guaranteed.	R/W
b31 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R

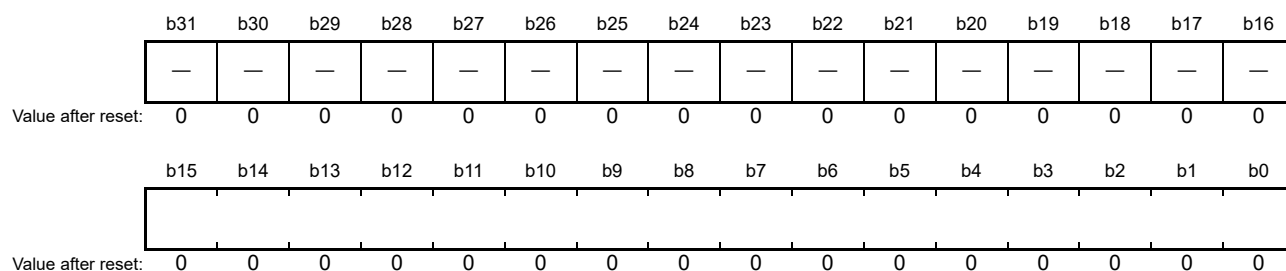
The GETINFOR register controls retention of the following statistical information.

- Registers LCCVRU, LCCVRM, and LCCVRL
- Registers PW10VRU, PW10VRM, and PW10VRL
- Registers MW10RU, MW10RM, and MW10RL

The only value that is writable to the INFO bit is 1. When setting a value in registers PW10VRU, PW10VRM, and PW10VRL, or registers MW10RU, MW10RM, and MW10RL, set the INFO bit to 1 only while the STMR.W10S bit is 1. If the INFO bit is set to 1 before acquisition of the worst-10 values is completed, whether the information retained in registers PW10VRU, PW10VRM, and PW10VRL and registers MW10RU, MW10RM, MW10RL is correct or not is not guaranteed. Use the GETW10R.GW10 bit to confirm that acquisition has been completed before setting the INFO bit to 1. The INFO bit is automatically returned to 0 on completion of information fetching.

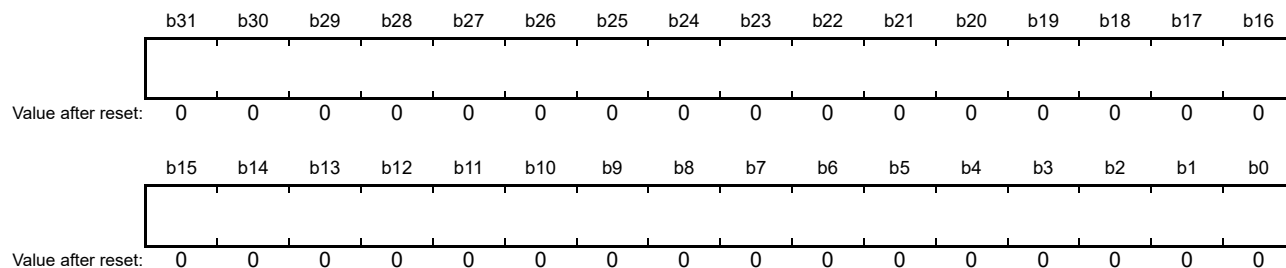
37.2.23 Local Clock Counters (LCCVRU, LCCVRM, LCCVRL)

Address(es): EPTPC.LCCVRU 000C 4170h



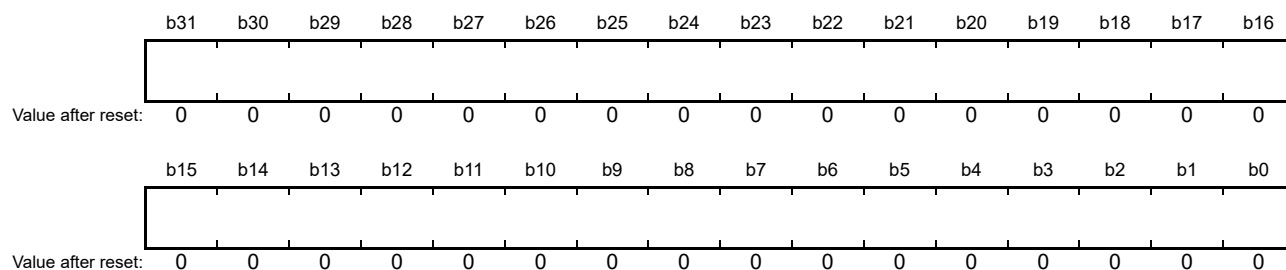
Bit	Symbol	Bit Name	Description	R/W
b15 to b0	—	—	These bits are for reading the higher-order 16 bits of the integer portion of the local clock counter's value.	R
b31 to b16	—	Reserved	These bits are read as 0.	R

Address(es): EPTPC.LCCVRM 000C 4174h



Bit	Symbol	Bit Name	Description	R/W
b31 to b0	—	—	These bits are for reading the lower-order 32 bits of the integer portion of the local clock counter's value.	R

Address(es): EPTPC.LCCVRL 000C 4178h



Bit	Symbol	Bit Name	Description	R/W
b31 to b0	—	—	These bits are for reading the fractional portion of the local clock counter's value (in nanoseconds).	R

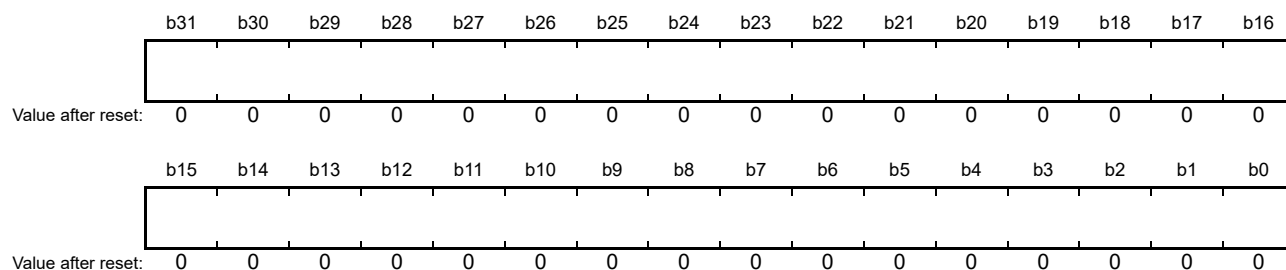
Registers LCCVRU, LCCVRM, and LCCVRL indicate the local clock counter's value.

When the GETINFOR.INFO bit is set to 1, the value of the local clock counter at that time is stored in registers LCCVRU, LCCVRM, and LCCVRL. The higher-order 16 bits of the integer portion in seconds are saved in LCCVRU, the lower-order 32 bits of the integer portion in seconds are saved in LCCVRM, and the fractional portion in nanoseconds is saved in LCCVRL.

For example, if the local time information is 14:25, 44 seconds, 10 milliseconds, 23 microseconds, and 39 nanoseconds, registers LCCVRU, LCCVRM, and LCCVRL have $14 \times 3600 + 25 \times 60 + 44 = 51944$ (s) = 0000 0000 CAE8h as the setting of the higher-order 48 bits and $10 \times 10^6 + 23 \times 10^3 + 39 = 10023039$ (ns) = 0098 F07Fh as the setting of the lower-order 32 bits.

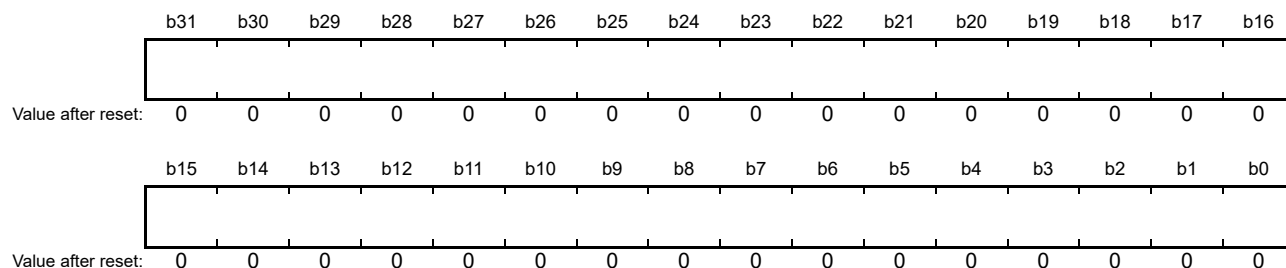
37.2.24 Positive Gradient Worst 10 Value Registers (PW10VRU, PW10VRM, PW10VRL)

Address(es): EPTPC.PW10VRU 000C 4210h



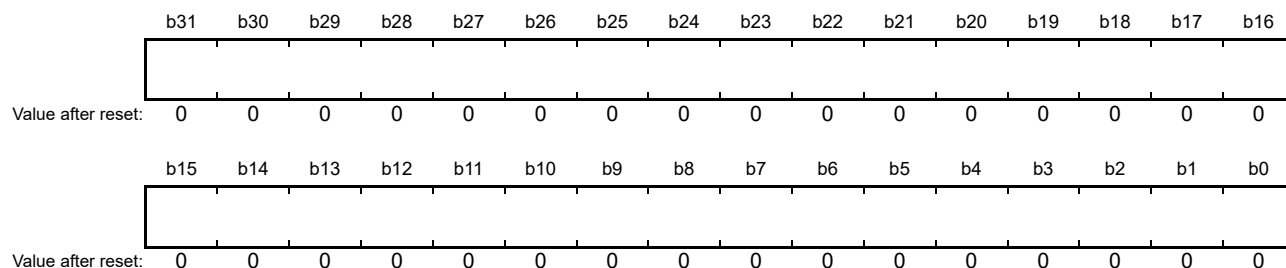
Bit	Symbol	Bit Name	Description	R/W
b31 to b0	—	—	These bits are for reading the higher-order 32 bits of the positive gradient value.	R

Address(es): EPTPC.PW10VRM 000C 4214h



Bit	Symbol	Bit Name	Description	R/W
b31 to b0	—	—	These bits are for reading the middle-order 32 bits of the positive gradient value.	R

Address(es): EPTPC.PW10VRL 000C 4218h



Bit	Symbol	Bit Name	Description	R/W
b31 to b0	—	—	These bits are for reading the lower-order 32 bits of the positive gradient value.	R

Registers PW10VRU, PW10VRM, and PW10VRL indicate the worst 10 of the positive gradient values.

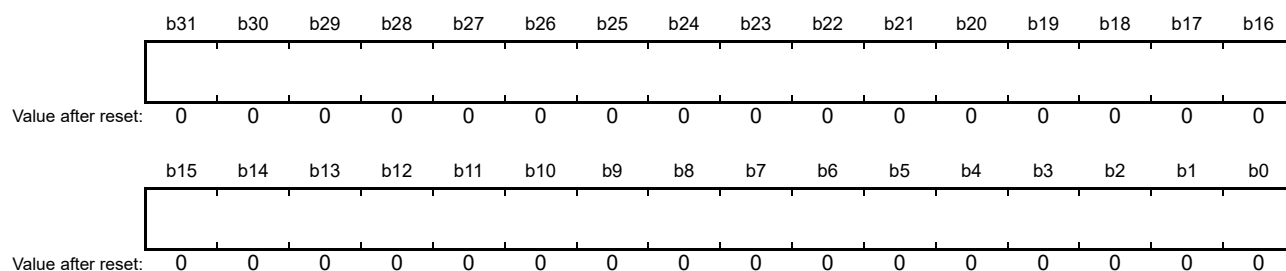
When the GETINFOR.INFO bit is set to 1, the worst 10 value at that time is stored in registers PW10VRU, PW10VRM, and PW10VRL.

The format of the worst 10 gradients stored in registers PW10VRU, PW10VRM, and PW10VRL are the same as for registers PLIMITRU, PLIMITRM, and PLIMITRL. For details on the format of the values, see the description of the PLIMITR registers.

Registers PW10VRU, PW10VRM, and PW10VRL are not used when the device is used as a master clock.

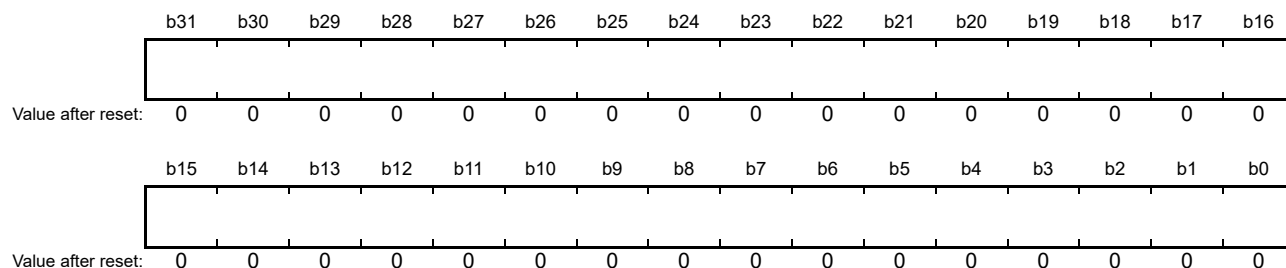
37.2.25 Negative Gradient Worst 10 Value Registers (MW10RU, MW10RM, MW10RL)

Address(es): EPTPC.MW10RU 000C 42D0h



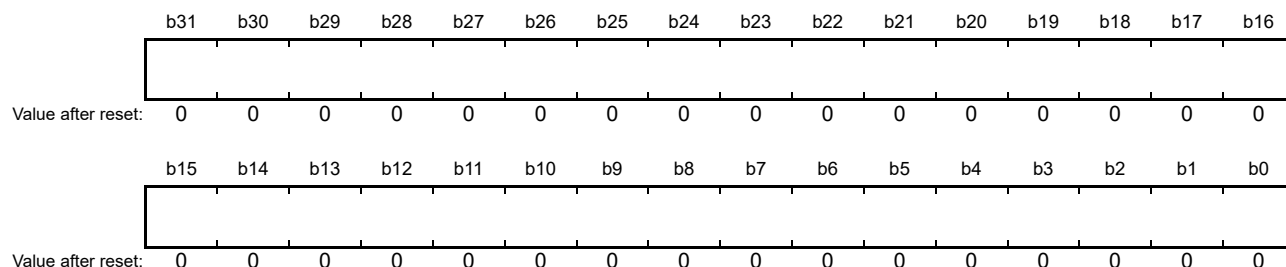
Bit	Symbol	Bit Name	Description	R/W
b31 to b0	—	—	These bits are for reading the higher-order 32 bits of the negative gradient value.	R

Address(es): EPTPC.MW10RM 000C 42D4h



Bit	Symbol	Bit Name	Description	R/W
b31 to b0	—	—	These bits are for reading the middle-order 32 bits of the negative gradient value.	R

Address(es): EPTPC.MW10RL 000C 42D8h



Bit	Symbol	Bit Name	Description	R/W
b31 to b0	—	—	These bits are for reading the lower-order 32 bits of the negative gradient value.	R

Registers MW10RU, MW10RM, and MW10RL indicate the worst 10 of the negative gradient values.

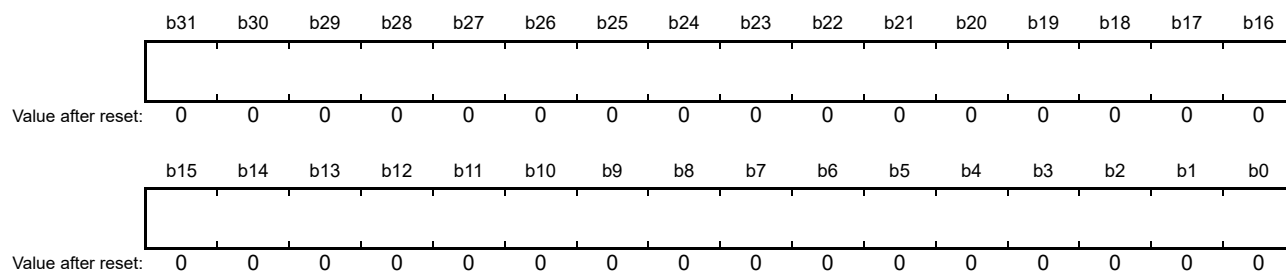
When the GETINFOR.INFO bit is set to 1, the worst 10 value at that time is stored in registers MW10RU, MW10RM, and MW10RL.

The format of the worst 10 gradients stored in registers MW10RU, MW10RM, and MW10RL are the same as for registers MLIMITRU, MLIMITRM, and MLIMITRL. For details on the format of the values, see the description of the MLIMITR registers.

Registers MW10RU, MW10RM, and MW10RL are not used when the device is used as a master clock.

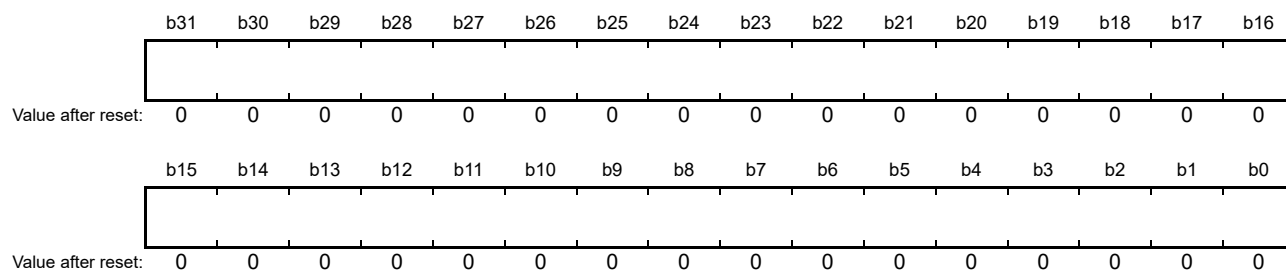
37.2.26 Timer Start Time Setting Registers m (TMSTTRUm, TMSTTRLm) (m = 0 to 5)

Address(es): EPTPC.TMSTTRU0 000C 4300h, EPTPC.TMSTTRU1 000C 4310h, EPTPC.TMSTTRU2 000C 4320h, EPTPC.TMSTTRU3 000C 4330h, EPTPC.TMSTTRU4 000C 4340h, EPTPC.TMSTTRU5 000C 4350h



Bit	Symbol	Bit Name	Description	R/W
b31 to b0	—	—	These bits hold the setting for the higher-order 32 bits of the start time of the pulse output timer in nanoseconds.	R/W

Address(es): EPTPC.TMSTTRL0 000C 4304h, EPTPC.TMSTTRL1 000C 4314h, EPTPC.TMSTTRL2 000C 4324h, EPTPC.TMSTTRL3 000C 4334h, EPTPC.TMSTTRL4 000C 4344h, EPTPC.TMSTTRL5 000C 4354h



Bit	Symbol	Bit Name	Description	R/W
b31 to b0	—	—	These bits hold the setting for the lower-order 32 bits of the start time of the pulse output timer in nanoseconds.	R/W

Registers TMSTTRUm and TMSTTRLm are used to set the start time of pulse output timer m.

Set the start time of pulse output timer m (64 bits) in units of nanoseconds. Though the setting is in units of nanoseconds, the start time of pulse output timer m depends on the resolution of the STCA clock.

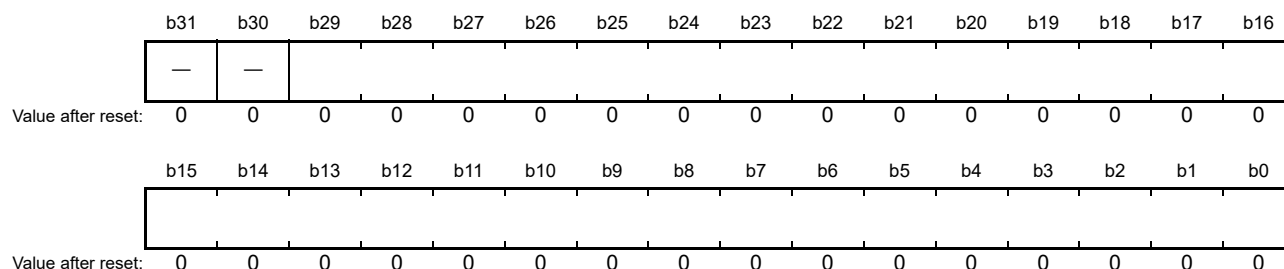
For example, if the STCA clock is running at 50 MHz, one cycle takes 20 ns, so the time at which the timer starts may differ from the time set in registers TMSTTRUm and TMSTTRLm by up to 20 ns.

When writing to registers TMSTTRUm and TMSTTRLm, write values consecutively in order of TMSTTRUm and then TMSTTRLm while the TMSTARTR.ENm bit is 0.

Note, however, that the format for setting times in registers TMSTTRUm and TMSTTRLm differs from that described in section 37.2.23, Local Clock Counters (LCCVRU, LCCVRM, LCCVRL).

37.2.27 Timer Cycle Setting Registers m (TMCYCRm) (m = 0 to 5)

Address(es): EPTPC.TMCYCR0 000C 4308h, EPTPC.TMCYCR1 000C 4318h, EPTPC.TMCYCR2 000C 4328h, EPTPC.TMCYCR3 000C 4338h, EPTPC.TMCYCR4 000C 4348h, EPTPC.TMCYCR5 000C 4358h



Bit	Symbol	Bit Name	Description	R/W
b29 to b0	—	—	These bits set the cycle of the pulse output timer in nanoseconds. Set a value that is equivalent to at least four cycles of the STCA clock.	R/W
b31, b30	—	Reserved	These bits are read as 0. The write value should be 0.	R

A TMCYCRm register specifies the period of the output signal generated by the corresponding pulse output timer m. Set a value in nanoseconds that is equivalent to at least four cycles of the STCA clock while the value of the TMSTARTR.ENm bit is 0.

Though the settings of TMCYCRm registers are in units of nanoseconds, the period of the output signal generated by pulse output timer m and the time at which the timer starts depend on the period of the STCA clock. For example, one cycle of the STCA clock running at 50 MHz takes 20 ns, so the clock source for counting by pulse output timer m may differ from the period set in the TMCYCRm registers by up to 19 ns. The SYNFP module handles calculations to correct this difference.

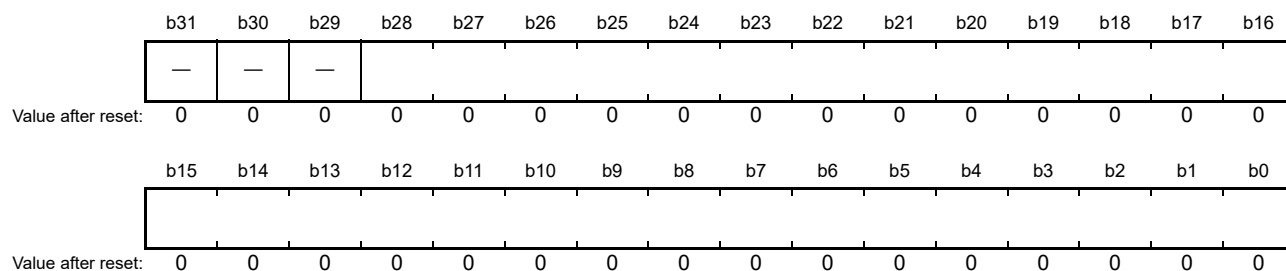
For example, if the setting for the timer period is 81 ns and the STCA clock is running at 50 MHz, the only available settings close to the actual timer period are for 80 ns or 100 ns. By setting the timer period in the SYNFP module to 80 ns for 19 and to 100 ns for 1 of every 20 cycles, we are able to adjust the average period to 81 ns.

$$(80 \text{ (ns)} \times 19 + 100 \text{ (ns)} \times 1) / 20 = 81 \text{ (ns)}$$

The minimum value that can be set in a TMCYCRm register is four cycles of the STCA clock. For example, if the STCA clock is running at 50 MHz, the minimum setting corresponds to 80 ns. Timer operation is not guaranteed if a value set in one of these registers is less than this value.

37.2.28 Timer Pulse Width Setting Register m (TMPLSRm) (m = 0 to 5)

Address(es): EPTPC.TMPLSR0 000C 430Ch, EPTPC.TMPLSR1 000C 431Ch, EPTPC.TMPLSR2 000C 432Ch,
EPTPC.TMPLSR3 000C 433Ch, EPTPC.TMPLSR4 000C 434Ch, EPTPC.TMPLSR5 000C 435Ch



Bit	Symbol	Bit Name	Description	R/W
b28 to b0	—	—	These bits set the width at high level of the pulse signal from the timer in nanoseconds. Set a value that is equivalent to at least two cycles of the STCA clock.	R/W
b31 to b29	—	Reserved	These bits are read as 0. The write value should be 0.	R

A TMPLSRm register specifies the width at high level of the output signal generated by the corresponding pulse output timer m.

When the TMSTARTR.ENm bit is 0, set a value corresponding to a time no shorter than two cycles of the STCA clock in units of nanoseconds.

The higher-order 3 bits of the TMPLSRm register are reserved. These bits are read as 000b. When writing, write 000b to these bits.

Though the settings of the TMPLSRm register are in units of nanoseconds, the width at high level of the signal from the timer depends on the period of the STCA clock. The method for correcting the width at high level of the signal from the timer is the same as that for correcting the timer periods set by the TMCYCRm register.

37.2.29 Timer Start Register (TMSTARTR)

Address(es): EPTPC.TMSTARTR 000C 437Ch

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	EN5	EN4	EN3	EN2	EN1	EN0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	EN0	Pulse Output Timer 0 Start	0: Stops pulse output timer 0. 1: Starts pulse output timer 0.	R/W
b1	EN1	Pulse Output Timer 1 Start	0: Stops pulse output timer 1. 1: Starts pulse output timer 1.	R/W
b2	EN2	Pulse Output Timer 2 Start	0: Stops pulse output timer 2. 1: Starts pulse output timer 2.	R/W
b3	EN3	Pulse Output Timer 3 Start	0: Stops pulse output timer 3. 1: Starts pulse output timer 3.	R/W
b4	EN4	Pulse Output Timer 4 Start	0: Stops pulse output timer 4. 1: Starts pulse output timer 4.	R/W
b5	EN5	Pulse Output Timer 5 Start	0: Stops pulse output timer 5. 1: Starts pulse output timer 5.	R/W
b31 to b6	—	Reserved	These bits are read as 0. The write value should be 0.	R

The TMSTARTR register is used to start and stop pulse output timer m.

37.2.30 PRC-TC Status Register (PRSR)

Address(es): EPTPC.PRSR 000C 4400h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	URE1	URE0	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	MACE	—	—	—	—	OVRE3	OVRE2	OVRE1	OVRE0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	OVRE0	Relay Packet Overflow Detection Flag 0	0: No overflow in transfer of data from SYNFP1 to PTPEDMAC 1: An overflow has been detected in transfer of data from SYNFP1 to PTPEDMAC.	R/W*1
b1	OVRE1	Relay Packet Overflow Detection Flag 1	0: No overflow in transfer of data from SYNFP0 to PTPEDMAC 1: An overflow has been detected in transfer of data from SYNFP0 to PTPEDMAC.	R/W*1
b2	OVRE2	Relay Packet Overflow Detection Flag 2	0: No overflow in transfer of data from SYNFP1 to SYNFP0 1: An overflow has been detected in transfer of data from SYNFP1 to SYNFP0.	R/W*1
b3	OVRE3	Relay Packet Overflow Detection Flag 3	0: No overflow in transfer of data from SYNFP0 to SYNFP1 1: An overflow has been detected in transfer of data from SYNFP0 to SYNFP1.	R/W*1
b7 to b4	—	Reserved	These bits are read as 0. The write value should be 0.	R
b8	MACE	Source MAC Address Mismatch Detection Flag	0: A MAC address mismatch has not been detected. 1: A MAC address mismatch has been detected.	R/W*1
b15 to b9	—	Reserved	These bits are read as 0. The write value should be 0.	R
b27 to b16	—	Reserved	These bits are read as undefined. The write value should be 0.	R/W
b28	URE0	Relay Packet Underflow Detection Flag 0	0: No underflow in transfer of data from SYNFP1 to SYNFP0 1: An underflow has been detected in transfer of data from SYNFP1 to SYNFP0.	R/W*1
b29	URE1	Relay Packet Underflow Detection Flag 1	0: No underflow in transfer of data from SYNFP0 to SYNFP1 1: An underflow has been detected in transfer of data from SYNFP0 to SYNFP1.	R/W*1
b31, b30	—	Reserved	These bits are read as 0. The write value should be 0.	R

Note 1. Writing 1 clears the flag. Writing 0 does not affect the flag value.

The PRSR register indicates the state of the PRC-TC module.

The PRC-TC module transmits and receives Ethernet packets through Ethernet port 0, Ethernet port 1, and the PTPEDMAC.

The PRSR register indicates the states as overflows, underflows, and non-matching MAC addresses in the transmission and reception of Ethernet packets.

OVRE0 Flag (Relay Packet Overflow Detection Flag 0)

This flag indicates that the FIFO buffer has overflowed while packets were being transferred from the SYNFP1 module to the PTPEDMAC. The data received in the PTPEDMAC may be incorrect if the setting of the OVRE0 flag is 1.

OVRE1 Flag (Relay Packet Overflow Detection Flag 1)

This flag indicates that the FIFO buffer has overflowed while packets were being transferred from the SYNFP0 module to the PTPEDMAC. The data received in the PTPEDMAC may be incorrect if the setting of the OVRE1 flag is 1.

OVRE2 Flag (Relay Packet Overflow Detection Flag 2)

This flag indicates that the FIFO buffer has overflowed while packets were being transferred from the SYNFP1 module to the SYNFP0 module.

OVRE3 Flag (Relay Packet Overflow Detection Flag 3)

This flag indicates that the FIFO buffer has overflowed while packets were being transferred from the SYNFP0 module to the SYNFP1 module.

MACE Flag (Source MAC Address Mismatch Detection Flag)

This flag becomes 1 if the source MAC address of a packet for transmission from the PTPEDMAC matches none of registers PRMACRU0, PRMACRL0, PRMACRU1, and PRMACRL1.

URE0 Flag (Relay Packet Underflow Detection Flag 0)

This flag indicates that the FIFO buffer has underflowed while packets were being transferred from the SYNFP1 module to the SYNFP0 module.

URE1 Flag (Relay Packet Underflow Detection Flag 1)

This flag indicates that the FIFO buffer has underflowed while packets were being transferred from the SYNFP0 module to the SYNFP1 module.

37.2.31 PRC-TC Status Notification Enable Register (PRIPR)

Address(es): EPTPC.PRIPR 000C 4404h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	URE1	URE0	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	MACE	—	—	—	—	OVRE3	OVRE2	OVRE1	OVRE0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

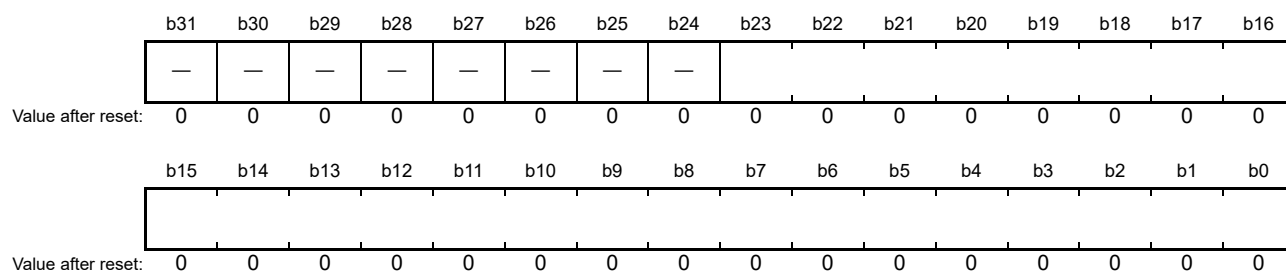
Bit	Symbol	Bit Name	Description	R/W
b0	OVRE0	PRSR.OVRE0 Status Notification Enable	0: Disables notification of the state of PRSR.OVRE0. 1: Enables notification of the state of PRSR.OVRE0.	R/W
b1	OVRE1	PRSR.OVRE1 Status Notification Enable	0: Disables notification of the state of PRSR.OVRE1. 1: Enables notification of the state of PRSR.OVRE1.	R/W
b2	OVRE2	PRSR.OVRE2 Status Notification Enable	0: Disables notification of the state of PRSR.OVRE2. 1: Enables notification of the state of PRSR.OVRE2.	R/W
b3	OVRE3	PRSR.OVRE3 Status Notification Enable	0: Disables notification of the state of PRSR.OVRE3. 1: Enables notification of the state of PRSR.OVRE3.	R/W
b7 to b4	—	Reserved	These bits are read as 0. The write value should be 0.	R
b8	MACE	PRSR.MACE Status Notification Enable	0: Disables notification of the state of PRSR.MACE 1: Enables notification of the state of PRSR.MACE	R/W
b27 to b9	—	Reserved	These bits are read as 0. The write value should be 0.	R
b28	URE0	PRSR.URE0 Status Notification Enable	0: Disables notification of the state of PRSR.URE0. 1: Enables notification of the state of PRSR.URE0.	R/W
b29	URE1	PRSR.URE1 Status Notification Enable	0: Disables notification of the state of PRSR.URE1. 1: Enables notification of the state of PRSR.URE1.	R/W
b31, b30	—	Reserved	These bits are read as 0. The write value should be 0.	R

Note: Refer to section 37.4, Interrupts, for details on interrupt system.

The PRIPR register specifies whether the MIESR.PRC flag does or does not reflect changes in the state of the PRC-TC module.

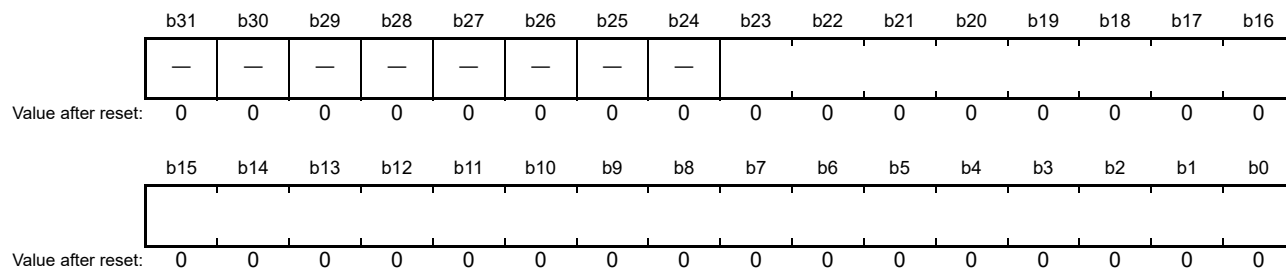
37.2.32 Channel 0 Local MAC Address Registers (PRMACRU0, PRMACRL0)

Address(es): EPTPC.PRMACRU0 000C 4410h



Bit	Symbol	Bit Name	Description	R/W
b23 to b0	—	—	These bits hold the setting for the higher-order 24 bits of the local MAC address for Ethernet port 0.	R/W
b31 to b24	—	Reserved	These bits are read as 0. The write value should be 0.	R

Address(es): EPTPC.PRMACRL0 000C 4414h



Bit	Symbol	Bit Name	Description	R/W
b23 to b0	—	—	These bits hold the setting for the lower-order 24 bits of the local MAC address for Ethernet port 0.	R/W
b31 to b24	—	Reserved	These bits are read as 0. The write value should be 0.	R

Registers PRMACRU0 and PRMACRL0 are used to set the local MAC address for Ethernet port 0. Set the higher-order 24 bits and the lower-order 24 bits of the MAC address in PRMACRU0 and PRMACRL0, respectively.

These registers are used in transmission from the PTPEDMAC to send frames from transmission sources with MAC addresses matching the value of this register to Ethernet port 0. Set registers PRMACRU0 and PRMACRL0 to the same value as registers SYMACRU and SYMACRL for the EPTPC0 module.

Rewrite registers PRMACRU0 and PRMACRL0 before starting the EDMAC, ETHERC, and PTPEDMAC. Do not change the settings while the EPTPC is operating.

37.2.33 Channel 1 Local MAC Address Registers (PRMACRU1, PRMACRL1)

Address(es): EPTPC.PRMACRU1 000C 4418h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—								
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b23 to b0	—	—	These bits hold the setting for the higher-order 24 bits of the local MAC address for Ethernet port 1.	R/W
b31 to b24	—	Reserved	These bits are read as 0. The write value should be 0.	R

Address(es): EPTPC.PRMACRL1 000C 441Ch

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—								
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b23 to b0	—	—	These bits hold the setting for the lower-order 24 bits of the local MAC address for Ethernet port 1.	R/W
b31 to b24	—	Reserved	These bits are read as 0. The write value should be 0.	R

Registers PRMACRU1 and PRMACRL1 are used to set the local MAC address for Ethernet port 1. Set the higher-order 24 bits and the lower-order 24 bits of the MAC address in registers PRMACRU1 and PRMACRL1, respectively.

These registers are used in transmission from the PTPEDMAC to send frames from transmission sources with MAC addresses matching the value of this register to Ethernet port 1. Set registers PRMACRU1 and PRMACRL1 to the same value as registers SYMACRU and SYMACRL for the EPTPC1 module.

Rewrite registers PRMACRU1 and PRMACRL1 before starting the EDMAC, ETHERC, or PTPEDMAC. Do not change the settings while the EPTPC is operating.

37.2.34 Packet Transmission Control Register (TRNDISR)

Address(es): EPTPC.TRNDISR 000C 4420h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	TDIS[1:0]	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b1, b0	TDIS[1:0]	Packet Transmission Control	b1 b0 0 0: PTP packets are transmitted through both Ethernet port 0 and Ethernet port 1. 0 1: PTP packets are only transmitted through Ethernet port 0. 1 0: PTP packets are only transmitted through Ethernet port 1. 1 1: Setting prohibited	R/W
b31 to b2	—	Reserved	These bits are read as 0. The write value should be 0.	R

The TRNDISR register controls transmission of PTP packets when Ethernet port 0 and Ethernet port 1 have the same local MAC address.

When registers PRMACRU0 and PRMACRL0 and registers PRMACRU1 and PRMACRL1 have the same local MAC address setting, the EPTPC can select whether PTP packets with the matching address are transmitted through both ports or are only transmitted through one port or the other.

If Ethernet port 0 and Ethernet port 1 have different local MAC address settings, set the TDIS[1:0] bits to 00b. If the setting is other than 00b, the transmission of frames with an source MAC address matching that of the Ethernet port may be blocked.

Rewrite the TRNDISR register before starting the EDMAC, ETHERC, or PTPEDMAC. Do not change the settings while the EPTPC is operating.

37.2.35 Relay Mode Register (TRNMR)

Address(es): EPTPC.TRNMR 000C 4430h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	FWD1	FWD0	—	—	—	—	—	—	—	MOD
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	MOD	Cut-Through Mode	0: Store-and-forward 1: Cut-through	R/W
b7 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R
b8	FWD0	Channel 0 Relay Enable	0: Unicast, multicast (other than PTP packets), and broadcast messages from the other station are not relayed from port 0 to port 1. 1: Unicast, multicast (other than PTP packets), and broadcast messages from the other station are relayed from port 0 to port 1.	R/W
b9	FWD1	Channel 1 Relay Enable	0: Unicast, multicast (other than PTP packets), and broadcast messages from the other station are not relayed from port 1 to port 0. 1: Unicast, multicast (other than PTP packets), and broadcast messages from the other station are relayed from port 1 to port 0.	R/W
b31 to b10	—	Reserved	These bits are read as 0. The write value should be 0.	R

The TRNMR register is used to control relaying of packets between Ethernet port 0 and Ethernet port 1 by the PRC-TC module. For example, as Figure 37.3 shows, if the network is in a daisy-chain configuration, packets can be transferred through all three devices by enabling relaying of packets between Ethernet port 0 and Ethernet port 1 in the center of the figure.

For recalculating the CRC value in packet transfer, in the case of cut-through mode, a packet that had an abnormal CRC value may be transferred with a normal CRC value. Use store-and-forward mode when abnormal packets need to be discarded. There is no difference in the latency of transfer in cut-through mode and store-and-forward mode when the length of the packets is shorter than 96 bytes.

Rewrite the TRNMR register before starting the EDMAC, ETHERC, or PTPEDMAC. Do not change the settings while the EPTPC is operating.

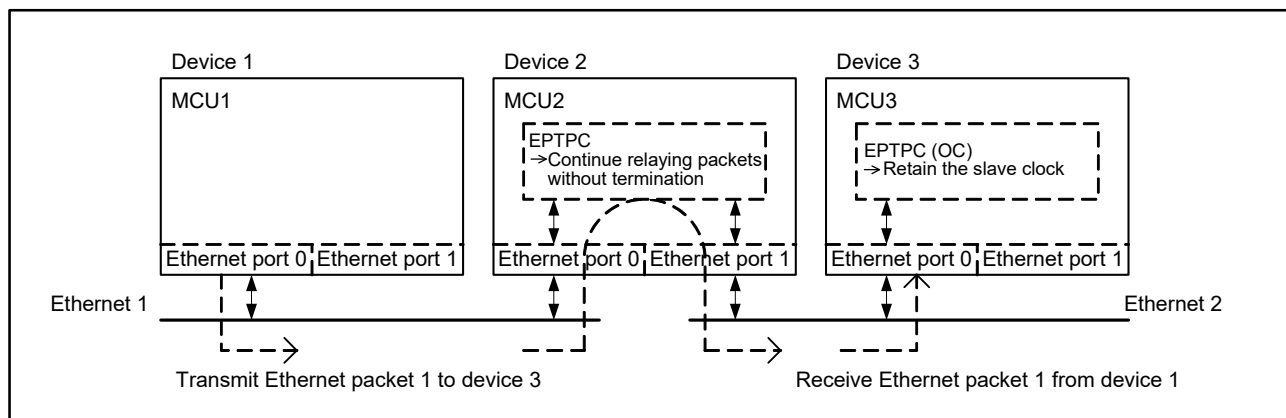
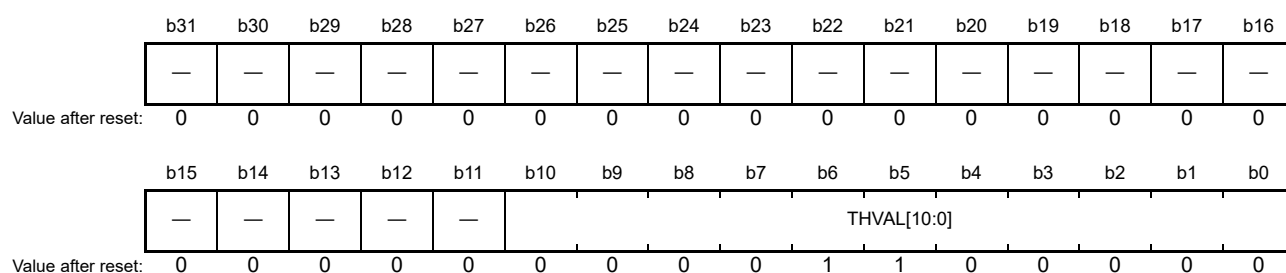


Figure 37.3 Network Configuration Example

37.2.36 Cut-Through Transfer Start Threshold Register (TRNCTTDR)

Address(es): EPTPC.TRNCTTDR 000C 4434h



Bit	Symbol	Bit Name	Description	R/W
b1, b0	THVAL[10:0]	FIFO Read Start Threshold	Threshold for starting to read data from the relay FIFO in cut-through mode (specified as the number of bytes)*1, *2	R
b10 to b2				R/W
b31 to b11	—	Reserved	These bits are read as 0. The write value should be 0.	R

Note 1. A value cannot be set in the lower-order 2 bits. These bits are fixed to 0.

Note 2. A value of less than 96 bytes cannot be set.

The TRNCTTDR register is used to set the threshold for the transmitting port starting to read data from the relay FIFO when the cut-through method is selected for relaying between the Ethernet ports.

When the TRNMR.MOD bit is 1, data transfer can start without waiting for all frame data to be stored in the relay FIFO. The setting of the THVAL[10:0] bits is the amount of data, in bytes, that must be stored in the relay FIFO before transmission starts. Set a multiple of four as the threshold value.

Reading from the relay FIFO starts when either of the following conditions is met.

- The relay FIFO holds at least as much data as the number of bytes specified by the THVAL[10:0] bits.
- The relay FIFO holds at least one frame.

Rewrite the TRNCTTDR register before starting the EDMAC, ETHERC, or PTPEDMAC. Do not change the settings while the EPTPC is operating.

37.2.37 SYNFP Status Register (SYSR)

Address(es): EPTPC0.SYSR 000C 4800h, EPTPC1.SYSR 000C 4C00h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	GENDN	RESDN
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	INFABT	—	RECLP	—	—	—	—	—	DRQOVR	INTDEV	DRPTO	—	MPDUD	INTCHG	OFMUD
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	OFMUD	offsetFromMaster Value Update Flag	0: The offsetFromMaster value has not been updated. 1: The offsetFromMaster value has been updated.	R/W*1
b1	INTCHG	Receive logMessageInterval Value Change Detection Flag	0: No change in the received logMessageInterval value. 1: A change in the received logMessageInterval value.	R/W*1
b2	MPDUD	meanPathDelay Value Update Flag	0: The meanPathDelay value has not been updated. 1: The meanPathDelay value has been updated.	R/W*1
b3	—	Reserved	This bit is read as 0. The write value should be 0.	R
b4	DRPTO	Delay_Resp/Pdelay_Resp Reception Timeout Detection Flag	0: A Delay_Resp/Pdelay_Resp timeout has not occurred. 1: A Delay_Resp/Pdelay_Resp timeout has occurred.	R/W*1
b5	INTDEV	Receive logMessageInterval Value Out-of-Range Flag	0: The received logMessageInterval value is within the range. 1: The received logMessageInterval value is out of the range.	R/W*1
b6	DRQOVR	Delay_Req Reception FIFO Overflow Detection Flag	0: The received Delay_Req has not caused the reception FIFO to overflow. 1: The received Delay_Req has caused the reception FIFO to overflow.	R/W*1
b11 to b7	—	Reserved	These bits are read as 0. The write value should be 0.	R
b12	RECLP	Loop Reception Detection Flag	0: A received message has not returned through a loop. 1: A received message has returned through a loop.	R/W*1
b13	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b14	INFABT	Control Information Abnormality Detection Flag	0: No abnormality in control information 1: Abnormality in control information	R/W*1
b15	—	Reserved	This bit is read as 0. The write value should be 0.	R
b16	RESDN	Response Stop Completion Detection Flag	0: Stopping responses has not been completed. 1: Stopping responses has been completed.	R/W*1
b17	GENDN	Generation Stop Completion Detection Flag	0: Stopping generation has not been completed. 1: Stopping generation has been completed.	R/W*1
b23 to b18	—	Reserved	These bits are read as undefined. The write value should be 0.	R/W
b31 to b24	—	Reserved	These bits are read as 0. The write value should be 0.	R

Note 1. Writing 1 clears the flag. Writing 0 does not affect the flag value.

The SYSR register indicates the state of the SYNFP module.

OFMUD Flag (offsetFromMaster Value Update Flag)

This flag indicates that the value of offsetFromMaster has been updated.

INTCHG Flag (Receive logMessageInterval Value Change Detection Flag)

This flag indicates that the logMessageInterval value of the Delay_Resp, Sync or Announce message differs from the previously received value.

MPDUD Flag (meanPathDelay Value Update Flag)

This flag indicates that the value of meanPathDelay has been updated.

DRPTO Flag (Delay_Resp/Pdelay_Resp Reception Timeout Detection Flag)

This flag indicates that a Delay_Resp or Pdelay_Resp message has not been received within the period set in the RSTOUTR register.

INTDEV Flag (Receive logMessageInterval Value Out-of-Range Flag)

This flag indicates the reception of a Delay_Resp message with a logMessageInterval value outside the range -7 to $+6$.

DRQOVR Flag (Delay_Req Reception FIFO Overflow Detection Flag)

This flag indicates that the FIFO buffer for storing information from received Delay_Req messages holds 32 or more entries.

RECLP Flag (Loop Reception Detection Flag)

This flag indicates that the value of the sourcePortIdentity field in a received PTP message matches the local PortIdentity (as set by registers SYCIDRU, SYCIDRL, and SYPNUMR).

INFABT Flag (Control Information Abnormality Detection Flag)

This flag indicates that the control information included a mismatch.

When this bit becomes 1, reset the EPTPC, PTPEDMAC, and the corresponding channels ETHERC and EDMAC. Refer to section 36.5.4, Procedure to Reset Ethernet Controller for the procedure to reset these modules.

RESDN Flag (Response Stop Completion Detection Flag)

This flag indicates the end of processing for transmission of a Delay_Resp or Pdelay_Resp as response messages when the handling of a received Delay_Req or Pdelay_Req by the SYNFP module has been disabled by the setting of the SYRFL1R or SYRVLDR register.

GENDN Flag (Generation Stop Completion Detection Flag)

This flag indicates the end of processing for transmission of messages of a type disabled in the SYTRENR or SYRVLDR register.

37.2.38 SYNFP Status Notification Enable Register (SYIPR)

Address(es): EPTPC0.SYIPR 000C 4804h, EPTPC1.SYIPR 000C 4C04h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	GEND N	RESDN
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

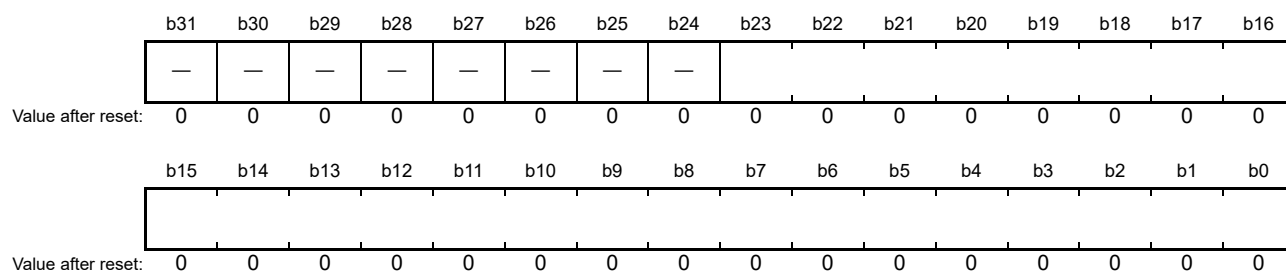
	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	INFABT	—	RECLP	—	—	—	—	—	DRQO VR	INTDE V	DRPTO	—	MPDU D	INTCH G	OFMU D
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	OFMUD	SYSR.OFMUD Status Notification Enable	0: Disables notification of the state of SYSR.OFMUD. 1: Enables notification of the state of SYSR.OFMUD.	R/W
b1	INTCHG	SYSR.INTCHG Status Notification Enable	0: Disables notification of the state of SYSR.INTCHG. 1: Enables notification of the state of SYSR.INTCHG.	R/W
b2	MPDUD	SYSR.MPDUD Status Notification Enable	0: Disables notification of the state of SYSR.MPDUD. 1: Enables notification of the state of SYSR.MPDUD.	R/W
b3	—	Reserved	This bit is read as 0. The write value should be 0.	R
b4	DRPTO	SYSR.DRPTO Status Notification Enable	0: Disables notification of the state of SYSR.DRPTO. 1: Enables notification of the state of SYSR.DRPTO.	R/W
b5	INTDEV	SYSR.INTDEV Status Notification Enable	0: Disables notification of the state of SYSR.INTDEV. 1: Enables notification of the state of SYSR.INTDEV.	R/W
b6	DRQOVR	SYSR.DRQOVR Status Notification Enable	0: Disables notification of the state of SYSR.DRQOVR. 1: Enables notification of the state of SYSR.DRQOVR.	R/W
b11 to b7	—	Reserved	These bits are read as 0. The write value should be 0.	R
b12	RECLP	SYSR.RECLP Status Notification Enable	0: Disables notification of the state of SYSR.RECLP. 1: Enables notification of the state of SYSR.RECLP.	R/W
b13	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b14	INFABT	SYSR.INFABT Status Notification Enable	0: Disables notification of the state of SYSR.INFABT. 1: Enables notification of the state of SYSR.INFABT.	R/W
b15	—	Reserved	This bit is read as 0. The write value should be 0.	R
b16	RESDN	SYSR.RESDN Status Notification Enable	0: Disables notification of the state of SYSR.RESDN. 1: Enables notification of the state of SYSR.RESDN.	R/W
b17	GENDN	SYSR.GENDN Status Notification Enable	0: Disables notification of the state of SYSR.GENDN. 1: Enables notification of the state of SYSR.GENDN.	R/W
b23 to b18	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b31 to b24	—	Reserved	These bits are read as 0. The write value should be 0.	R

The SYIPR register specifies whether the MIESR.SYn flag does or does not reflect changes in the state of the SYNFPn module.

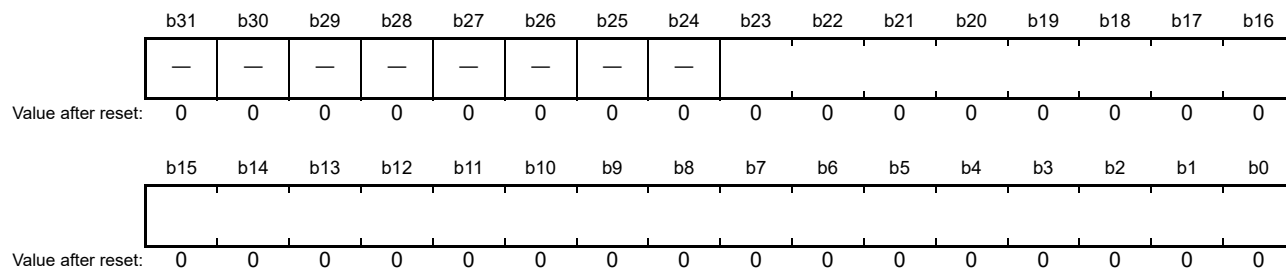
37.2.39 SYNFP MAC Address Registers (SYMACRU, SYMACRL)

Address(es): EPTPC0.SYMACRU 000C 4810h, EPTPC1.SYMACRU 000C 4C10h



Bit	Symbol	Bit Name	Description	R/W
b23 to b0	—	—	These bits hold the setting for the higher-order 24 bits of the local MAC address.	R/W
b31 to b24	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Address(es): EPTPC0.SYMACRL 000C 4814h, EPTPC1.SYMACRL 000C 4C14h



Bit	Symbol	Bit Name	Description	R/W
b23 to b0	—	—	These bits hold the setting for the lower-order 24 bits of the local MAC address.	R/W
b31 to b24	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Registers SYMACRU and SYMACRL are used to specify the local MAC address for Ethernet ports 0 and 1. Rewrite registers SYMACRU and SYMACRL before starting the EDMAC, ETHERC, or PTPEDMAC. Do not change the settings while the EPTPC is operating.

37.2.40 SYNFP LLC-CTL Value Register (SYLLCCTLR)

Address(es): EPTPC0.SYLLCCTLR 000C 4818h, EPTPC1.SYLLCCTLR 000C 4C18h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	CTL[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1

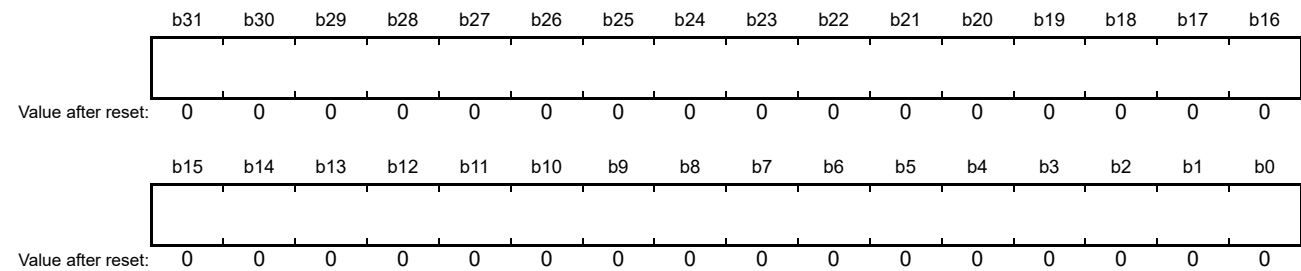
Bit	Symbol	Bit Name	Description	R/W
b7 to b0	CTL[7:0]	LLC-CTL Field	Set a value used for the control field in the LLC sublayer when generating IEEE802.3 frames.	R/W
b31 to b8	—	Reserved	These bits are read as 0. The write value should be 0.	R

The SYLLCCTLR register is used to set the control field (LLC-CTL) value of LCC frames generated by the SYNFP module.

Rewrite the SYLLCCTLR register before starting the EDMAC, ETHERC, or PTPEDMAC. Do not change the settings while the EPTPC is operating.

37.2.41 SYNFP Local IP Address Register (SYIPADDRR)

Address(es): EPTPC0.SYIPADDRR 000C 481Ch, EPTPC1.SYIPADDRR 000C 4C1Ch



Bit	Symbol	Bit Name	Description	R/W
b31 to b0	—	—	These bits hold the setting for the local IP address.	R/W

The SYIPADDRR register is used to specify the local IP address for Ethernet ports 0 and 1. Rewrite the SYIPADDRR register before starting the EDMAC, ETHERC, or PTPEDMAC. Do not change the settings while the EPTPC is operating.

37.2.42 SYNFP Specification Version Setting Register (SYSPVRR)

Address(es): EPTPC0.SYSPVRR 000C 4840h, EPTPC1.SYSPVRR 000C 4C40h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	TRSP[3:0]				VER[3:0]			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0

Bit	Symbol	Bit Name	Description	R/W
b3 to b0	VER[3:0]	versionPTP Field Value	These bits are used to set the versionPTP field value of the PTP v2 header. When a message is received, this value is compared with the versionPTP field of the received frame. In generating messages, the value is used for the versionPTP field of the frame for transmission. Set these bits to 0010b (PTP v2).	R/W
b7 to b4	TRSP[3:0]	transportSpecific Field Value	These bits are used to set the transportSpecific field value of the PTP v2 header. When a message is received, this value is compared with the transportSpecific field of the received frame. In generating messages, the value is used for the transportSpecific field of the frame for transmission. Set these bits to 0000b (IEEE 1588).	R/W
b31 to b8	—	Reserved	These bits are read as 0. The write value should be 0.	R

The SYSPVRR register is used to set the transportSpecific and versionPTP field values of the PTP v2 message header. The value should not be modified while reception or transmission of PTP messages is enabled.

37.2.43 SYNFP Domain Number Setting Register (SYDOMR)

Address(es): EPTPC0.SYDOMR 000C 4844h, EPTPC1.SYDOMR 000C 4C44h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	DNUM[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b7 to b0	DNUM[7:0]	domainNumber Field Value Setting	These bits are used to set the domainNumber field value of the PTP v2 header. When a message is received, this value is compared with the domainNumber field of the received frame as a condition for PTP reception processing. In generating messages, the value is used for the domainNumber field of the frame for transmission.	R/W
b31 to b8	—	Reserved	These bits are read as 0. The write value should be 0.	R

The SYDOMR register is used to set the domainNumber field value of the PTP v2 message header. The value should not be modified while reception or transmission of PTP messages is enabled.

37.2.44 Announce Message Flag Field Setting Register (ANFR)

Address(es): EPTPC0.ANFR 000C 4850h, EPTPC1.ANFR 000C 4C50h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	FLAG1 ₄	FLAG1 ₃	—	—	FLAG1 ₀	—	FLAG8	—	—	FLAG5	FLAG4	FLAG3	FLAG2	FLAG1	FLAG0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	FLAG0	leap61	This bit is used to set the logical value of the leap61 member of timePropertiesDS. 0: leap61 is set to FALSE. 1: leap61 is set to TRUE.	R/W
b1	FLAG1	leap59	This bit is used to set the logical value of the leap59 member of timePropertiesDS. 0: leap59 is set to FALSE. 1: leap59 is set to TRUE.	R/W
b2	FLAG2	currentUtcOffsetValid	This bit is used to set the logical value of the currentUtcOffsetValid member of timePropertiesDS. 0: currentUtcOffsetValid is set to FALSE. 1: currentUtcOffsetValid is set to TRUE.	R/W
b3	FLAG3	ptpTimescale	This bit is used to set the logical value of the ptpTimescale member of timePropertiesDS. 0: ptpTimescale is set to FALSE. 1: ptpTimescale is set to TRUE.	R/W
b4	FLAG4	timeTraceable	This bit is used to set the logical value of the timeTraceable member of timePropertiesDS. 0: timeTraceable is set to FALSE. 1: timeTraceable is set to TRUE.	R/W
b5	FLAG5	frequencyTraceable	This bit is used to set the logical value of the frequencyTraceable member of timePropertiesDS. 0: frequencyTraceable is set to FALSE. 1: frequencyTraceable is set to TRUE.	R/W
b7, b6	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b8	FLAG8	alternateMasterFlag	0: alternateMasterFlag is set to FALSE. 1: alternateMasterFlag is set to TRUE.	R/W
b9	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b10	FLAG10	unicastFlag	0: unicastFlag is set to FALSE. 1: unicastFlag is set to TRUE.	R/W
b12, b11	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b13	FLAG13	PTP profile Specific 1	0: PTP profile Specific 1 is set to FALSE. 1: PTP profile Specific 1 is set to TRUE.	R/W
b14	FLAG14	PTP profile Specific 2	0: PTP profile Specific 2 is set to FALSE. 1: PTP profile Specific 2 is set to TRUE.	R/W
b31 to b15	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

The ANFR register is used to set the flagField section of the header when the SYNFP module is to generate an Announce message.

The value specified by the ANFR register is only reflected in the SYNFP module after the SYRVLDR.ANUP bit is set to 1.

37.2.45 Sync Message Flag Field Setting Register (SYNFR)

Address(es): EPTPC0.SYNFR 000C 4854h, EPTPC1.SYNFR 000C 4C54h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	FLAG1 ₄	FLAG1 ₃	—	—	FLAG1 ₀	FLAG9	FLAG8	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b7 to b0	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b8	FLAG8	alternateMasterFlag	0: alternateMasterFlag is set to FALSE. 1: alternateMasterFlag is set to TRUE.	R/W
b9	FLAG9	twoStepFlag	Set this bit to 0 (FALSE).	R/W
b10	FLAG10	unicastFlag	0: unicastFlag is set to FALSE. 1: unicastFlag is set to TRUE.	R/W
b12, b11	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b13	FLAG13	PTP profile Specific 1	0: PTP profile Specific 1 is set to FALSE. 1: PTP profile Specific 1 is set to TRUE.	R/W
b14	FLAG14	PTP profile Specific 2	0: PTP profile Specific 2 is set to FALSE. 1: PTP profile Specific 2 is set to TRUE.	R/W
b31 to b15	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

The SYNFR register is used to set the flagField section of the header when the SYNFP module is to generate a Sync message.

The value specified by the SYNFR register is only reflected in the SYNFP module after the SYRVLDR.STUP bit is set to 1.

37.2.46 Delay_Req Message Flag Field Setting Register (DYRQFR)

Address(es): EPTPC0.DYRQFR 000C 4858h, EPTPC1.DYRQFR 000C 4C58h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	FLAG1 ₄	FLAG1 ₃	—	—	FLAG1 ₀	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b9 to b0	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b10	FLAG10	unicastFlag	0: unicastFlag is set to FALSE. 1: unicastFlag is set to TRUE.	R/W
b12, b11	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b13	FLAG13	PTP profile Specific 1	0: PTP profile Specific 1 is set to FALSE. 1: PTP profile Specific 1 is set to TRUE.	R/W
b14	FLAG14	PTP profile Specific 2	0: PTP profile Specific 2 is set to FALSE. 1: PTP profile Specific 2 is set to TRUE.	R/W
b31 to b15	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Register DYRQFR is used to set the flagField section of the header when the SYNFP module is to generate a Delay_Req or Pdelay_Req message.

The value specified by DYRQFR is only reflected in the SYNFP module after the SYRVLDR.STUP bit is set to 1.

37.2.47 Delay_Resp Message Flag Field Setting Register (DYRPFR)

Address(es): EPTPC0.DYRPFR 000C 485Ch, EPTPC1.DYRPFR 000C 4C5Ch

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	FLAG1 ₄	FLAG1 ₃	—	—	FLAG1 ₀	FLAG9	FLAG8	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b7 to b0	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b8	FLAG8	alternateMasterFlag* ¹	0: alternateMasterFlag is set to FALSE. 1: alternateMasterFlag is set to TRUE.	R/W
b9	FLAG9	twoStepFlag* ²	Set this bit to 0 (FALSE).	R/W
b10	FLAG10	unicastFlag	0: unicastFlag is set to FALSE. 1: unicastFlag is set to TRUE.	R/W
b12, b11	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b13	FLAG13	PTP profile Specific 1	0: PTP profile Specific 1 is set to FALSE. 1: PTP profile Specific 1 is set to TRUE.	R/W
b14	FLAG14	PTP profile Specific 2	0: PTP profile Specific 2 is set to FALSE. 1: PTP profile Specific 2 is set to TRUE.	R/W
b31 to b15	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Note 1. This bit is reserved for Pdelay_Resp messages. Set the bit to 0.

Note 2. This bit is reserved for Delay_Resp messages.

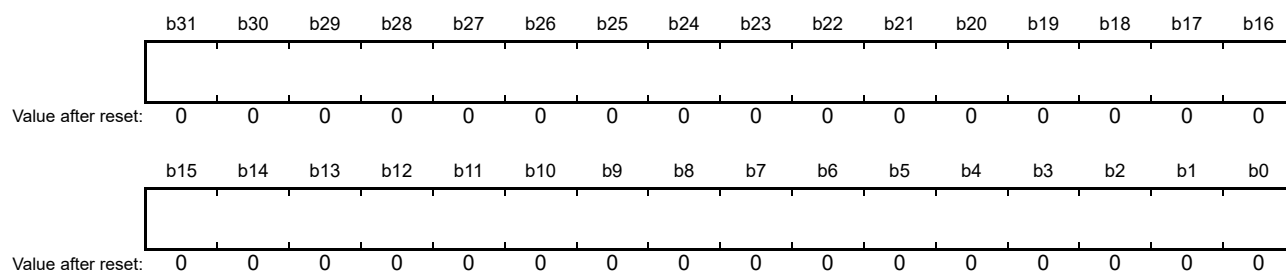
The DYRPFR register is used to set the flagField section of the header when the SYNFP module is to generate a Delay_Resp or PDelay_Resp message.

The value specified by the DYRPFR register is only reflected in the SYNFP module after the SYRVLDR.STUP bit is set to 1.

Do not change the value of the DYRPFR register while processing for the transmission of Delay_Resp or Pdelay_Resp messages is enabled. Furthermore, after disabling this processing for transmission, do not change the value of the DYRPFR register until the SYSR.RESDN flag becomes 1.

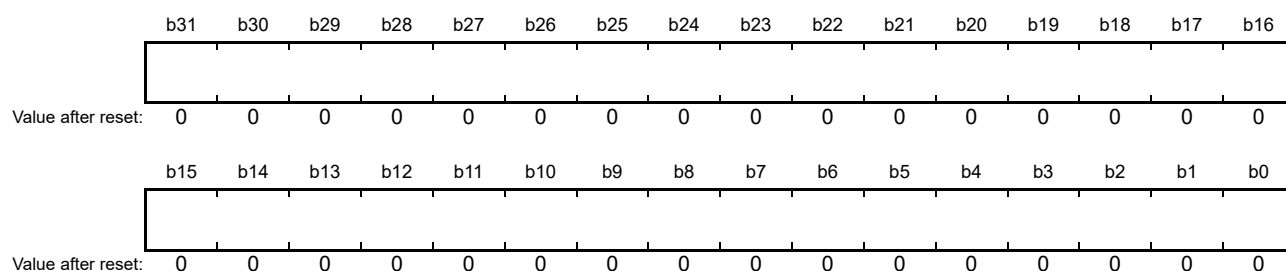
37.2.48 SYNFP Local Clock ID Registers (SYCIDRU, SYCIDRL)

Address(es): EPTPC0.SYCIDRU 000C 4860h, EPTPC1.SYCIDRU 000C 4C60h



Bit	Symbol	Bit Name	Description	R/W
b31 to b0	—	—	These bits hold the setting for the higher-order 32 bits of the clock-ID of the local port.	R/W

Address(es): EPTPC0.SYCIDRL 000C 4864h, EPTPC1.SYCIDRL 000C 4C64h



Bit	Symbol	Bit Name	Description	R/W
b31 to b0	—	—	These bits hold the setting for the lower-order 32 bits of the clock-ID of the local port.	R/W

The SYCIDR registers are used to set the clock-ID of the local port.

The registers are used for the clockIdentity section in the sourcePortIdentity field of the header when the SYNFP module is to generate a PTP message.

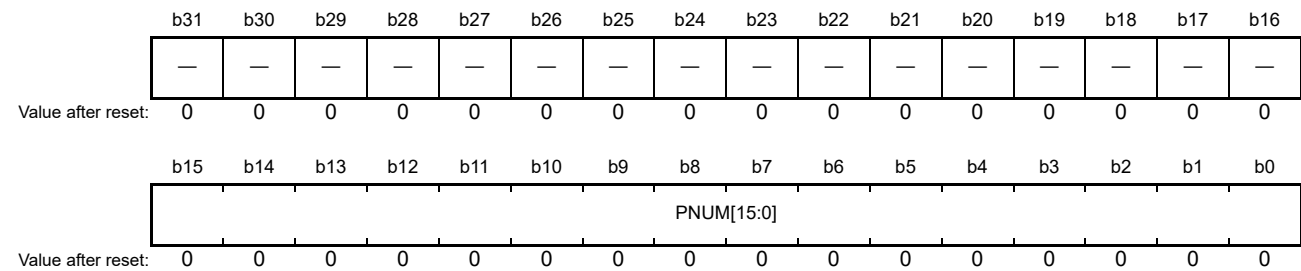
When a PTP message is received, the value of these registers is compared with the clockIdentity section in the sourcePortIdentity field of the PTP message to determine whether the message is one that was transmitted by you.

Usually, the setting should be the same as the value of portDS.portIdentity.clockIdentity.

Do not change the value of these registers while processing for the reception or transmission of PTP messages is enabled.

37.2.49 SYNFP Local Port Number Register (SYPNUMR)

Address(es): EPTPC0.SYPNUMR 000C 4868h, EPTPC1.SYPNUMR 000C 4C68h



Bit	Symbol	Bit Name	Description	R/W
b15 to b0	PNUM[15:0]	Local Port Number Setting	These bits hold the setting for the port number of the local port.	R/W
b31 to b16	—	Reserved	These bits are read as 0. The write value should be 0.	R

The SYPNUMR register is used to set the port number of the local port.

This register is used for the portNumber section in the sourcePortIdentity field of the header when the SYNFP module is to generate a PTP message.

When a PTP message is received, the value of this register is compared with the portNumber section in the sourcePortIdentity field of the PTP message to determine whether the message is one that was transmitted by the local device.

Usually, the setting should be the same as the value of portDS.portIdentity.portNumber.

Do not change the value of this register while processing for the reception or transmission of PTP messages is enabled.

37.2.50 SYNFP Register Value Load Directive Register (SYRVLDR)

Address(es): EPTPC0.SYRVLDR 000C 4880h, EPTPC1.SYRVLDR 000C 4C80h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	—	—	ANUP	STUP	BMUP
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	BMUP	BMC Update	When this bit is set to 1, the SYNFP module simultaneously reflects values of registers storing the MasterClock identifying information.	W
b1	STUP	State Update	When this bit is set to 1, the SYNFP module simultaneously reflects register values for PTP message reception and transmission.	W
b2	ANUP	Announce Message Generation Information Update	When this bit is set to 1, the Announce message generation block simultaneously reflects register values required for generating Announce messages.	W
b31 to b3	—	Reserved	These bits are read as 0. The write value should be 0.	R

The SYRVLDR register is used for simultaneously updating multiple register values in the SYNFP module.

BMUP Bit (BMC Update)

When the BMUP bit is set to 1, the SYNFP module simultaneously reflects the values of registers storing the MasterClock identifying information listed below.

- Registers MTCIDU and MTCIDL
- MTPID register

STUP Bit (State Update)

When the STUP bit is set to 1, the SYNFP module simultaneously reflects the values of registers and bits for PTP message reception and transmission listed below.

- SYNFR register
- DYRQFR register
- SYTLIR.SYNC[7:0] bits
- SYTLIR.DREQ[7:0] bits
- RSTOUTR register
- SYRFL1R register
- SYRFL2R register
- SYTRENR register

ANUP Bit (Announce Message Generation Information Update)

When the ANUP bit is set to 1, the Announce message generation block simultaneously reflects the values of registers and bits required for generating Announce messages listed below.

- ANFR register
- SYTLIR.ANCE[7:0] bits
- GMPR register
- GMCQR register
- Registers GMIDRU and GMIDRL
- CUOTSR register
- SRR register

37.2.51 SYNFP Reception Filter Register 1 (SYRFL1R)

Address(es): EPTPC0.SYRFL1R 000C 4890h, EPTPC1.SYRFL1R 000C 4C90h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	PDFUP[2:0]			—	PDRP[2:0]			—	PDRQ[2:0]			—	DRP[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	DRQ[2:0]			—	FUP[2:0]			—	SYNC[2:0]			—	—	ANCE[1:0]	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	ANCE[1:0]	Announce Message Processing	0: Messages are not transferred to the PTPEDMAC. 1: Messages are transferred to the PTPEDMAC.	R/W
b1			0: The PRC-TC does not relay messages between ports 0 and 1. 1: The PRC-TC relays messages between ports 0 and 1.	R/W
b3, b2			These bits are read as 0. The write value should be 0.	R
b4	SYNC[2:0]	Sync Message Processing	0: Messages are not transferred to the PTPEDMAC. 1: Messages are transferred to the PTPEDMAC.	R/W
b5			0: The PRC-TC does not relay messages between ports 0 and 1. 1: The PRC-TC relays messages between ports 0 and 1.	R/W
b6			0: The SYNFP does not process messages. 1: The SYNFP processes messages.	R/W
b7	—	Reserved	This bit is read as 0. The write value should be 0.	R
b8	FUP[2:0]	Follow_Up Message Processing	0: Messages are not transferred to the PTPEDMAC. 1: Messages are transferred to the PTPEDMAC.	R/W
b9			0: The PRC-TC does not relay messages between ports 0 and 1. 1: The PRC-TC relays messages between ports 0 and 1.	R/W
b10			0: The SYNFP does not process messages. 1: The SYNFP processes messages.	R/W
b11	—	Reserved	This bit is read as 0. The write value should be 0.	R
b12	DRQ[2:0]	Delay_Req Message Processing	0: Messages are not transferred to the PTPEDMAC. 1: Messages are transferred to the PTPEDMAC.	R/W
b13			0: The PRC-TC does not relay messages between ports 0 and 1. 1: The PRC-TC relays messages between ports 0 and 1.	R/W
b14			0: The SYNFP does not process messages. 1: The SYNFP processes messages.	R/W
b15	—	Reserved	This bit is read as 0. The write value should be 0.	R
b16	DRP[2:0]	Delay_Resp Message Processing	0: Messages are not transferred to the PTPEDMAC. 1: Messages are transferred to the PTPEDMAC.	R/W
b17			0: The PRC-TC does not relay messages between ports 0 and 1. 1: The PRC-TC relays messages between ports 0 and 1.	R/W
b18			0: The SYNFP does not process messages. 1: The SYNFP processes messages.	R/W
b19	—	Reserved	This bit is read as 0. The write value should be 0.	R
b20	PDRQ[2:0]	Pdelay_Req Message Processing	0: Messages are not transferred to the PTPEDMAC. 1: Messages are transferred to the PTPEDMAC.	R/W
b21			0: The PRC-TC does not relay messages between ports 0 and 1. 1: The PRC-TC relays messages between ports 0 and 1.	R/W
b22			0: The SYNFP does not process messages. 1: The SYNFP processes messages.	R/W

Bit	Symbol	Bit Name	Description	R/W
b23	—	Reserved	This bit is read as 0. The write value should be 0.	R
b24	PDRP[2:0]	Pdelay_Resp Message Processing	0: Messages are not transferred to the PTPEDMAC. 1: Messages are transferred to the PTPEDMAC.	R/W
b25			0: The PRC-TC does not relay messages between ports 0 and 1. 1: The PRC-TC relays messages between ports 0 and 1.	R/W
b26			0: The SYNFP does not process messages. 1: The SYNFP processes messages.	R/W
b27	—	Reserved	This bit is read as 0. The write value should be 0.	R
b28	PDFUP[2:0]	Pdelay_Resp_Follow_Up Message Processing	0: Messages are not transferred to the PTPEDMAC. 1: Messages are transferred to the PTPEDMAC.	R/W
b29			0: The PRC-TC does not relay messages between ports 0 and 1. 1: The PRC-TC relays messages between ports 0 and 1.	R/W
b30			0: The SYNFP does not process messages. 1: The SYNFP processes messages.	R/W
b31	—	Reserved	This bit is read as 0. The write value should be 0.	R

The SYRFL1R register is used to set up filtering for the reception of PTP messages.

Multiple bits corresponding to different types of messages can be set to 1. Setting all bits for a type of message to 0 leads to all messages of the given type being discarded.

The value specified by the SYRFL1R register is only reflected in the SYNFP module after the SYRVLDR.STUP bit is set to 1.

37.2.52 SYNFP Reception Filter Register 2 (SYRFL2R)

Address(es): EPTPC0.SYRFL2R 000C 4894h, EPTPC1.SYRFL2R 000C 4C94h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	ILL[1:0]	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	SIG[1:0]	—	—	—	MAN[1:0]	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	MAN[1:0]	Management Message Processing Setting	0: Messages are not transferred to the PTPEDMAC. 1: Messages are transferred to the PTPEDMAC.	R/W
b1			0: The PRC-TC does not relay messages between ports 0 and 1. 1: The PRC-TC relays messages between ports 0 and 1.	R/W
b3, b2	—	Reserved	These bits are read as 0. The write value should be 0.	R
b4	SIG[1:0]	Signaling Message Processing Setting	0: Messages are not transferred to the PTPEDMAC. 1: Messages are transferred to the PTPEDMAC.	R/W
b5			0: The PRC-TC does not relay messages between ports 0 and 1. 1: The PRC-TC relays messages between ports 0 and 1.	R/W
b27 to b6	—	Reserved	These bits are read as 0. The write value should be 0.	R
b28	ILL[1:0]	Illegal Message Processing Setting*1	0: Messages are not transferred to the PTPEDMAC. 1: Messages are transferred to the PTPEDMAC.	R/W
b29			0: The PRC-TC does not relay messages between ports 0 and 1. 1: The PRC-TC relays messages between ports 0 and 1.	R/W
b31, b30	—	Reserved	These bits are read as 0. The write value should be 0.	R

Note 1. PTP messages other than PTP v2 messages and messages of undefined type are handled as illegal messages.

The SYRFL2R register is used to set up filtering for the reception of PTP messages.

Multiple bits corresponding to different types of messages can be set to 1. Setting all bits for a type of message to 0 leads to all messages of the given type being discarded.

The value specified by the SYRFL2R register is only reflected in the SYNFP module after the SYRVLDR.STUP bit is set to 1.

37.2.53 SYNFP Transmission Enable Register (SYTRENR)

Address(es): EPTPC0.SYTRENr 000C 4898h, EPTPC1.SYTRENr 000C 4C98h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	PDRQ	—	—	—	DRQ	—	—	—	SYNC	—	—	—	ANCE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	ANCE	Announce Message Transmission Enable	0: Announce messages are not transmitted. 1: Announce messages are transmitted.	R/W
b3 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R
b4	SYNC	Sync Message Transmission Enable	0: Sync messages are not transmitted. 1: Sync messages are transmitted.	R/W
b7 to b5	—	Reserved	These bits are read as 0. The write value should be 0.	R
b8	DRQ	Delay_Req Message Transmission Enable	0: Delay_Req messages are not transmitted. 1: Delay_Req messages are transmitted.	R/W
b11 to b9	—	Reserved	These bits are read as 0. The write value should be 0.	R
b12	PDRQ	Pdelay_Req Message Transmission Enable	0: Pdelay_Req messages are not transmitted. 1: Pdelay_Req messages are transmitted.	R/W
b31 to b13	—	Reserved	These bits are read as 0. The write value should be 0.	R

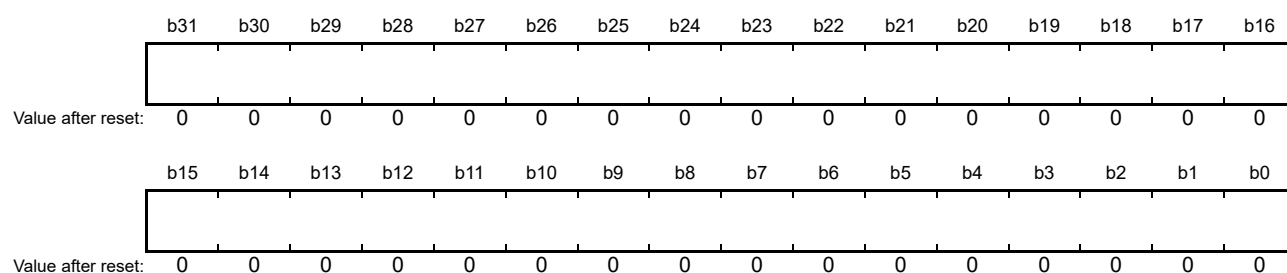
The SYTRENR register is used to enable or disable transmission of PTP messages.

The PDRQ and DRQ bits should not be set to 1 at the same time. Operation is not guaranteed when both bits are set to 1.

The value specified by the SYTRENR register is only reflected in the SYNFP module after the SYRVLD.R.STUP bit is set to 1.

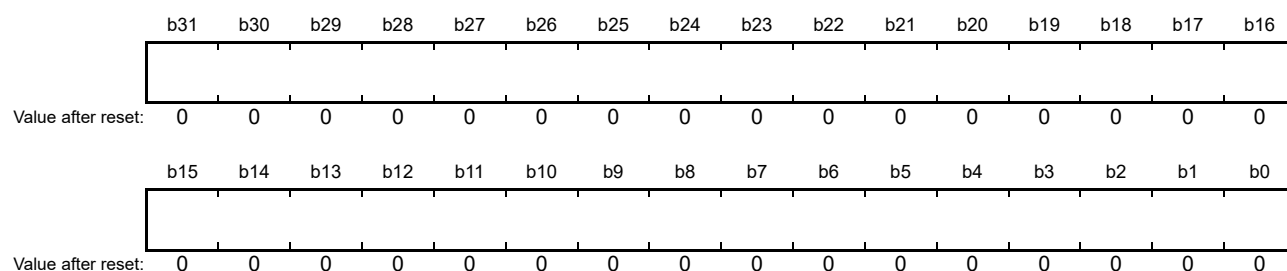
37.2.54 Master Clock ID Registers (MTCIDU, MTCIDL)

Address(es): EPTPC0.MTCIDU 000C 48A0h, EPTPC1.MTCIDU 000C 4CA0h



Bit	Symbol	Bit Name	Description	R/W
b31 to b0	—	—	These bits hold the setting for the higher-order 32 bits of the clock-ID of the master clock.	R/W

Address(es): EPTPC0.MTCIDL 000C 48A4h, EPTPC1.MTCIDL 000C 4CA4h



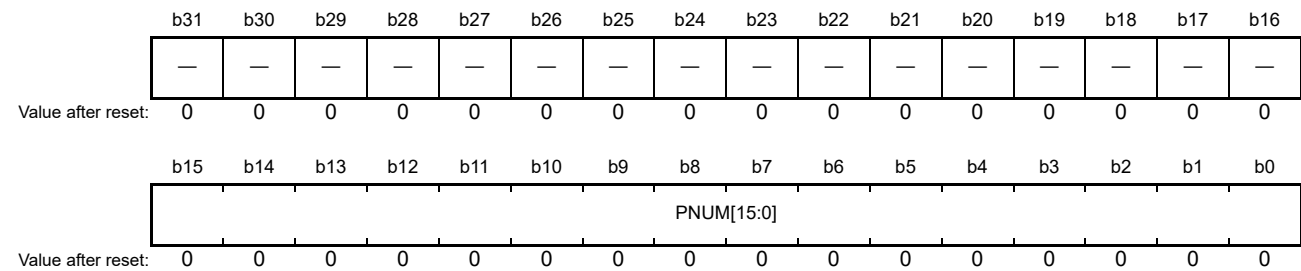
Bit	Symbol	Bit Name	Description	R/W
b31 to b0	—	—	These bits hold the setting for the lower-order 32 bits of the clock-ID of the master clock.	R/W

Registers MTCIDU and MTCIDL are used to set the clock-ID of the master for synchronization.

The value specified by registers MTCIDU and MTCIDL is only reflected in the SYNFP module after the SYRVLDR.BMUP bit is set to 1.

37.2.55 Master Clock Port Number Register (MTPID)

Address(es): EPTPC0.MTPID 000C 48A8h, EPTPC1.MTPID 000C 4CA8h



Bit	Symbol	Bit Name	Description	R/W
b15 to b0	PNUM[15:0]	Master Clock Port Number Setting	These bits hold the setting for the port number of the master clock.	R/W
b31 to b16	—	Reserved	These bits are read as 0. The write value should be 0.	R

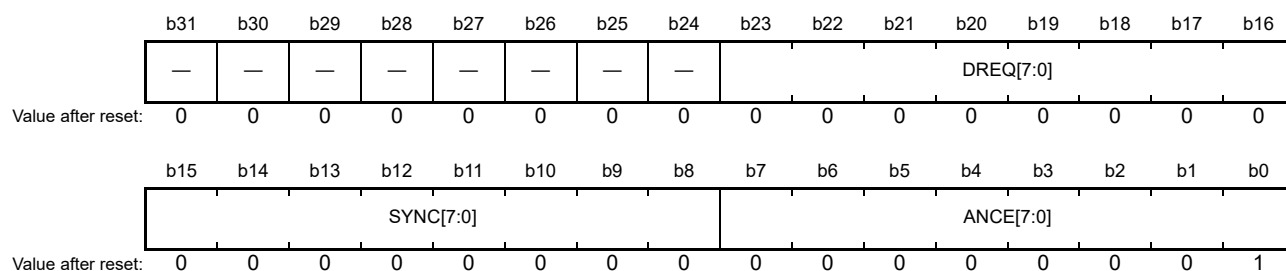
The MTPID register is used to set the port number of the master for synchronization.

The value specified by the MTPID register is only reflected in the SYNFP module after the SYRVLDR.BMUP bit is set to 1.

In normal usage, set the value of parentDS.parentPortIdentity.portNumber in this register.

37.2.56 SYTFP Transmission Interval Setting Register (SYTLIR)

Address(es): EPTPC0.SYTLIR 000C 48C0h, EPTPC1.SYTLIR 000C 4CC0h



Bit	Symbol	Bit Name	Description	R/W
b7 to b0	ANCE[7:0]	Announce Message Transmission Interval Setting	These bits set the interval for the transmission of Announce messages.	R/W
b15 to b8	SYNC[7:0]	Sync Message Transmission Interval Setting	These bits set the interval for the transmission of Sync messages. The setting is also placed in the logMessageInterval field of transmitted Sync messages.	R/W
b23 to b16	DREQ[7:0]	Delay_Req Transmission Interval Average Value/ Pdelay_Req Transmission Interval Setting	The bits set the average interval for the transmission of Delay_Req messages and the interval for the transmission of Pdelay_Req messages. The setting is also placed in the logMessageInterval field of Delay_Resp messages.	R/W
b31 to b24	—	Reserved	These bits are read as 0. The write value should be 0.	R

The SYTLIR register is used to set the interval for the transmission of messages generated by the SYNFP module. The setting is an integer logarithm in base 2 ($\log_2(x)$) and determines a value x in seconds.

In other words, the interval for transmission is 2^n (s), where n is the setting. The available settings are from -7 (F9h) to $+6$ (06h).

Example:

If the setting is 06h, then the interval for transmission is $2^6 = 64$ (s).

If the setting is 00h, then the interval for transmission is $2^0 = 1$ (s).

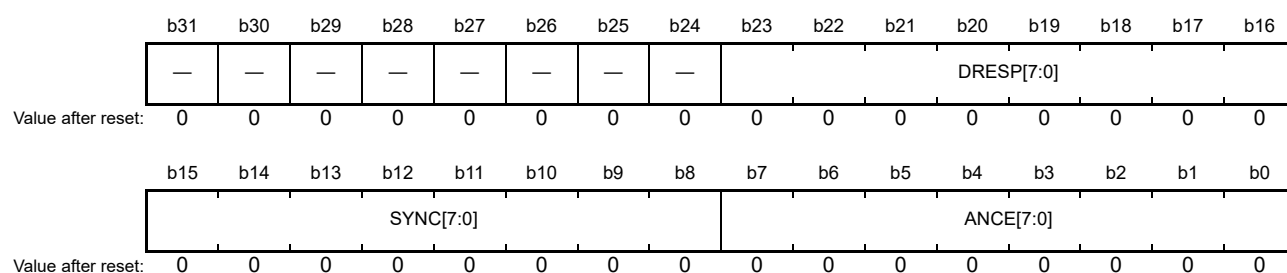
If the setting is FFh, then the interval for transmission is $2^{-1} = 0.5$ (s) = 500 (ms).

If the setting is F9h, then the interval for transmission is $2^{-7} = 0.0078125$ (s) = 7.8125 (ms).

The value set in the ANCE[7:0] bits is only reflected in the SYNFP module after the SYRVLDR.ANUP bit is set to 1. The value set in the DREQ[7:0] and SYNC[7:0] bits is only reflected in the SYNFP module after the SYRVLDR.STUP bit is set to 1.

37.2.57 SYNFP Received logMessageInterval Value Indication Register (SYRLIR)

Address(es): EPTPC0.SYRLIR 000C 48C4h, EPTPC1.SYRLIR 000C 4CC4h

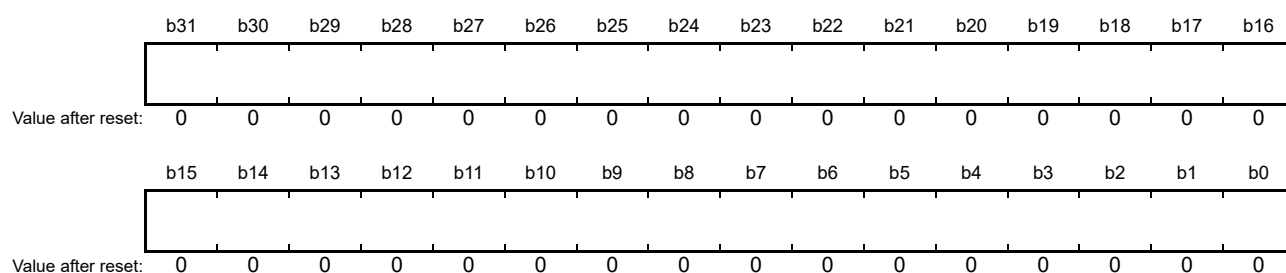


Bit	Symbol	Bit Name	Description	R/W
b7 to b0	ANCE[7:0]	Announce Message logMessageInterval Field Indication Flag	These bits indicate the logMessageInterval field value of a received Announce message.	R
b15 to b8	SYNC[7:0]	Sync Message logMessageInterval Field Indication Flag	These bits indicate the logMessageInterval field value of a received Sync message.	R
b23 to b16	DRESP[7:0]	Delay_Resp Message logMessageInterval Field Indication Flag	These bits indicate the logMessageInterval field value of a received Delay_Resp message.	R
b31 to b24	—	Reserved	These bits are read as 0.	R

The SYRLIR register indicates the logMessageInterval field values of received PTP messages.

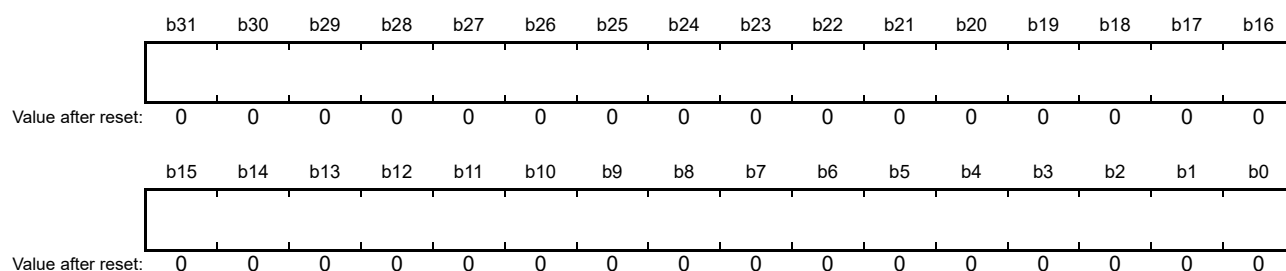
37.2.58 offsetFromMaster Value Registers (OFMRU, OFMRL)

Address(es): EPTPC0.OFMRU 000C 48C8h, EPTPC1.OFMRU 000C 4CC8h



Bit	Symbol	Bit Name	Description	R/W
b31 to b0	—	—	These bits indicate the higher-order 32 bits of the calculated offsetFromMaster value.	R

Address(es): EPTPC0.OFMRL 000C 48CCh, EPTPC1.OFMRL 000C 4CCCh



Bit	Symbol	Bit Name	Description	R/W
b31 to b0	—	—	These bits indicate the lower-order 32 bits of the calculated offsetFromMaster value.	R

Registers OFMRU and OFMRL indicate the calculated offsetFromMaster value.

The value is in units of nanoseconds, expressed in two's-complement representation. Note that the numeric representation*1 differs from that of the offsetFromMaster member of the current data set (currentDS).

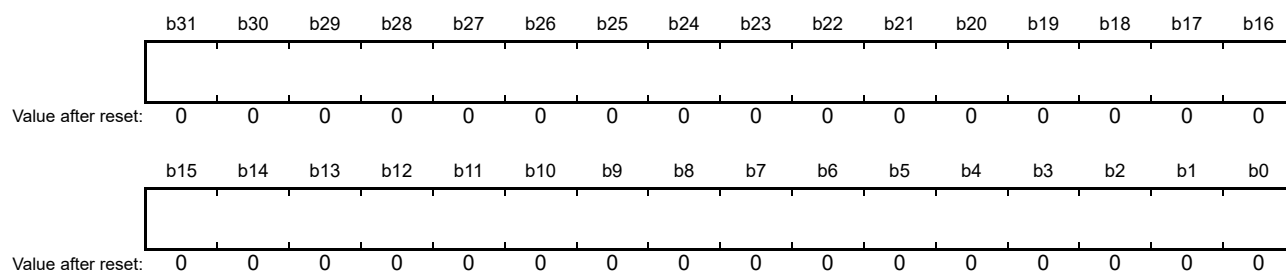
When read, access to the registers should be in order of OFMRU and then OFMRL.

Note 1. The value of currentDS.offsetFromMaster is multiplied by 2^{16} .

Example: 2.5 (ns) = 00000000_00028000h

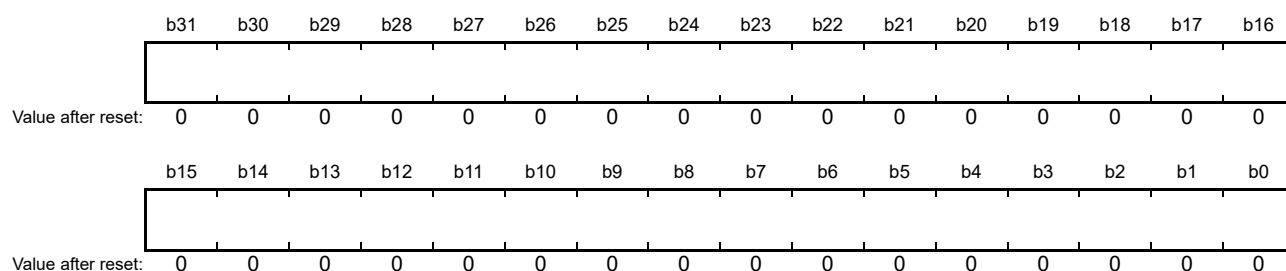
37.2.59 meanPathDelay Value Registers (MPDRU, MPDRL)

Address(es): EPTPC0.MPDRU 000C 48D0h, EPTPC1.MPDRU 000C 4CD0h



Bit	Symbol	Bit Name	Description	R/W
b31 to b0	—	—	These bits indicate the higher-order 32 bits of the calculated meanPathDelay value.	R

Address(es): EPTPC0.MPDRL 000C 48D4h, EPTPC1.MPDRL 000C 4CD4h



Bit	Symbol	Bit Name	Description	R/W
b31 to b0	—	—	These bits indicate the lower-order 32 bits of the calculated meanPathDelay value.	R

Registers MPDRU and MPDRL indicate the calculated meanPathDelay value.

The value is in units of nanoseconds, expressed in two's-complement representation. Note that the numeric representation*1 differs from that of the meanPathDelay member of the current data set (currentDS).

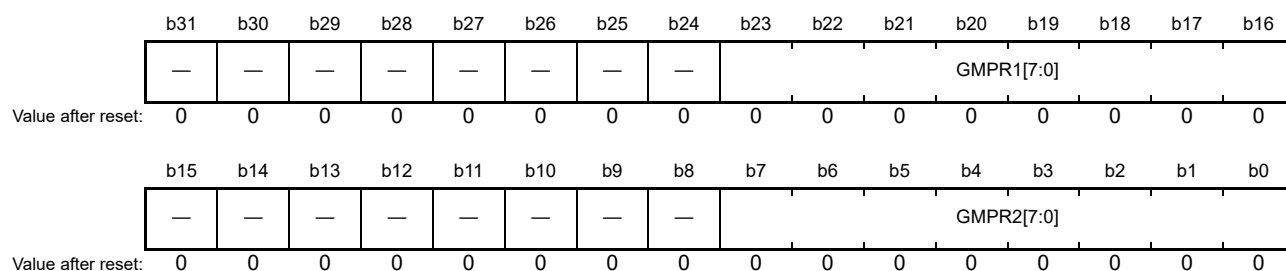
When read, access to the registers should be in order of MPDRU and then MPDRL.

Note 1. The value of currentDS.meanPathDelay is multiplied by 2^{16} .

Example: 2.5 (ns) = 00000000_00028000h

37.2.60 grandmasterPriority Field Setting Register (GMPR)

Address(es): EPTPC0.GMPR 000C 48E0h, EPTPC1.GMPR 000C 4CE0h



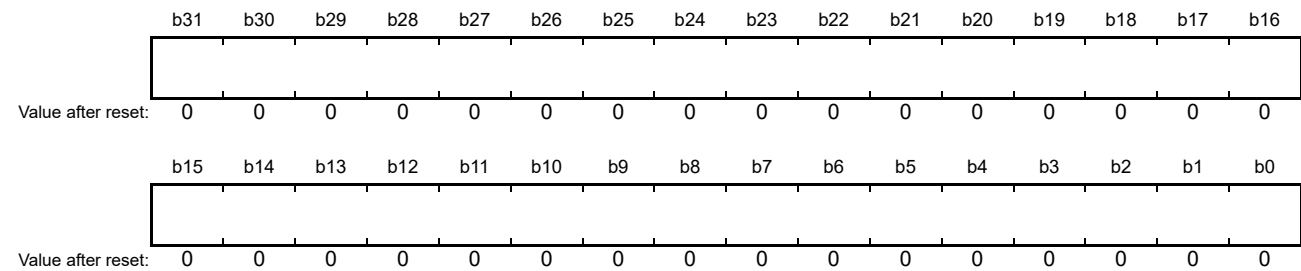
Bit	Symbol	Bit Name	Description	R/W
b7 to b0	GMPR2[7:0]	grandmasterPriority2 Field Value Setting	These bits are used to set the value of the grandmasterPriority2 fields of Announce messages.	R/W
b15 to b8	—	Reserved	These bits are read as 0. The write value should be 0.	R
b23 to b16	GMPR1[7:0]	grandmasterPriority1 Field Value Setting	These bits are used to set the value of the grandmasterPriority1 fields of Announce messages.	R/W
b31 to b24	—	Reserved	These bits are read as 0. The write value should be 0.	R

The GMPR register is used to specify the grandmasterPriority1 and grandmasterPriority2 field values of Announce messages generated by the SYNFP module.

The value specified by the GMPR register is only reflected in the SYNFP module after the SYRVLDR.ANUP bit is set to 1.

37.2.61 grandmasterClockQuality Field Setting Register (GMCQR)

Address(es): EPTPC0.GMCQR 000C 48E4h, EPTPC1.GMCQR 000C 4CE4h



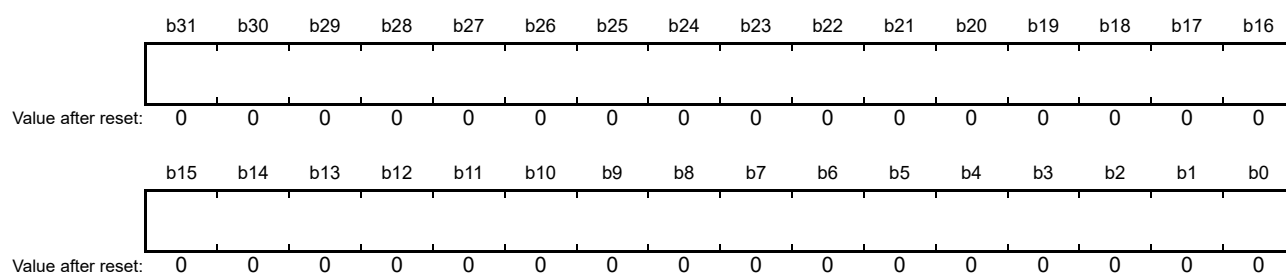
Bit	Symbol	Bit Name	Description	R/W
b31 to b0	—	—	These bits are used to set the value of the grandmasterClockQuality fields of Announce messages. The correspondence between bits and the grandmasterClockQuality fields is as listed below. b31 to b24: clockClass b23 to b16: clockAccuracy b15 to b0: offsetScaledLogVariance	R/W

The GMCQR register is used to specify the grandmasterClockQuality field value of Announce messages generated by the SYNFP module.

The value specified by the GMCQR register is only reflected in the SYNFP module after the SYRVLDR.ANUP bit is set to 1.

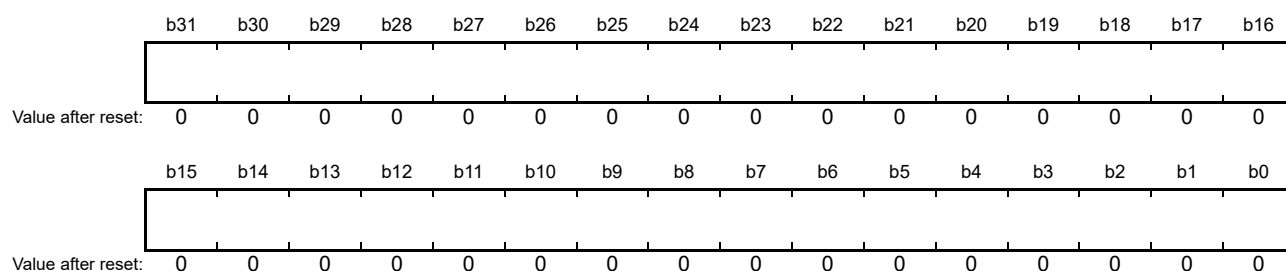
37.2.62 grandmasterIdentity Field Setting Registers (GMIDRU, GMIDRL)

Address(es): EPTPC0.GMIDRU 000C 48E8h, EPTPC1.GMIDRU 000C 4CE8h



Bit	Symbol	Bit Name	Description	R/W
b31 to b0	—	—	These bits hold the setting for the higher-order 32 bits of the value of the grandmasterIdentity fields of Announce messages.	R/W

Address(es): EPTPC0.GMIDRL 000C 48ECh, EPTPC1.GMIDRL 000C 4CECh



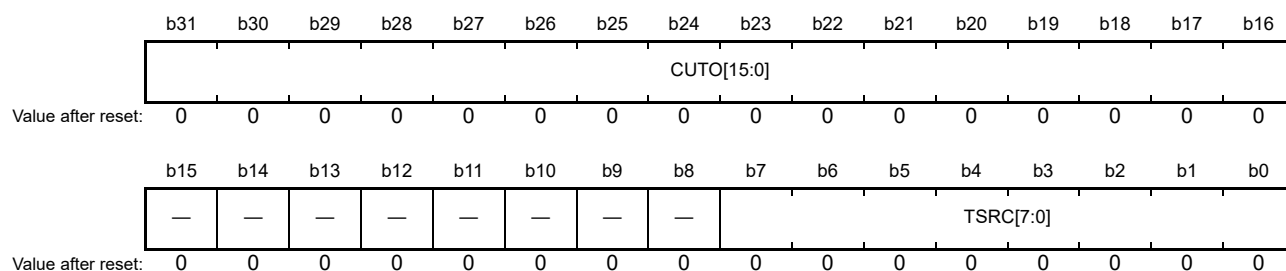
Bit	Symbol	Bit Name	Description	R/W
b31 to b0	—	—	These bits hold the setting for the lower-order 32 bits of the value of the grandmasterIdentity fields of Announce messages.	R/W

Registers GMIDRU and GMIDRL are used to specify the grandmasterIdentity field value of Announce messages generated by the SYNFP module.

The value specified by registers GMIDRU and GMIDRL is only reflected in the SYNFP module after the SYRVLDR.ANUP bit is set to 1.

37.2.63 currentUtcOffset/timeSource Field Setting Register (CUOTSR)

Address(es): EPTPC0.CUOTSR 000C 48F0h, EPTPC1.CUOTSR 000C 4CF0h



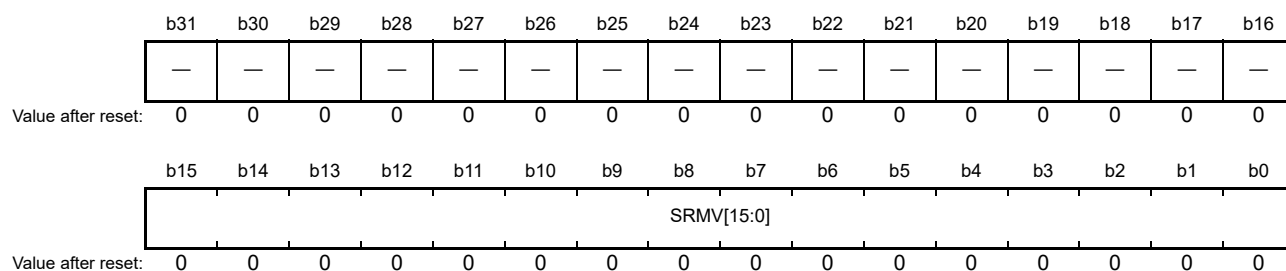
Bit	Symbol	Bit Name	Description	R/W
b7 to b0	TSRC[7:0]	timeSource Field Setting	These bits set the value of the timeSource fields of Announce messages.	R/W
b15 to b8	—	Reserved	These bits are read as 0. The write value should be 0.	R
b31 to b16	CUTO[15:0]	currentUtcOffset Field Setting	These bits set the value of the currentUtcOffset fields of Announce messages.	R/W

The CUOTSR register is used to specify the currentUtcOffset and timeSource field values of Announce messages generated by the SYNFP module.

The value specified by the CUOTSR register is only reflected in the SYNFP module after the SYRVLDR.ANUP bit is set to 1.

37.2.64 stepsRemoved Field Setting Register (SRR)

Address(es): EPTPC0.SRR 000C 48F4h, EPTPC1.SRR 000C 4CF4h



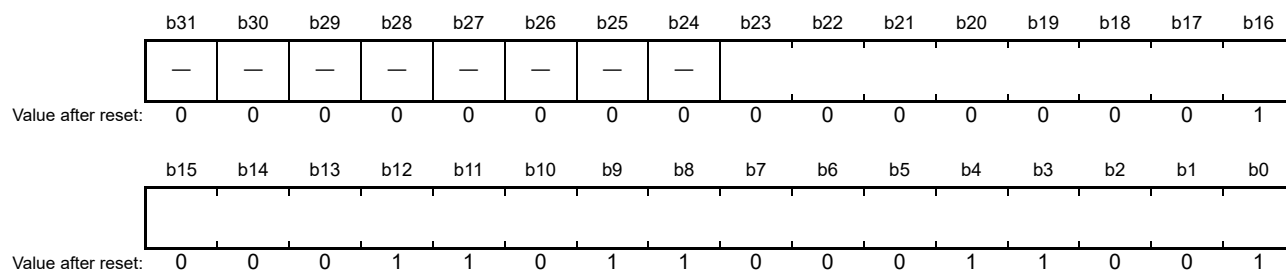
Bit	Symbol	Bit Name	Description	R/W
b15 to b0	SRMV[15:0]	stepsRemoved Field Value Setting	These bits set the value of the stepsRemoved fields of Announce messages.	R/W
b31 to b16	—	Reserved	These bits are read as 0. The write value should be 0.	R

The SRR register is used to specify the stepsRemoved field value of Announce messages generated by the SYNFP module.

The value specified by the SRR register is only reflected in the SYNFP module after the SYRVLDR.ANUP bit is set to 1.

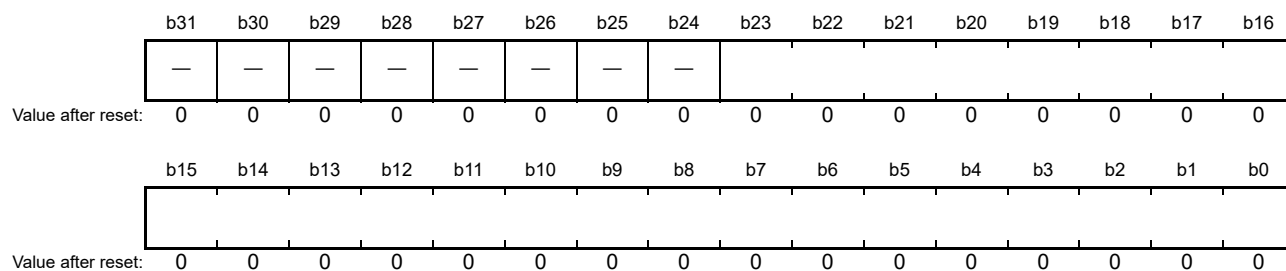
37.2.65 PTP-primary Message Destination MAC Address Setting Registers (PPMACRU, PPMACRL)

Address(es): EPTPC0.PPMACRU 000C 4900h, EPTPC1.PPMACRU 000C 4D00h



Bit	Symbol	Bit Name	Description	R/W
b23 to b0	—	—	These bits hold the setting for the higher-order 24 bits of the destination MAC address for PTP-primary messages.	R/W
b31 to b24	—	Reserved	These bits are read as 0. The write value should be 0.	R

Address(es): EPTPC0.PPMACRL 000C 4904h, EPTPC1.PPMACRL 000C 4D04h



Bit	Symbol	Bit Name	Description	R/W
b23 to b0	—	—	These bits hold the setting for the lower-order 24 bits of the destination MAC address for PTP-primary messages.	R/W
b31 to b24	—	Reserved	These bits are read as 0. The write value should be 0.	R

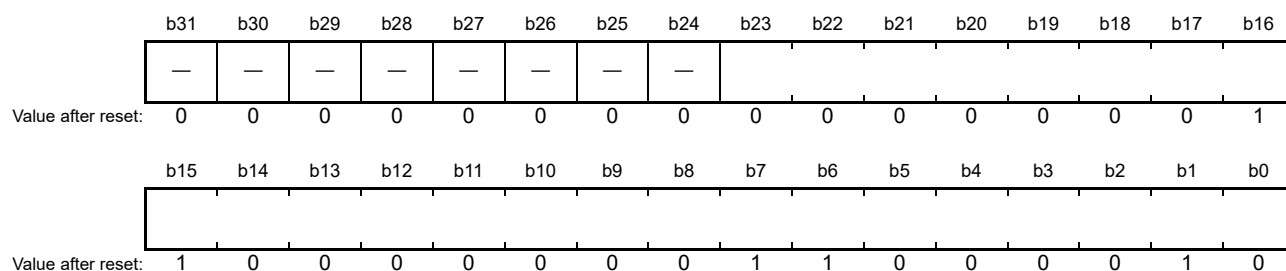
The PPMACR registers are used to specify the destination MAC address for PTP-primary messages. In normal usage, set 01:1B:19:00:00:00 in these registers.

The value is used in the destination MAC address field when generating an Ethernet frame for a PTP-primary message. It is also used as a determining condition for received frames carrying PTP messages.

Set a value in these registers before starting the EDMAC, ETHERC, or PTPEDMAC. Do not change the settings while the EPTPC is operating.

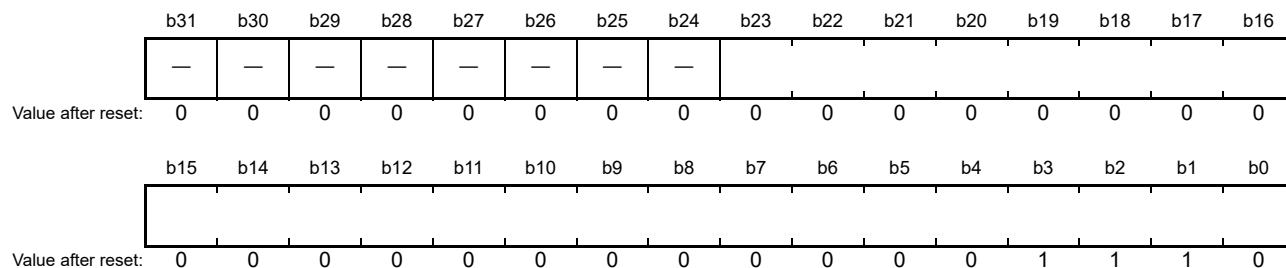
37.2.66 PTP-pdelay Message MAC Address Setting Registers (PDMACRU, PDMACRL)

Address(es): EPTPC0.PDMACRU 000C 4908h, EPTPC1.PDMACRU 000C 4D08h



Bit	Symbol	Bit Name	Description	R/W
b23 to b0	—	—	These bits hold the setting for the higher-order 24 bits of the destination MAC address for PTP-pdelay messages.	R/W
b31 to b24	—	Reserved	These bits are read as 0. The write value should be 0.	R

Address(es): EPTPC0.PDMACRL 000C 490Ch, EPTPC1.PDMACRL 000C 4D0Ch



Bit	Symbol	Bit Name	Description	R/W
b23 to b0	—	—	These bits hold the setting for the lower-order 24 bits of the destination MAC address for PTP-pdelay messages.	R/W
b31 to b24	—	Reserved	These bits are read as 0. The write value should be 0.	R

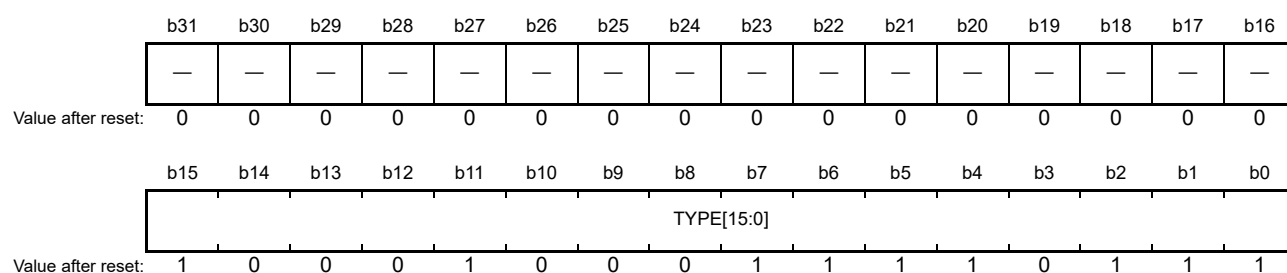
The PDMACR registers are used to specify the destination MAC address for PTP-pdelay messages. In normal usage, set 01:80:C2:00:00:0E in these registers.

The value is used in the destination MAC address field when generating frames carrying PTP-pdelay messages in the Ethernet format. It is also used as a determining condition for received frames carrying PTP messages.

Set a value in these registers before starting the EDMAC, ETHERC, or PTPEDMAC. Do not change the settings while the EPTPC is operating.

37.2.67 PTP Message Ethertype Setting Register (PETYPER)

Address(es): EPTPC0.PETYPER 000C 4910h, EPTPC1.PETYPER 000C 4D10h



Bit	Symbol	Bit Name	Description	R/W
b15 to b0	TYPE[15:0]	PTP Message Ethertype Value Setting	These bits hold the setting for the Ethertype field value for frames in the Ethernet II format.	R/W
b31 to b16	—	Reserved	These bits are read as 0. The write value should be 0.	R

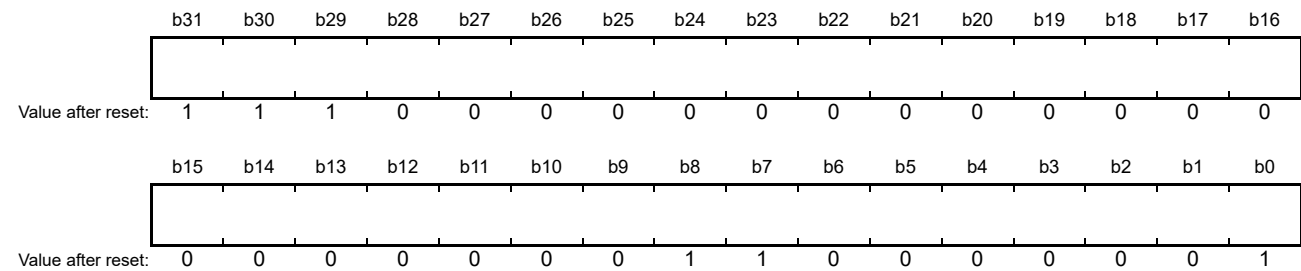
The PETYPER register is used to specify the Ethertype field for frames carrying the PTP messages. In normal usage, set 0000 88F7h in this register.

The value is used in the Ethertype field when generating frames carrying PTP messages in the Ethernet II format. It is also used as a determining condition for received frames carrying PTP messages.

Set a value in these registers before starting the EDMAC, ETHERC, or PTPEDMAC. Do not change the settings while the EPTPC is operating.

37.2.68 PTP-primary Message Destination IP Address Setting Register (PPIPR)

Address(es): EPTPC0.PPIPR 000C 4920h, EPTPC1.PPIPR 000C 4D20h



Bit	Symbol	Bit Name	Description	R/W
b31 to b0	—	—	These bits hold the setting for the destination IP address for PTP-primary messages.	R/W

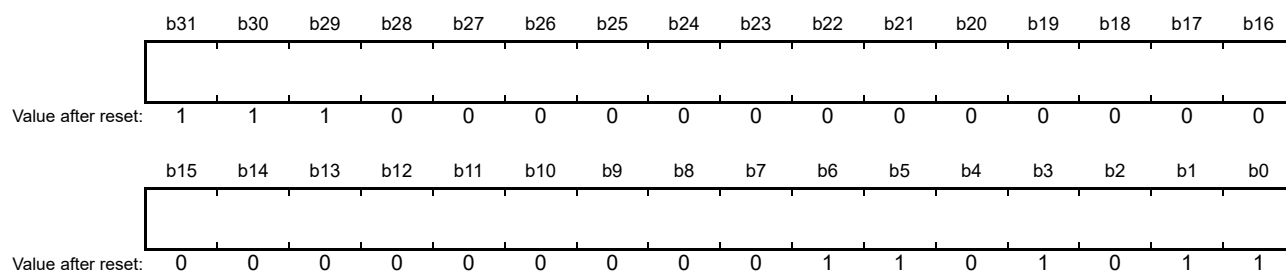
The PPIPR register is used to specify the destination IP address for PTP messages. In normal usage, set E000 0181h (224.0.1.129) in this register.

The value is used in the destination IP address field when generating frames carrying PTP-primary messages in the IPv4 format. The lower-order 23 bits are also used in the destination MAC address field for Ethernet frames. The value is also used as a determining condition for received frames carrying PTP messages.

Set a value in this register before starting the EDMAC, ETHERC, or PTPEDMAC. Do not change the settings while the EPTPC is operating.

37.2.69 PTP-pdelay Message Destination IP Address Setting Register (PDIPR)

Address(es): EPTPC0.PDIPR 000C 4924h, EPTPC1.PDIPR 000C 4D24h



Bit	Symbol	Bit Name	Description	R/W
b31 to b0	—	—	These bits hold the setting for the destination IP address for PTP-pdelay messages.	R/W

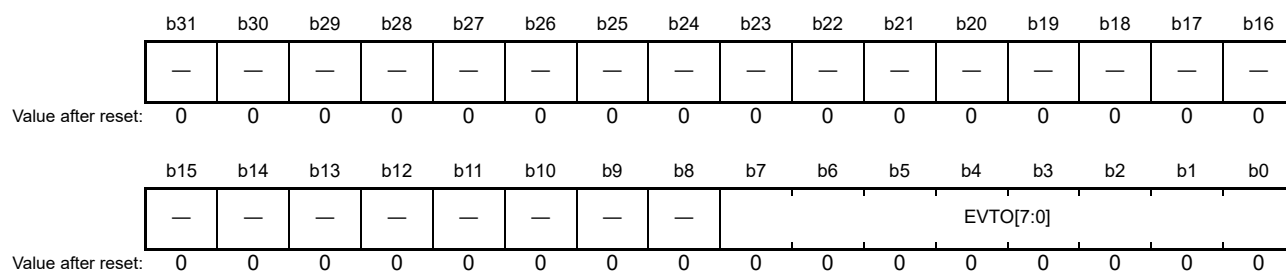
The PDIPR register is used to specify the destination IP address for PTP-pdelay messages. In normal usage, set E000 006Bh (224.0.0.107) in this register.

The value is used in the destination IP address field when generating frames carrying PTP-pdelay messages in the IPv4 format. The lower-order 23 bits are also used in the destination MAC address field for Ethernet frames. The value is also used as a determining condition for received frames carrying PTP messages.

Set a value in this register before starting the EDMAC, ETHERC, or PTPEDMAC. Do not change the settings while the EPTPC is operating.

37.2.70 PTP Event Message TOS Setting Register (PETOSR)

Address(es): EPTPC0.PETOSR 000C 4928h, EPTPC1.PETOSR 000C 4D28h



Bit	Symbol	Bit Name	Description	R/W
b7 to b0	EVTO[7:0]	PTP Event Message TOS Field Value Setting	These bits hold the setting for the value of the TOS field within the IPv4 headers of PTP event messages.	R/W
b31 to b8	—	Reserved	These bits are read as 0. The write value should be 0.	R

The PETOSR register is used to set the TOS (type of service) field value within the IPv4 headers of PTP event messages. Set a value in this register before starting the EDMAC, ETHERC, or PTPEDMAC. Do not change the settings while the EPTPC is operating.

37.2.71 PTP general Message TOS Setting Register (PGTOSR)

Address(es): EPTPC0.PGTOSR 000C 492Ch, EPTPC1.PGTOSR 000C 4D2Ch

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	GETO[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b7 to b0	GETO[7:0]	PTP general Message TOS Field Value Setting	These bits hold the setting for the value of the TOS field within the IPv4 headers of PTP general messages.	R/W
b31 to b8	—	Reserved	These bits are read as 0. The write value should be 0.	R

The PGTOSR register is used to set the TOS (type of service) field value within the IPv4 headers of PTP general messages.

Set a value in this register before starting the EDMAC, ETHERC, or PTPEDMAC. Do not change the settings while the EPTPC is operating.

37.2.72 PTP-primary Message TTL Setting Register (PPTTLR)

Address(es): EPTPC0.PPTTLR 000C 4930h, EPTPC1.PPTTLR 000C 4D30h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

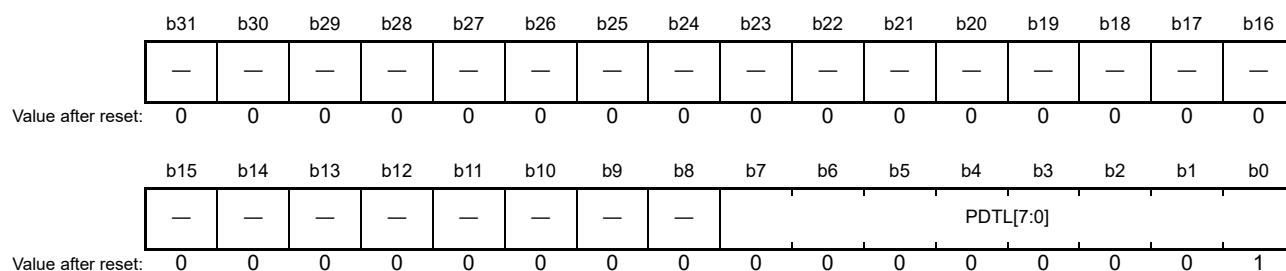
	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	PRTL[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b7 to b0	PRTL[7:0]	PTP-primary Message TTL Field Value Setting	These bits hold the setting for the value of the TTL field within the IPv4 headers of PTP-primary messages.	R/W
b31 to b8	—	Reserved	These bits are read as 0. The write value should be 0.	R

The PPTTLR register is used to set the TTL (time to live) field value within the IPv4 headers of PTP-primary messages. Set a value in this register before starting the EDMAC, ETHERC, or PTPEDMAC. Do not change the settings while the EPTPC is operating.

37.2.73 PTP-pdelay Message TTL Setting Register (PDTTLR)

Address(es): EPTPC0.PDTTLR 000C 4934h, EPTPC1.PDTTLR 000C 4D34h

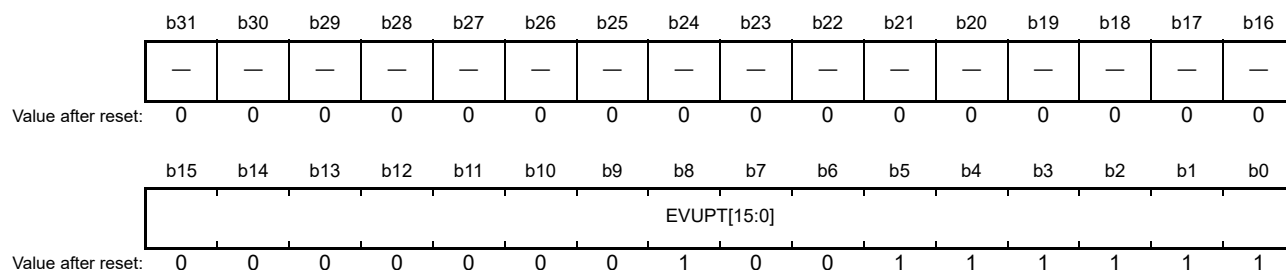


Bit	Symbol	Bit Name	Description	R/W
b7 to b0	PDTL[7:0]	PTP-pdelay Message TTL Field Value	These bits hold the setting for the value of the TTL field within the IPv4 headers of PTP-pdelay messages.	R/W
b31 to b8	—	Reserved	These bits are read as 0. The write value should be 0.	R

The PDTTLR register is used to set the TTL field value within the IPv4 headers of PTP-pdelay messages. Set a value in this register before starting the EDMAC, ETHERC, or PTPEDMAC. Do not change the settings while the EPTPC is operating.

37.2.74 PTP Event Message UDP Destination Port Number Setting Register (PEUDPR)

Address(es): EPTPC0.PEUDPR 000C 4938h, EPTPC1.PEUDPR 000C 4D38h

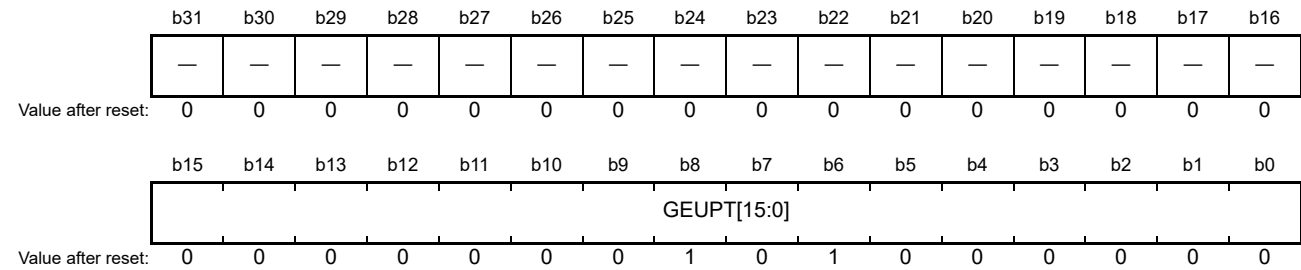


Bit	Symbol	Bit Name	Description	R/W
b15 to b0	EVUPT[15:0]	PTP Event Message Destination Port Number Setting	These bits hold the setting for the value of the destination port number field within the UDP headers of PTP event messages.	R/W
b31 to b16	—	Reserved	These bits are read as 0. The write value should be 0.	R

The PEUDPR register is used to set the destination port number field value within the UDP headers of PTP event messages. In normal usage, set 013Fh (319) in this register. Set a value in this register before starting the EDMAC, ETHERC, or PTPEDMAC. Do not change the settings while the EPTPC is operating.

37.2.75 PTP general Message UDP Destination Port Number Setting Register (PGUDPR)

Address(es): EPTPC0.PGUDPR 000C 493Ch, EPTPC1.PGUDPR 000C 4D3Ch



Bit	Symbol	Bit Name	Description	R/W
b15 to b0	GEUPT[15:0]	PTP general Message Destination Port Number	These bits hold the setting for the value of the destination port number field within the UDP headers of PTP general messages.	R/W
b31 to b16	—	Reserved	These bits are read as 0. The write value should be 0.	R

The PGUDPR register is used to set the destination port number field value within the UDP headers of PTP general messages. In normal usage, set 0140h (320) in this register.

Set a value in this register before starting the EDMAC, ETHERC, or PTPEDMAC. Do not change the settings while the EPTPC is operating.

37.2.76 Frame Reception Filter Setting Register (FFLTR)

Address(es): EPTPC0.FFLTR 000C 4940h, EPTPC1.FFLTR 000C 4D40h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	EXTPRM
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	—	—	ENB	PRT	SEL
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	SEL	Receive MAC Address Select*1	These bits select how filtering is handled when multicast frames other than PTP messages are received.	R/W
b1	PRT	Frame Reception Enable*1	b2 b0 0 0 0: Filtering is disabled (all multicast frames are received). 0 0 1: Filtering is disabled (all multicast frames are received). 0 1 0: Filtering is disabled (all multicast frames are received). 0 1 1: Filtering is disabled (all multicast frames are received). 1 0 0: Do not receive multicast frames. 1 0 1: Do not receive multicast frames. 1 1 0: Only receive multicast frames matching the MAC address setting in registers FMAC0RU and FMAC0RL. 1 1 1: Only receive multicast frames matching the MAC address setting in registers FMAC1RU and FMAC1RL.	R/W
b2	ENB	Reception Filter Enable*1		R/W
b15 to b3	—	Reserved	These bits are read as 0. The write value should be 0.	R
b16	EXTPRM	Extended Promiscuous Mode Setting	0: Normal operation (unicast frames addressed to the EPTPC are received, filtering of PTP frames is applied, multicast filtering is applied, and all broadcast frames are received). 1: Extended promiscuous mode (all frames are received)	R/W
b31 to b17	—	Reserved	These bits are read as 0. The write value should be 0.	R

Note 1. The setting of these bits is only effective when the EXTPRM bit is 0.

The FFLTR register is used to switch extended promiscuous mode on or off and to select how filtering is handled when multicast frames other than PTP messages are received.

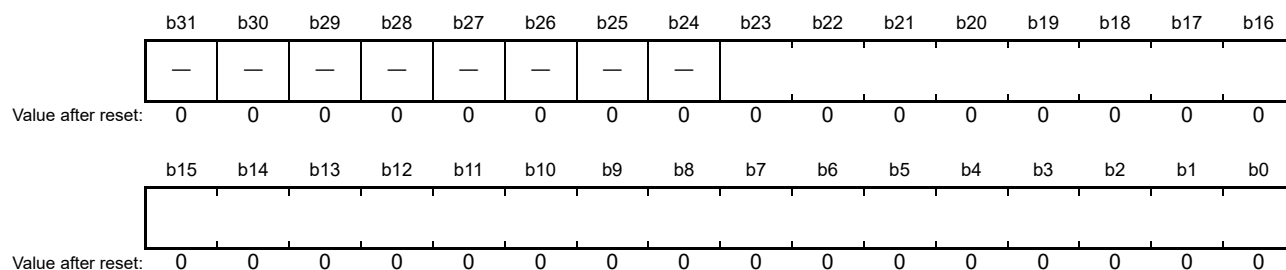
The filter for the reception of multicast frames other than PTP messages is enabled by setting the ENB, PRT, and SEL bits to 110b or 111b. Frames passed by the filter are then transferred by EDMACn.

Relaying of multicast frames other than PTP messages is in accordance with the TRNMR register and relaying and reception of PTP messages is in accordance with the setting of registers SYRFL1R and SYRFL2R.

Set a value in this register before starting the EDMAC, ETHERC, or PTPEDMAC. Do not change the settings while the EPTPC is operating.

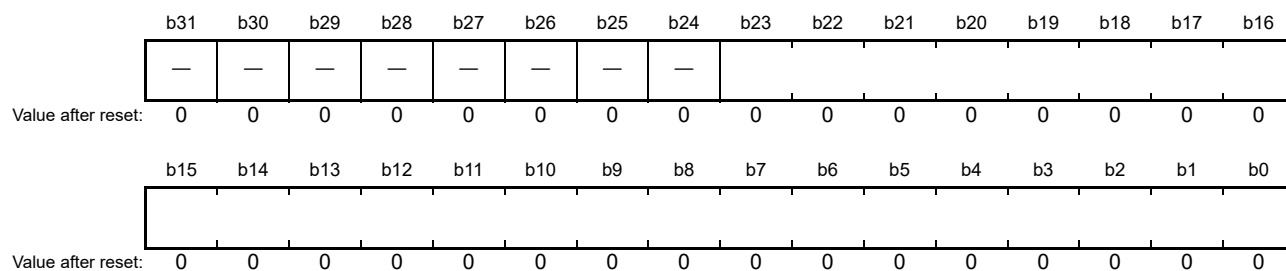
37.2.77 Frame Reception Filter MAC Address 0 Setting Registers (FMAC0RU, FMAC0RL)

Address(es): EPTPC0.FMAC0RU 000C 4960h, EPTPC1.FMAC0RU 000C 4D60h



Bit	Symbol	Bit Name	Description	R/W
b23 to b0	—	—	These bits hold the setting for the higher-order 24 bits of the destination MAC address for received multicast frames.	R/W
b31 to b24	—	Reserved	These bits are read as 0. The write value should be 0.	R

Address(es): EPTPC0.FMAC0RL 000C 4964h, EPTPC1.FMAC0RL 000C 4D64h



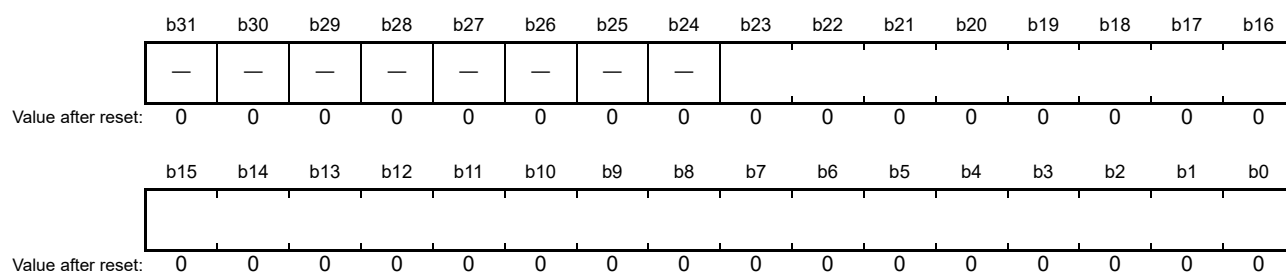
Bit	Symbol	Bit Name	Description	R/W
b23 to b0	—	—	These bits hold the setting for the lower-order 24 bits of the destination MAC address for received multicast frames.	R/W
b31 to b24	—	Reserved	These bits are read as 0. The write value should be 0.	R

Registers FMAC0RU and FMAC0RL are used to specify the MAC address for filtering in the reception of multicast frames other than PTP messages.

Set a value in this register before starting the EDMAC, ETHERC, or PTPEDMAC. Do not change the settings while the EPTPC is operating.

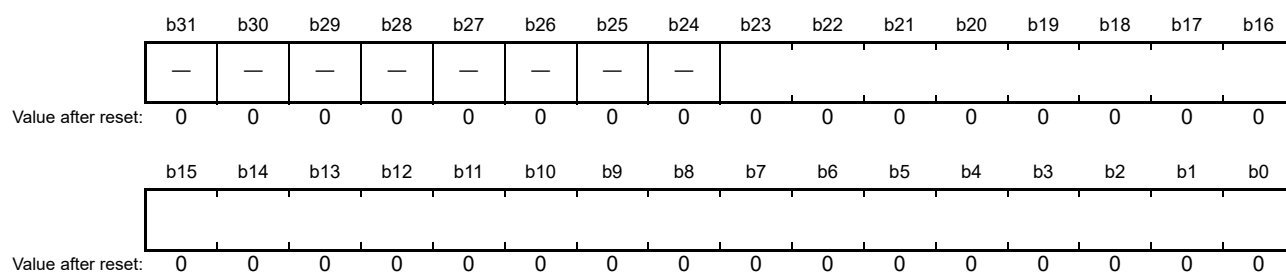
37.2.78 Frame Reception Filter MAC Address 1 Setting Registers (FMAC1RU, FMAC1RL)

Address(es): EPTPC0.FMAC1RU 000C 4968h, EPTPC1.FMAC1RU 000C 4D68h



Bit	Symbol	Bit Name	Description	R/W
b23 to b0	—	—	These bits hold the setting for the higher-order 24 bits of the destination MAC address for received multicast frames.	R/W
b31 to b24	—	Reserved	These bits are read as 0. The write value should be 0.	R

Address(es): EPTPC0.FMAC1RL 000C 496Ch, EPTPC1.FMAC1RL 000C 4D6Ch



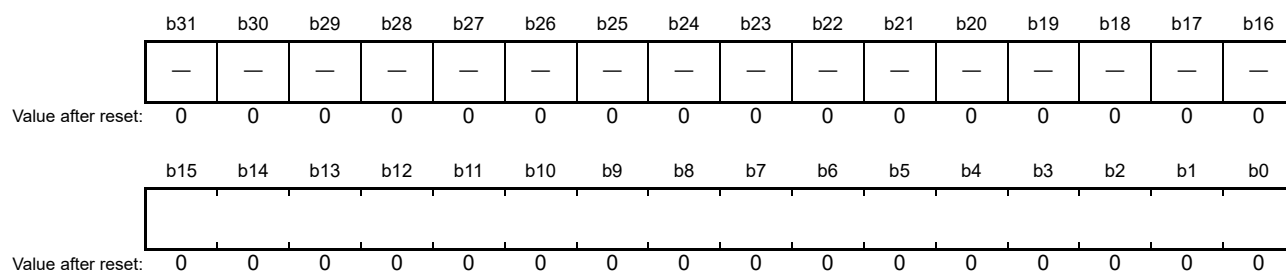
Bit	Symbol	Bit Name	Description	R/W
b23 to b0	—	—	These bits hold the setting for the lower-order 24 bits of the destination MAC address for received multicast frames.	R/W
b31 to b24	—	Reserved	These bits are read as 0. The write value should be 0.	R

Registers FMAC1RU and FMAC1RL are used to specify the MAC address for filtering in the reception of multicast frames other than PTP messages.

Set a value in this register before starting the EDMAC, ETHERC, or PTPEDMAC. Do not change the settings while the EPTPC is operating.

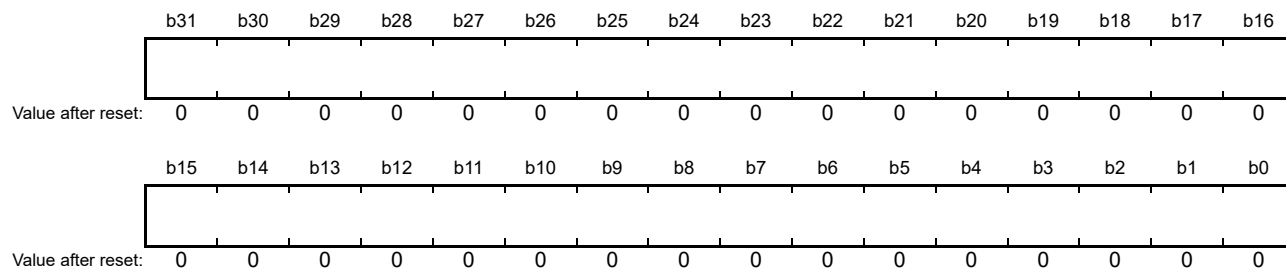
37.2.79 Asymmetric Delay Setting Registers (DASYMRU, DASYMRL)

Address(es): EPTPC0.DASYMRU 000C 49C0h, EPTPC1.DASYMRU 000C 4DC0h



Bit	Symbol	Bit Name	Description	R/W
b15 to b0	—	—	These bits hold the setting for the higher-order 16 bits of the asymmetric delay value. Set them to 0000h in this MCU.	R/W
b31 to b16	—	Reserved	These bits are read as 0. The write value should be 0.	R

Address(es): EPTPC0.DASYMRL 000C 49C4h, EPTPC1.DASYMRL 000C 4DC4h



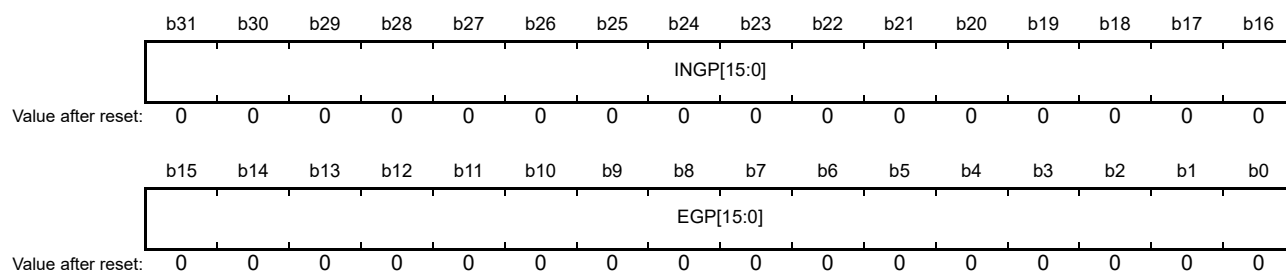
Bit	Symbol	Bit Name	Description	R/W
b31 to b0	—	—	These bits hold the setting for the lower-order 32 bits of the asymmetric delay value. Set them to 0000 0000h in this MCU.	R/W

Registers DASYMRU and DASYMRL are used to set the asymmetric delay value (delayAsymmetry).

Set the registers DASYMRU and DASYMRL to 0000 0000h in this MCU.

37.2.80 Timestamp Latency Setting Register (TSLATR)

Address(es): EPTPC0.TSLATR 000C 49C8h, EPTPC1.TSLATR 000C 4DC8h



Bit	Symbol	Bit Name	Description	R/W
b15 to b0	EGP[15:0]	Output Port Timestamp Latency Setting	These bits hold the setting for the time stamp latency (ns) for the output ports.	R/W
b31 to b16	INGP[15:0]	Input Port Timestamp Latency Setting	These bits hold the setting for the time stamp latency (ns) for the input ports.	R/W

The TSLATR register is used to set the amount of latency in timestamp acquisition in nanoseconds.

The value should not be modified while reception or transmission of PTP messages is enabled.

EGP[15:0] Bits (Output Port Timestamp Latency Setting)

Set the corresponding fixed value in Table 37.10. The timestamp latency differs with the link transfer rate (100 Mbps or 10 Mbps) and the frequency of the STCA clock (20, 25, 50, or 100 MHz).

Table 37.10 Settings of the EGP[15:0] Bits (ns)

Link Transfer Rate		STCA Clock Frequency			
		20 MHz	25 MHz	50 MHz	100 MHz
MII	100 Mbps	590	625	695	730
	10 Mbps	7430	7465	7535	7570
RMII	100 Mbps	770	805	875	910
	10 Mbps	9230	9265	9335	9370

INGP[15:0] Bits (Input Port Timestamp Latency Setting)

Set the corresponding fixed value in Table 37.11. The timestamp latency differs with the link transfer rate (100 Mbps or 10 Mbps) and the frequency of the STCA clock (20, 25, 50, or 100 MHz).

Table 37.11 Settings of the INGP[15:0] Bits (ns)

Link Transfer Rate		STCA Clock Frequency			
		20 MHz	25 MHz	50 MHz	100 MHz
MII	100 Mbps	980	945	875	840
	10 Mbps	8180	8145	8075	8015
RMII	100 Mbps	1060	1025	955	920
	10 Mbps	8980	8945	8875	8815

37.2.81 SYNFP Operation Setting Register (SYCONFR)

Address(es): EPTPC0.SYCONFR 000C 49CCh, EPTPC1.SYCONFR 000C 4DCCh

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	TCMOD	—	—	—	FILDIS
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	SBDIS	—	—	—	—	TCYC[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b7 to b0	TCYC[7:0]	PTP Message Transmission Interval Setting	These bits are used to set the time from the completion of one transmission to the start of the next in cycles of the transmission clock. A value n in these bits means that a transmission interval of n cycles will be secured. No interval is secured if the setting is 00h. The recommended value is 28h (40 cycles).	R/W
b11 to b8	—	Reserved	These bits are read as 0. The write value should be 0.	R
b12	SBDIS	Sync Message Transmission Bandwidth Securing Disable	0: Securing of the bandwidth for the transmission of SYNC messages is enabled (transfer by the EDMAC is given lower priority). 1: Securing of the bandwidth for the transmission of SYNC messages is disabled (transfer by the EDMAC is given higher priority).	R/W
b15 to b13	—	Reserved	These bits are read as 0. The write value should be 0.	R
b16	FILDIS	Receive Message domainNumber Filter Disable	0: Filtering conditions for the reception of PTP messages include comparison with the domainNumber field. 1: Filtering conditions for the reception of PTP messages do not include comparison with the domainNumber field.	R/W
b19 to b17	—	Reserved	These bits are read as 0. The write value should be 0.	R
b20	TCMOD	TC Mode Setting	0: E2E TC 1: P2P TC	R/W
b31 to b21	—	Reserved	These bits are read as 0. The write value should be 0.	R

The SYCONFR register controls operation of the SYNFP module.

Set a value in this register before starting the EDMAC, ETHERC, or PTPEDMAC. Do not change the settings while the EPTPC is operating.

TCYC[7:0] Bits (PTP Message Transmission Interval Setting)

The TCYC[7:0] bits are used to secure the waiting time between packets to keep a fixed transmission delay time.

The setting defines the interval from input of the transmission completed signal from the ETHERC to output of the next transmission request as a number of cycles of the transmission clock (which runs at 2.5 MHz if the link transfer rate is 10 Mbps and at 25 MHz if the rate is 100 Mbps).

SBDIS Bit (Sync Message Transmission Bandwidth Securing Disable)

This bit disables securing of bandwidth to increase accuracy of the interval for the transmission of SYNC messages.

TCMOD Bit (TC Mode Setting)

This bit sets the delay mechanism for operation as a transparent clock (TC). The bit setting changes the method of calculating the values used in the correctionField field during operation as a TC.

37.2.82 SYNFP Frame Format Setting Register (SYFORMR)

Address(es): EPTPC0.SYFORMR 000C 49D0h, EPTPC1.SYFORMR 000C 4DD0h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	FORM1	FORM0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

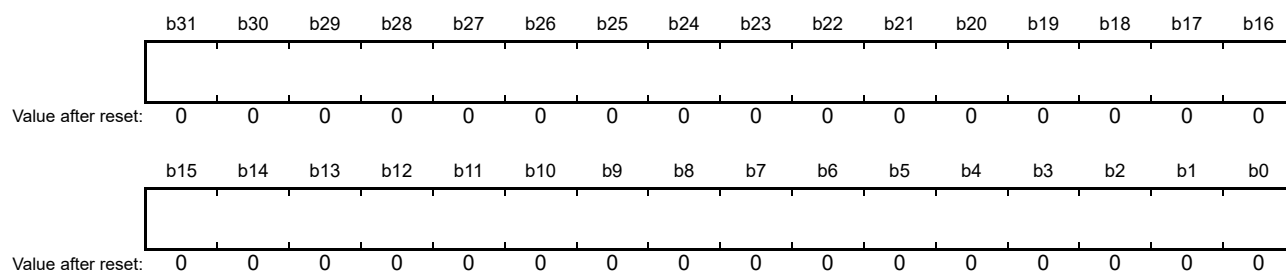
Bit	Symbol	Bit Name	Description	R/W
b0	FORM0	Ethernet Frame Format Setting	0: Ethernet II frame format 1: IEEE802.3 frame format	R/W
b1	FORM1	Ethernet/UDP Encapsulation	0: PTP directly over Ethernet 1: PTP over UDP/IPv4	R/W
b31 to b2	—	Reserved	These bits are read as 0. The write value should be 0.	R

The SYFORMR register is used to set the format for frame generation by the SYNFP module.

Set a value in this register before starting the EDMAC, ETHERC, or PTPEDMAC. Do not change the settings while the EPTPC is operating.

37.2.83 Response Message Reception Timeout Register (RSTOUTR)

Address(es): EPTPC0.RSTOUTR 000C 49D4h, EPTPC1.RSTOUTR 000C 4DD4h



Bit	Symbol	Bit Name	Description	R/W
b31 to b0	—	Response Message Reception Timeout Time Setting	A response message not being received within $n \times 1024$ (ns), where n is the setting, is judged to represent a timeout.	R/W

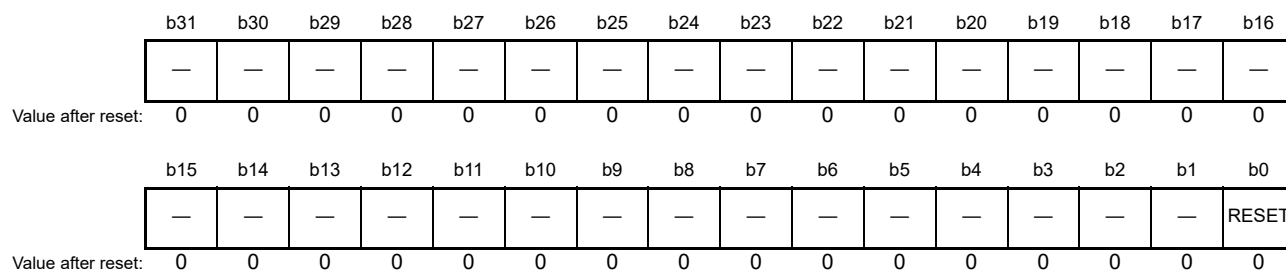
The RSTOUTR register is used to set the time for detection of a timeout in the reception of PTP response messages (Delay_Resp and Pdelay_Resp).

If a Delay_Resp or Pdelay_Resp message is not received within the time specified by this register after transmission of a Delay_Req or Pdelay_Req message, the SYSR.DRPTO flag becomes 1.

The value specified by this register is only reflected in the SYNFP module after the SYRVLDR.STUP bit is set to 1.

37.2.84 PTP Reset Register (PTRSTR)

Address(es): EPTPC.PTRSTR 000C 0500h



Bit	Symbol	Bit Name	Description	R/W
b0	RESET	EPTPC Software Reset	0: The EPTPC is not reset. 1: The EPTPC is reset.*1	R/W
b31 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R

Note 1. Do not access the EPTPC-related registers other than this register while a software reset is being issued.

The PTRSTR register is used to reset the EPTPC.

It takes 64 cycles of the peripheral module clock (PCLKA) until initialization of the EPTPC is completed. After the RESET bit is set to 1, wait for 64 cycles of the PCLKA before setting its value back to 0.

After the RESET bit is set back to 0, wait for 256 cycles of the PCLKA until the EPTPC is released from the reset.

37.2.85 STCA Clock Select Register (STCSELR)

Address(es): EPTPC.STCSELR 000C 0504h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	SCLKSEL[2:0]		—	—	—	—	—	—	SCLKDIV[2:0]		—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0

Bit	Symbol	Bit Name	Description	R/W
b2 to b0	SCLKDIV[2:0]	PCLKA Clock Frequency Division	b2 b0 0 0 1: 1 0 1 0: 1/2 0 1 1: 1/3 1 0 0: 1/4 1 0 1: 1/5 1 1 0: 1/6 Settings other than above are prohibited.	R/W
b7 to b3	—	Reserved	These bits are read as 0. The write value should be 0.	R
b10 to b8	SCLKSEL[2:0]	STCA Clock Select	b10 b8 0 0 0: PCLKA clock divided by 1 to 6 0 1 0: Input clock from the REF50CK0 pin 0 1 1: Input clock from the REF50CK1 pin Settings other than above are prohibited.	R/W
b31 to b11	—	Reserved	These bits are read as 0. The write value should be 0.	R

The STCSELR register selects the STCA clock signal for use in the EPTPC.

Set a value in this register before starting the EDMAC, ETHERC, or PTPEDMAC. Do not change the settings while the EPTPC is operating.

SCLKDIV[2:0] Bits (PCLKA Clock Frequency Division)

These bits set the division ratio of PCLKA.

When the setting of the SCLKSEL[2:0] bits is 000b, the frequency-divided PCLKA is used as the STCA clock signal.

SCLKSEL[2:0] Bits (STCA Clock Select)

These bits select the STCA clock signal for use in the EPTPC.

37.2.86 SYNFP Bypass Register (SYBYPSR)

Address(es): EPTPC.SYBYPSR 000C 0508h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	BYPAS S1
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	BYPAS S0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

Bit	Symbol	Bit Name	Description	R/W
b0	BYPASS0	SYNFP0 Bypass	0: SYNFP0 module is used. 1: SYNFP0 module is bypassed.	R/W
b15 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R
b16	BYPASS1	SYNFP1 Bypass	0: SYNFP1 module is used. 1: SYNFP1 module is bypassed.	R/W
b31 to b17	—	Reserved	These bits are read as 0. The write value should be 0.	R

The SYBYPSR register controls the operations of the synchronization frame processing units.

Set a value in this register before starting the EDMAC, ETHERC, or PTPEDMAC. Do not change the settings while the EPTPC is operating.

BYPASSn Bit (SYNFPn Bypass) (n = 0, 1)

When setting this bit to 1, the functions of SYNFPn module are stopped and all frames received by the ETHERCn are transferred to the EDMACn. In this case, the time synchronization of the Ethernet port n cannot be used.

Set this bit to 1 when the EPTPC is not to be used.

When this bit is 1, the registers in the corresponding SYNFPn module cannot be accessed. For details, refer to section 37.6.1, Setting the Module-Stop Function.

37.3 Operation

The EPTPC is set not to receive (analyze) and transmit (generate) PTP messages after release from the reset state.

Accordingly, the EPTPC has no effect on the transmission and reception of frames by the ETHERC and EDMAC at that time.

Settings of the EPTPC registers are required for the ETHERC and EDMAC to be able to use packet filtering by MAC address in the SYNFP modules and for relaying between Ethernet ports by the PRC-TC module. That is, the transmission and reception of PTP messages requires setting of the EPTPC.

Figure 37.4 is a block diagram of the modules involved in transferring and relaying frames.

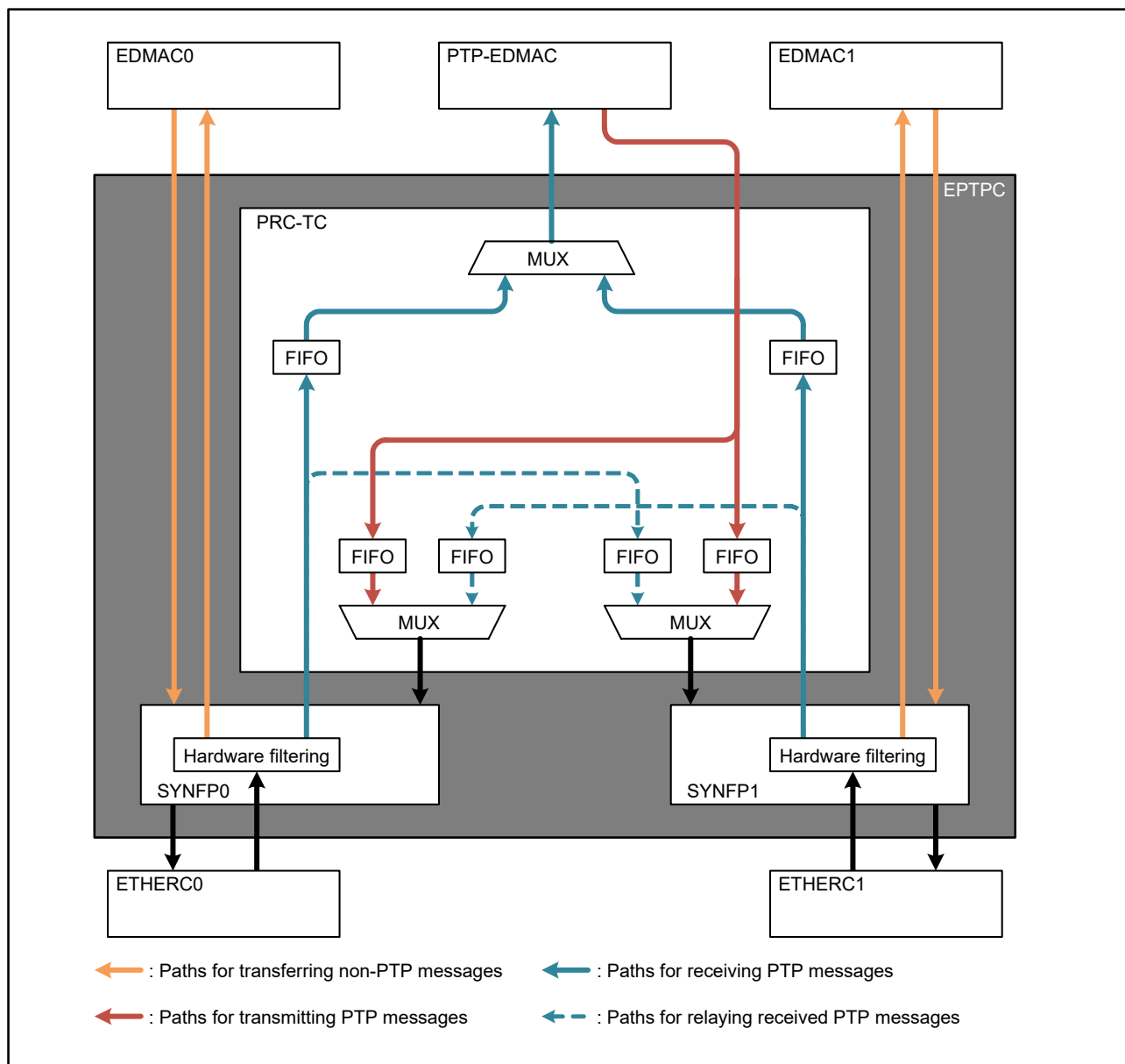


Figure 37.4 Block Diagram of the Modules Involved in Transferring and Relaying Frames

37.3.1 Transmission and Reception and Relaying of Non-PTP Messages

The EPTPC operates in extended promiscuous mode if the setting of the FFLTR.EXTPRM bit is 1 (extended promiscuous mode). In this mode, all frames received by the Ethernet ports are transferred to the EDMAC without filtering.

The EPTPC operates in normal mode if the setting of the FFLTR.EXTPRM bit is 0 (normal operation). In this mode, the SYNFP module applies its hardware filtering function to filter frames received by the Ethernet ports.

Frames for the other station can also be relayed to the Ethernet port on the other side in accordance with the settings of the TRNMR.FWD0 and FWD1 bits.

The EPTPC and EDMAC transfer received unicast frames if they are for the given station. When a received frame is for the other station, the EPTPC is able to relay received frames to the other port if relaying between Ethernet ports is enabled.

Operation when multicast frames are received can be selected from the following: frames are transferred to the EDMAC, frames are not transferred to the EDMAC, or frames are transferred to the EDMAC only when the address matches the specified MAC address. The EPTPC also relays multicast frames to the other Ethernet port if relaying between Ethernet ports is enabled.

The EPTPC transfers received broadcast frames to the EDMAC for the receiving Ethernet port. The EPTPC also relays broadcast frames to the other Ethernet port if relaying between Ethernet ports is enabled.

The PRC-TC module, which handles the relaying of frames received by one Ethernet port to the other Ethernet port, has a relaying FIFO buffer for use in this process. Either of two methods for reading out from the relaying FIFO is selectable: store and forward or cut-through. With the cut-through method, reading out from the relaying FIFO starts when it holds more than one frame or more data than a specified threshold value, or when both of these conditions are satisfied. The TRNCTTDR.THVAL[10:0] bits specify the threshold value for reading out from the relaying FIFO.

With the cut-through method, when a frame is handled as an error frame due to a fault (such as malformed data being read) after a read operation starts, there is no procedure for notifying the ETHERC module of the error. Accordingly, the ETHERC may send such error frames over the line as if they were normal frames. Avoid situations of this kind by using the store-and-forward method with the relaying FIFO. With the store-and-forward method, the PRC-TC module internally discards error frames.

37.3.2 Paths for the Transfer of Non-PTP Messages

Messages received through the Ethernet port are transferred to the EDMAC. The PRC-TC module is capable of relaying messages between the Ethernet ports.

Figure 37.5 is a diagram of paths for the transmission and reception and relaying of non-PTP messages.

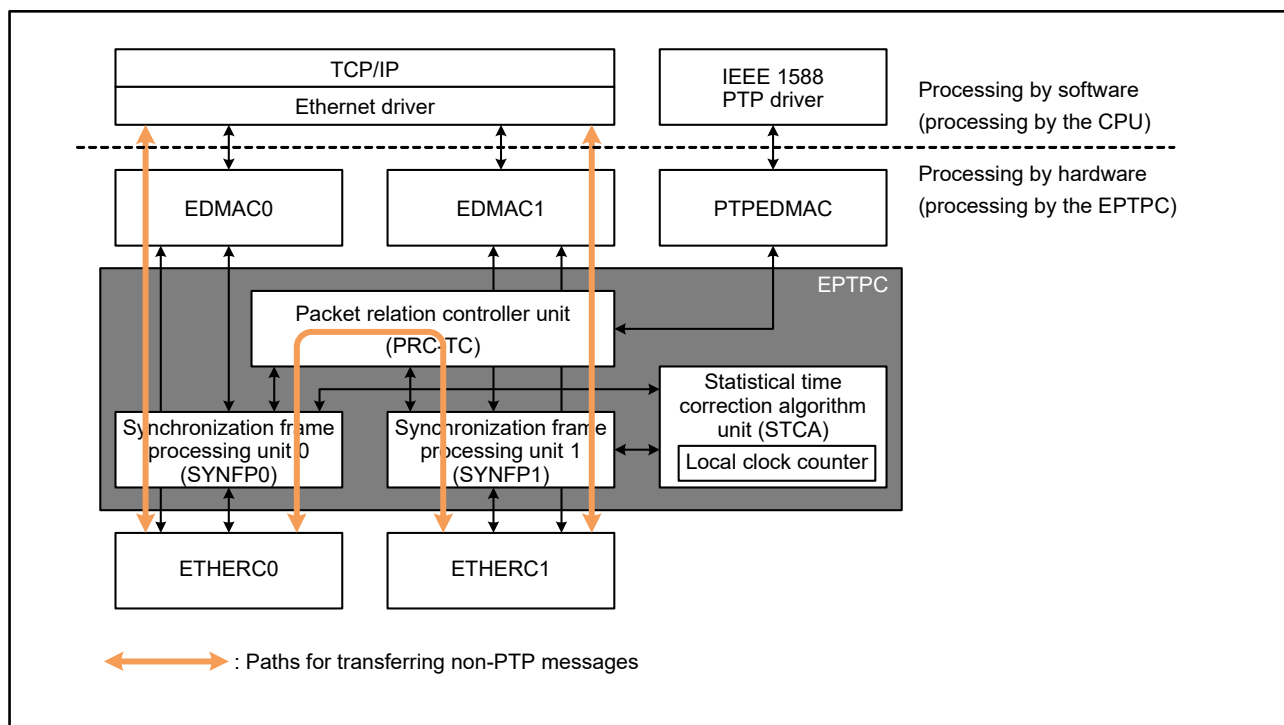


Figure 37.5 Paths for the Transmission and Reception and Relaying of Non-PTP Messages

37.3.3 Transmission and Reception and Relaying of PTP Messages

Hardware of the EPTPC automatically handles analysis of and the extraction of fields from received PTP messages, and generation and transmission of PTP messages. However, software must still handle the transmission of certain PTP messages. Table 37.12 shows the specifications for control over the transmission and reception of the various PTP messages.

Table 37.12 Control over the Transmission and Reception of PTP Messages

Message Type	Message	OC (Ordinary Clock)/BC (Boundary Clock)		TC (Transparent Clock)
		Master	Slave	
Event	Sync	Generation (automatic)	Reception (automatic)	Relaying (automatic)
	Delay_Req	Generation (automatic)	Reception (automatic)	Relaying (automatic)
	Pdelay_Req	Generation and reception (automatic)	Generation and reception (automatic)	Generation and reception (automatic)
	Pdelay_Resp	Generation and reception (automatic)	Generation and reception (automatic)	Generation and reception (automatic)
General	Announce	Generation (automatic)	Reception (software)	Reception (software)
	Follow_Up	—*1	Reception (automatic)	Relaying (automatic)
	Delay_Resp	Packet generation	Reception (automatic)	Relaying (automatic)
	Pdelay_Resp_Follow_Up	—*1	Reception (automatic)	Relaying (automatic)
	Management	Transmission and reception (software)		Transmission and reception (software) Relaying (automatic)
	Signaling	Transmission and reception (software)		Transmission and reception (software) Relaying (automatic)

Note 1. Control is not required as the clock in this case is a one-step clock.

37.3.4 Paths for the Transfer of PTP Messages

Paths for the transfer of PTP messages differ according to whether transfer requires processing by software or is automatically processed by hardware.

37.3.4.1 Paths for the Transfer of PTP Messages Requiring Processing by Software

Paths for the transfer of PTP messages where transfer requires processing by software are shown in Figure 37.6. The figure shows paths for message, clock-type, and process combinations for which “(software)” is indicated in Table 37.12.

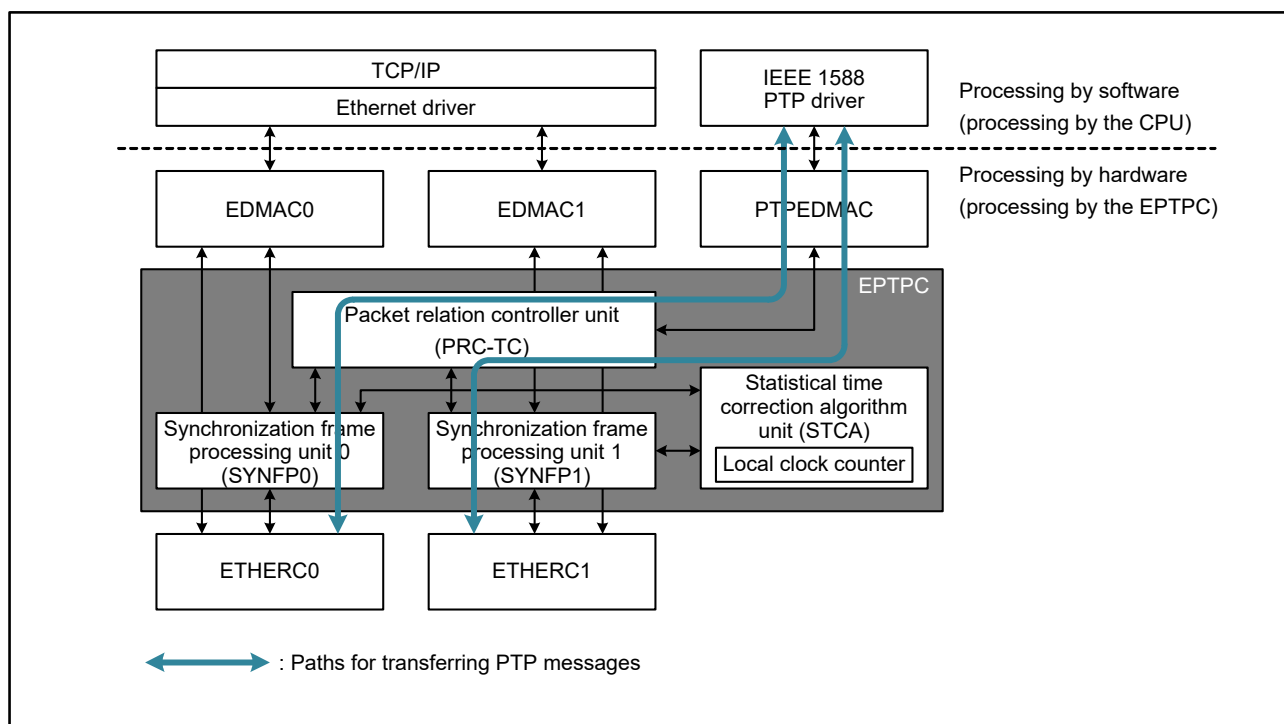


Figure 37.6 Paths for the Transfer of PTP Messages Requiring Processing by Software

37.3.4.2 Paths for the Transfer of PTP Messages Automatically Handled by Hardware

In the case of PTP messages for which hardware automatically handles the processing, the SYNFP modules handle transmission and reception, and the PRC-TC module handles relaying.

(1) Generation of and Response to PTP Messages by Hardware

Paths for transfer in the automatic generation of and response to PTP messages by the SYNFP modules are shown in Figure 37.7. The paths in the figure are used for operations “Generation (automatic)”, “Reception (automatic)”, and “Generation and reception (automatic)” indicated in Table 37.12.

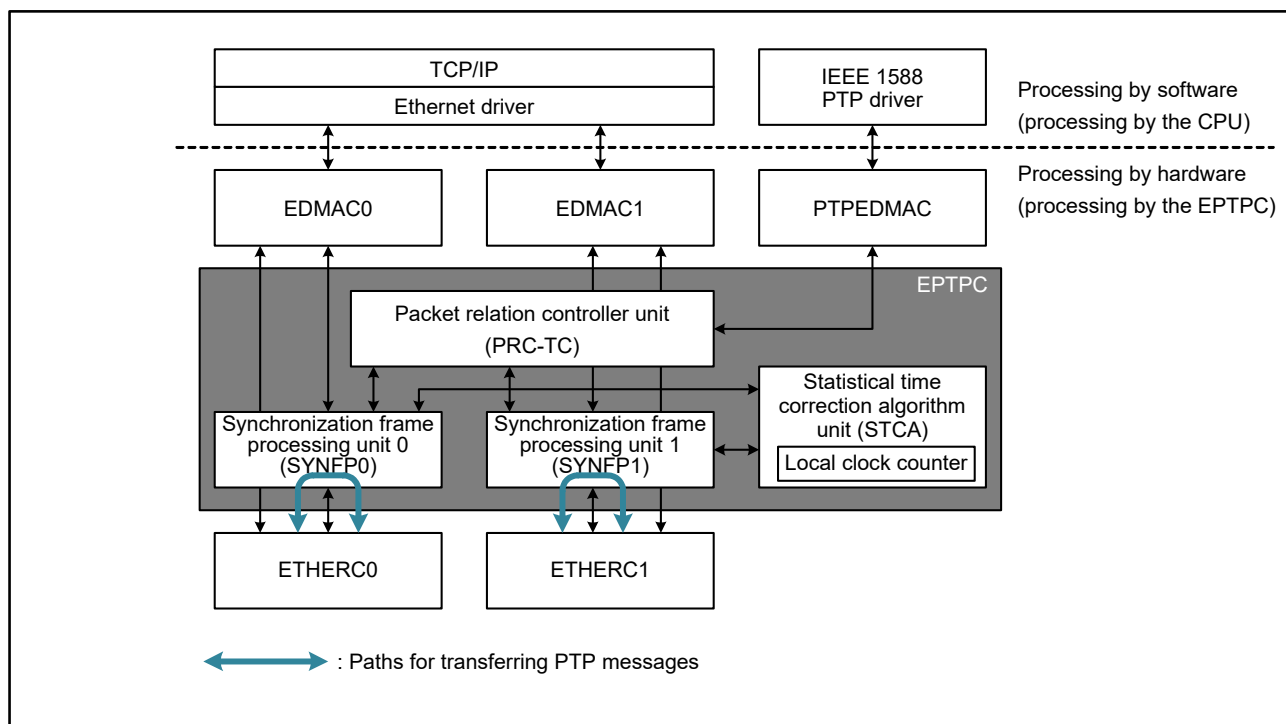


Figure 37.7 Paths for the Generation of and Response to PTP Messages by Hardware

(2) Relaying of PTP Messages by Hardware

Figure 37.8 shows paths for the relaying of PTP messages by the PRC-TC module. The paths in the figure are used for the “Relaying” operation indicated in Table 37.12.

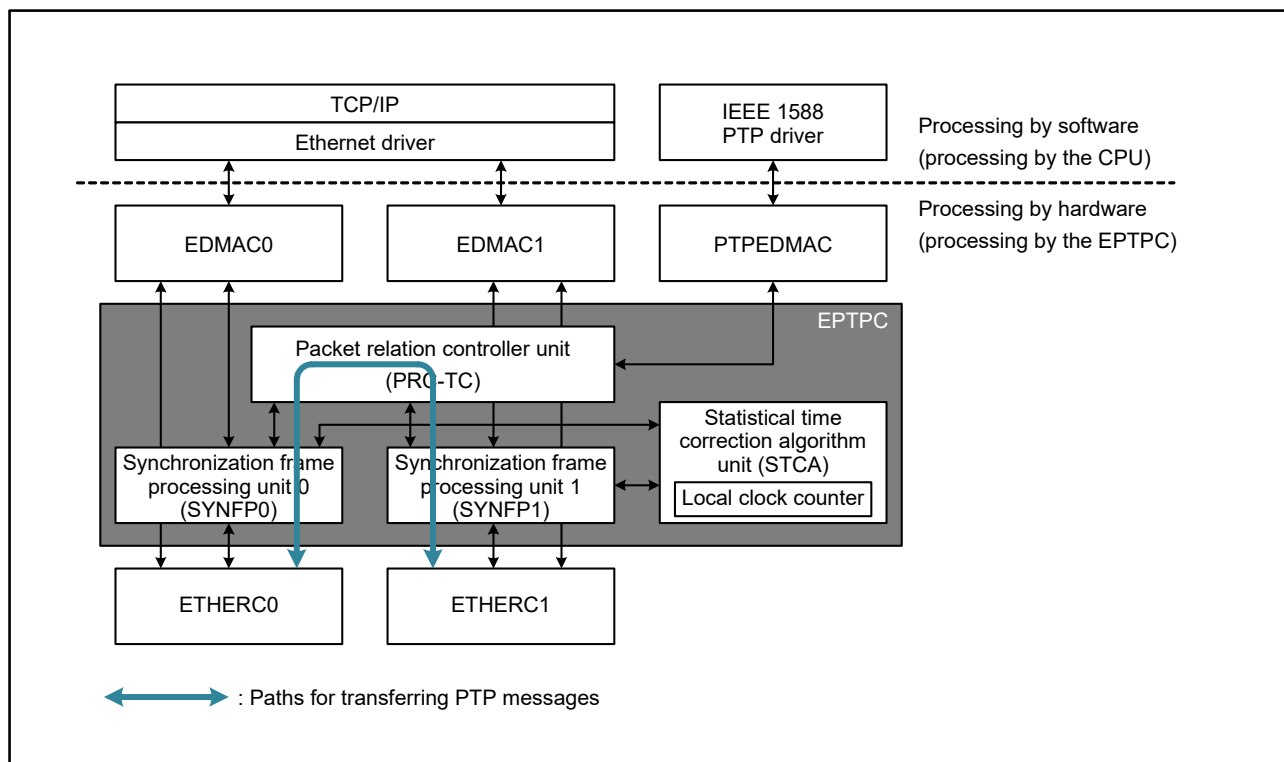


Figure 37.8 Paths for the Relaying of PTP Messages by Hardware

37.3.5 Clock Devices

The EPTPC can operate as the clock device defined in IEEE 1588.

37.3.5.1 End-to-End (E2E)

(1) Master

PTP messages are transmitted and received as listed below in operation as an end-to-end (E2E) master.

Table 37.13 List of PTP Message Processing (by an E2E Master)

Message Type	Message	Details of Processing
Event	Sync	Sync messages are transmitted at the fixed interval specified by the SYTLIR.SYNC[7:0] bits.
	Delay_Req	When this message is received, a Delay_Resp message is transmitted in response.
	Pdelay_Req	—
	Pdelay_Resp	—
General	Announce	Announce messages are transmitted at the fixed interval specified by the SYTLIR.ANCE[7:0] bits.
	Follow_Up	—
	Delay_Resp	This is transmitted as the response to a received Delay_Req messages.
	Pdelay_Resp_Follow_Up	—
	Management	Management messages are transmitted and received by software via the PTPEDMAC.
	Signaling	Signaling messages are transmitted and received by software via the PTPEDMAC.

(2) Slave

PTP messages are transmitted and received as listed below in operation as an E2E slave, and the calculated offsetFromMaster is used to correct the local time information.

Table 37.14 List of PTP Message Processing (by an E2E Slave)

Message Type	Message	Details of Processing
Event	Sync	The offsetFromMaster value is calculated when this message is received if twoStepFlag in flagField was FALSE (one-step clock).
	Delay_Req	Delay_Req messages are transmitted at random intervals from 0 to the time specified by the SYTLIR.DREQ[7:0] bits × 2.
	Pdelay_Req	—
	Pdelay_Resp	—
General	Announce	Announce messages are transmitted by software via the PTPEDMAC.
	Follow_Up	The offsetFromMaster value is calculated when this message is received if twoStepFlag in flagField of the most recently received Sync message was TRUE (two-step clock).
	Delay_Resp	The meanPathDelay value is calculated when this message is received.
	Pdelay_Resp_Follow_Up	—
	Management	Management messages are transmitted and received by software via the PTPEDMAC.
	Signaling	Signaling messages are transmitted and received by software via the PTPEDMAC.

37.3.5.2 Peer-to-Peer (P2P)

(1) Master

PTP messages are transmitted and received as listed below in operation as a peer-to-peer (P2P) master.

Table 37.15 List of PTP Message Processing (by a P2P Master)

Packet Type	Message	Details of Processing
Event	Sync	Timestamps for transmission are transmitted at the fixed interval specified by the SYTLIR.SYNC[7:0] bits.
	Delay_Req	—
	Pdelay_Req	Pdelay_Req messages are transmitted at the fixed interval specified by the SYTLIR.DREQ[7:0] bits. When this message is received, a Pdelay_Resp message is transmitted in response.
	Pdelay_Resp	This is transmitted as the response to a received Pdelay_Req messages. The meanPathDelay value is calculated when this message is received if twoStepFlag in flagField was FALSE (one-step clock).
General	Announce	Announce messages are transmitted at the fixed interval specified by the SYTLIR.ANCE[7:0] bits.
	Follow_Up	—
	Delay_Resp	—
	Pdelay_Resp_Follow_Up	The meanPathDelay value is calculated when this message is received if twoStepFlag in flagField of the most recently received Pdelay_Resp message was TRUE (two-step clock).
	Management	Management messages are transmitted by software via the PTPEDMAC.
	Signaling	Signaling messages are transmitted by software via the PTPEDMAC.

(2) Slave

PTP messages are transmitted and received as listed below in operation as a P2P slave, and the calculated offsetFromMaster is used to correct the local time information.

Table 37.16 List of PTP Message Processing (by a P2P Slave)

Packet Type	Message	Details of Processing
Event	Sync	The offsetFromMaster value is calculated when this message is received if twoStepFlag in flagField was FALSE (one-step clock).
	Delay_Req	—
	Pdelay_Req	Pdelay_Req messages are transmitted at the fixed interval specified by the SYTLIR.DREQ[7:0] bits. When this message is received, a Pdelay_Resp message is transmitted in response.
	Pdelay_Resp	This is transmitted as the response to a received Pdelay_Req messages. The meanPathDelay value is calculated when this message is received if twoStepFlag in flagField was FALSE (one-step clock).
General	Announce	Announce messages are transmitted by software via the PTPEDMAC.
	Follow_Up	The offsetFromMaster value is calculated when this message is received if twoStepFlag in flagField of the most recently received Sync message was TRUE (two-step clock).
	Delay_Resp	—
	Pdelay_Resp_Follow_Up	The meanPathDelay value is calculated when this message is received if twoStepFlag in flagField of the most recently received Pdelay_Resp message was TRUE (two-step clock).
	Management	Management messages are transmitted and received by software via the PTPEDMAC.
	Signaling	Signaling messages are transmitted and received by software via the PTPEDMAC.

37.3.5.3 Ordinary Clock (OC)

PTP messages are transmitted and received through one Ethernet port in operation as an ordinary clock.

An ordinary clock operates as the grand master clock or as a slave clock in the master-slave hierarchy.

For operation as an E2E master, E2E slave, P2P master, or P2P slave, see section 37.3.7, Operation as an E2E Master, section 37.3.8, Operation as an E2E Slave, section 37.3.10, Operation as a P2P Master, and section 37.3.11, Operation as a P2P Slave.

37.3.5.4 Boundary Clock (BC)

PTP messages are transmitted and received through both ports in operation as a boundary clock.

One port operates as a slave in synchronization with the root master clock and the other operates as the master that delivers time information synchronized with the master clock.

Both ports can also operate as masters.

For operation as an E2E master, E2E slave, P2P master, or P2P slave, see section 37.3.7, Operation as an E2E Master, section 37.3.8, Operation as an E2E Slave, section 37.3.10, Operation as a P2P Master, and section 37.3.11, Operation as a P2P Slave.

37.3.5.5 Transparent Clock (TC)

(1) E2E TC

In operation as an E2E transparent clock, received PTP-primary and PTP-pdelay messages are relayed.

Table 37.17 List of Packet Processing (by an E2E TC)

Message Type	Message	Details of Processing
Event	Sync	The packet residence time in the clock device is added when these messages are relayed.
	Delay_Req	
	Pdelay_Req	
	Pdelay_Resp	
General	Announce	These messages are relayed.
	Follow_Up	
	Delay_Resp	
	Pdelay_Resp_Follow_Up	
	Management	These messages are relayed.
	Signaling	Messages are also transmitted and received by software via the PTPEDMAC.

(2) P2P TC

In operation as a P2P transparent clock, PTP-primary messages other than Delay_Req and Delay_Resp are relayed.

Table 37.18 List of Packet Processing (by a P2P TC)

Message Type	Message	Details of Processing
Event	Sync	The packet residence time in the clock device is added when this message is relayed.
	Delay_Req	—
	Pdelay_Req	Pdelay_Req messages are transmitted at the fixed interval specified by the SYTLIR.DREQ[7:0] bits. When this message is received, a Pdelay_Resp message is transmitted in response.
	Pdelay_Resp	This is transmitted as the response to a received Pdelay_Req messages. The meanPathDelay value is calculated when this message is received if twoStepFlag in flagField was FALSE (one-step clock).
General	Announce	These messages are relayed.
	Follow_Up	
	Delay_Resp	—
	Pdelay_Resp_Follow_Up	The meanPathDelay value is calculated when this message is received if twoStepFlag in flagField of the most recently received Pdelay_Resp message was TRUE (two-step clock).
	Management	These messages are relayed.
	Signaling	Messages are also transmitted and received by software via the PTPEDMAC.

37.3.6 EPTPC Initialization

Transmitting and receiving PTP messages requires settings in various registers of the EPTPC.

Table 37.19 is a list of the registers for which settings are required. Set the registers corresponding to the Ethernet port used. Also set the registers listed in Table 37.20 if the UDP and IPv4 are used for the frame format of the PTP messages.

Table 37.19 List of the Registers for which Settings are Required

Register Name	Settings	Description
STCFR	Example: 0x00000002	The value for 50 MHz is given as an example. Three other settings are also available.
PRMACRU0, PRMACRL0	As desired	This setting is not required If Ethernet port 0 is not used.
PRMACRU1, PRMACRL1	As desired	This setting is not required If Ethernet port 1 is not used.
SYCONFR	Example: 0x00000002	The setting differs with the type of PTP clock operation.
SYMACRU, SYMACRL	As desired	
SYSPVRR	0x00000002	transportSpecific and version fields
SYDOMR	As desired	
SYCIDRU, SYCIDRL	As desired	Set the same value for Ethernet ports 0 and 1.
SYPNUMR	0x00000001 or 0x00000002	If the PTP clock operates as an OC, the setting is 0x00000001. If it operates as a BC or TC, set 0x00000001 for one port and 0x00000002 for the other.
PPMACRU, PPMACRL	01:1B:19:00:00:00	MAC address for PTP-primary messages
PDMACRU, PDMACRL	01:80:C2:00:00:0E	MAC address for PTP-pdelay messages
DASYMRU, DASYMRL	0x00000000	
TSLATR	As desired	Depends on the link transfer rate and STCA clock frequency.
SYFORMR	As desired	Four settings are available.
SYLLCCTLR	0x00000003	LLC-CTL field value for Ethernet frames
PETYPER	0x000088F7	Ethertype for PTP messages

Table 37.20 List of the Registers for which Settings are Required (Additional Settings Required when UDP or IPv4 is to be Used)

Register Name	Settings	Description
SYIPADDRR	As desired	Local IP address
PRIPR	0xE0000181	IP address for PTP-primary messages
PETOSR	As desired	Set the highest allowable traffic class selector codepoint as the value for the differentiated service (DS) field.
PGTOSR	As desired	
PPTTLR	As desired	TTL field value for PTP-primary messages
PEUDPR	0x0000013F	UDP port number for event messages
PGUDPR	0x00000140	UDP port number for general messages
PDIPR	0xE000006B	IP address for PTP-pdelay messages
PDTTLR	0x00000001	TTL field value for PTP-pdelay messages

In operation as an OC or BC, set registers as shown below in order to transfer received Announce, Management, and Signaling messages to the PTPEDMAC.

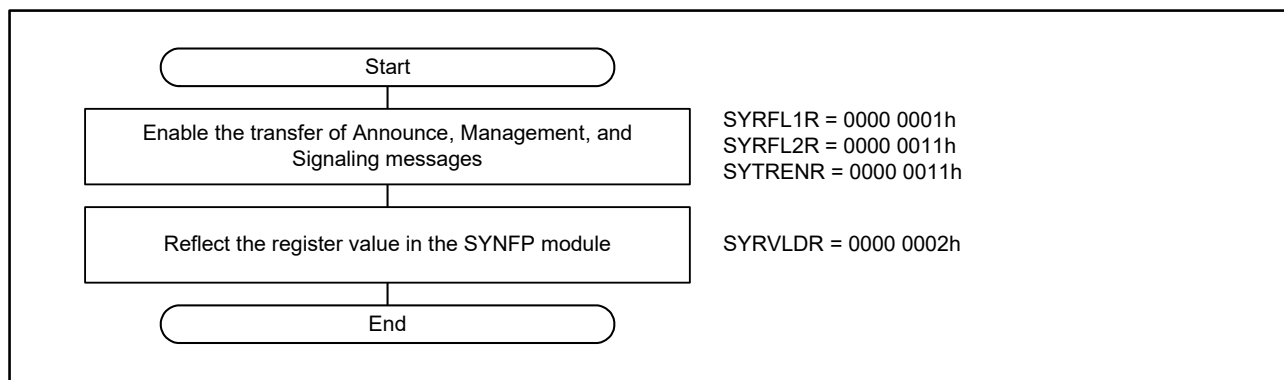


Figure 37.9 Common Settings for PTP Devices

37.3.7 Operation as an E2E Master

37.3.7.1 Preparatory Setting

Table 37.21 lists the registers for use in operation as an E2E master.

If the EPTPC operates as an OC or as a BC with both ports serving as masters, set the initial value of the time information in advance. Refer to section 37.2.18, Local Clock Counter Initial Value Registers (LCIVRU, LCIVRM, LCIVRL) for the initial value of the time information. Reflecting the values set in the registers requires setting the SYRVLDR.STUP or ANUP bit to 1.

Table 37.21 Registers for Use in Operation as an E2E Master

Register Name	SYRVLDR Register Bits Used for Loading Direction	Settings	Description
SYNFR	STUP	0x00000000	flagField for Sync messages
SYTLIR	STUP ANUP	Example: 0x00000001	Delay_Resp: 1 s Sync: 1 s Announce: 2 s
ANFR	ANUP	0x00000000	flagField for Announce messages
GMPR	ANUP	As desired	
GMCQR	ANUP	As desired	
GMIDR	ANUP	As desired	
CUOTSR	ANUP	As desired	timeSource: Internal Oscillator
SRR	ANUP	As desired	If the EPTPC operates as an OC or as a BC with both ports serving as masters, set this register to 0x00000000. If the EPTPC operates as a BC in the combination of a slave and master, set this register to the StepsRemoved field value of Announce messages received by the slave plus one.
SYRFL1R	STUP	0x00004001	This enables the processing of Delay_Req messages by the SYNFP module.
SYRFL2R	STUP	0x00000011	This enables the transfer of Signaling and Management messages to the PTPEDMAC.
SYTRENR	STUP	0x00000011	This enables the transmission of Sync and Announce messages.

37.3.7.2 Procedure for Starting Operations

Figure 37.10 shows the procedure for settings to start operation as an E2E master.

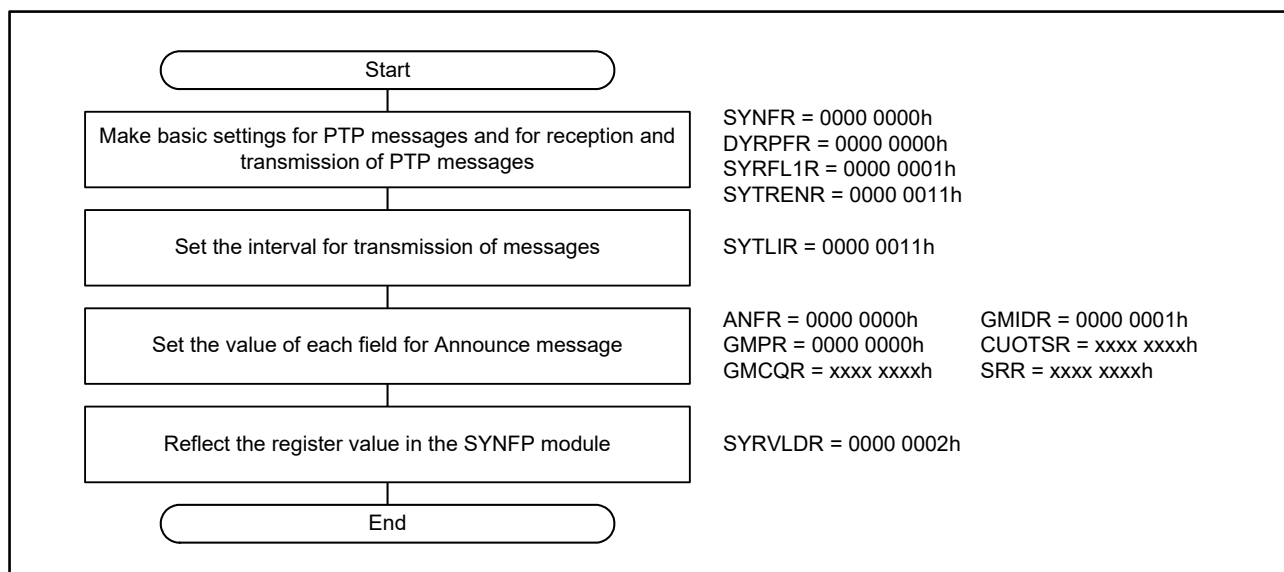


Figure 37.10 Procedure for Starting Operation as an E2E Master

37.3.7.3 Procedure for Changing the Settings

Increases in the frequency of receiving Delay_Req messages due to network conditions may lead to an overflow of the FIFO buffer which is receiving the Delay_Req messages. In such cases, change the value of the logMessageInterval field of Delay_Resp messages so that the slave sending the Delay_Req messages lengthens the interval between the messages. Figure 37.11 shows the procedure for changing the value of the logMessageInterval field.

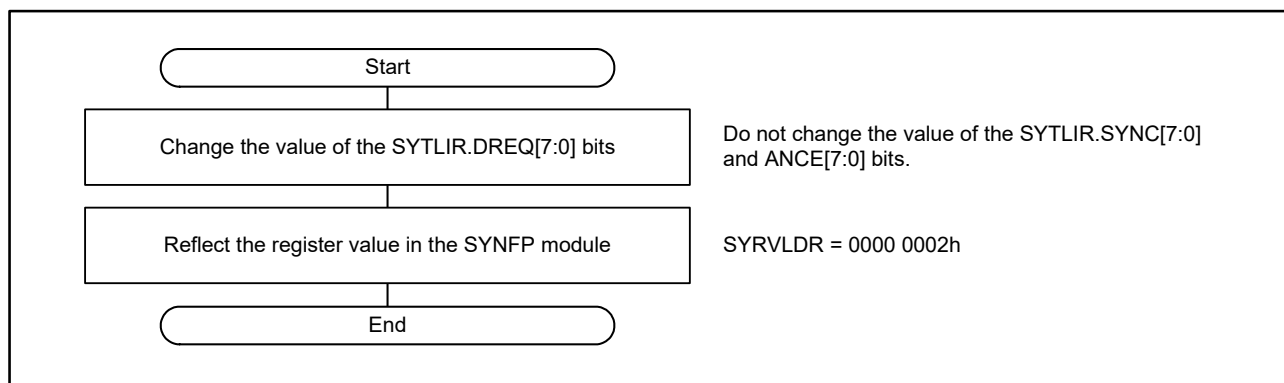


Figure 37.11 Procedure for Changing the Value of the logMessageInterval Field for Delay_Resp Messages

37.3.7.4 Procedure for Stopping Operations

Figure 37.12 shows the procedure for stopping operation as an E2E master. To confirm that the operation as the E2E master is completely stopped, read the SYSR.GENDN flag and RESDN flag to check that generating messages and sending responses are completely stopped.

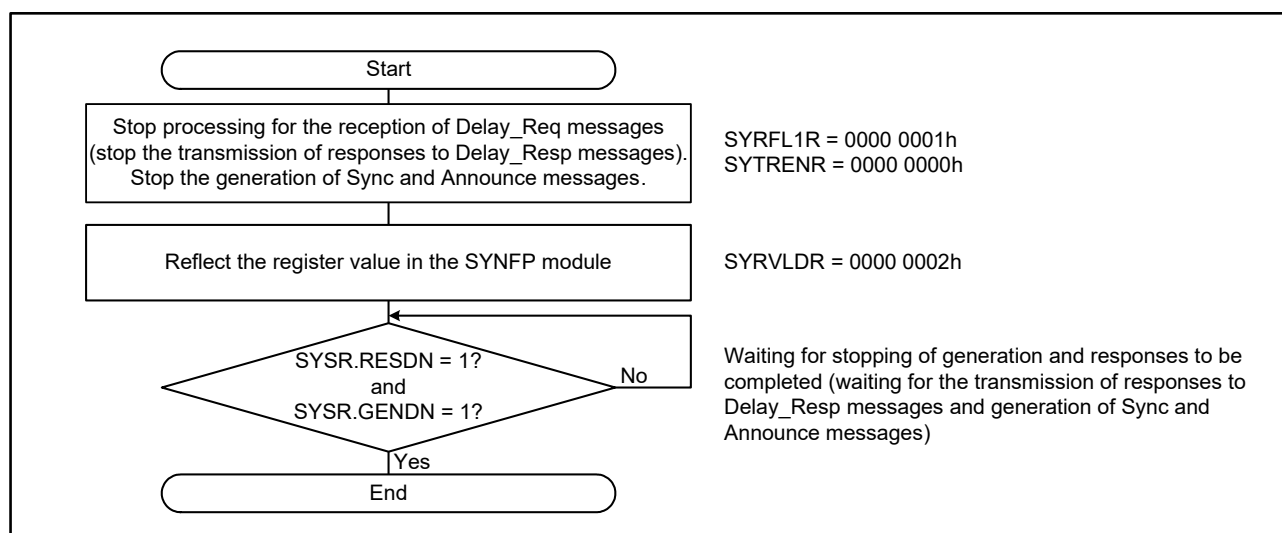


Figure 37.12 Procedure for Stopping Operation as an E2E Master

37.3.8 Operation as an E2E Slave

37.3.8.1 Preparatory Setting

Table 37.22 lists the registers for use in operation as an E2E slave.

Reflecting the value set in the register in SYNFP operations requires setting the SYRVLDR.STUP, ANUP, or BMUP bit to 1.

Table 37.22 Registers for Use in Operation as an E2E Slave

Register Name	SYRVLDR Register Bits Used for Loading Direction	Settings	Description
MTCID	BMUP	As desired	This is the clockIdentity value of the master clock that provides synchronization.
MTPID	BMUP	As desired	This is the portNumber value of the master clock that provides synchronization.
SYTLIR	ANUP BMUP	0x00000000	Delay_Resp: 1 s*1
RSTOUTR	STUP	As desired	
SYNTOR	—	As desired	
SYRFL1R	STUP	0x00040441	This enables the reception of Delay_Resp, Follow_Up, and Sync messages and the transfer of Announce messages to the PTPEDMAC.
SYRFL2R	STUP	0x00000011	This enables the transfer of Signaling and Management messages to the PTPEDMAC.
SYTRENR	STUP	0x00000100	This enables the generation of Delay_Req messages.

Note 1. In the reception of Delay_Resp messages by an E2E slave, the SYTLIR.DREQ[7:0] bits must be adjusted if the value of the SYRLIR.DRESP[7:0] flags is to be altered. The SYTLIR.DREQ[7:0] bits specify a value in the range from –7 to 6. Set the SYTLIR.DREQ[7:0] bits to –7 if the value indicated by the SYRLIR.DRESP[7:0] flags is less than or equal to –8 and to 6 if the value indicated by the SYRLIR.DRESP[7:0] flags is greater than or equal to 7.

37.3.8.2 Procedure for Starting Operations

Figure 37.13 shows the procedure for settings to start operation as an E2E slave.

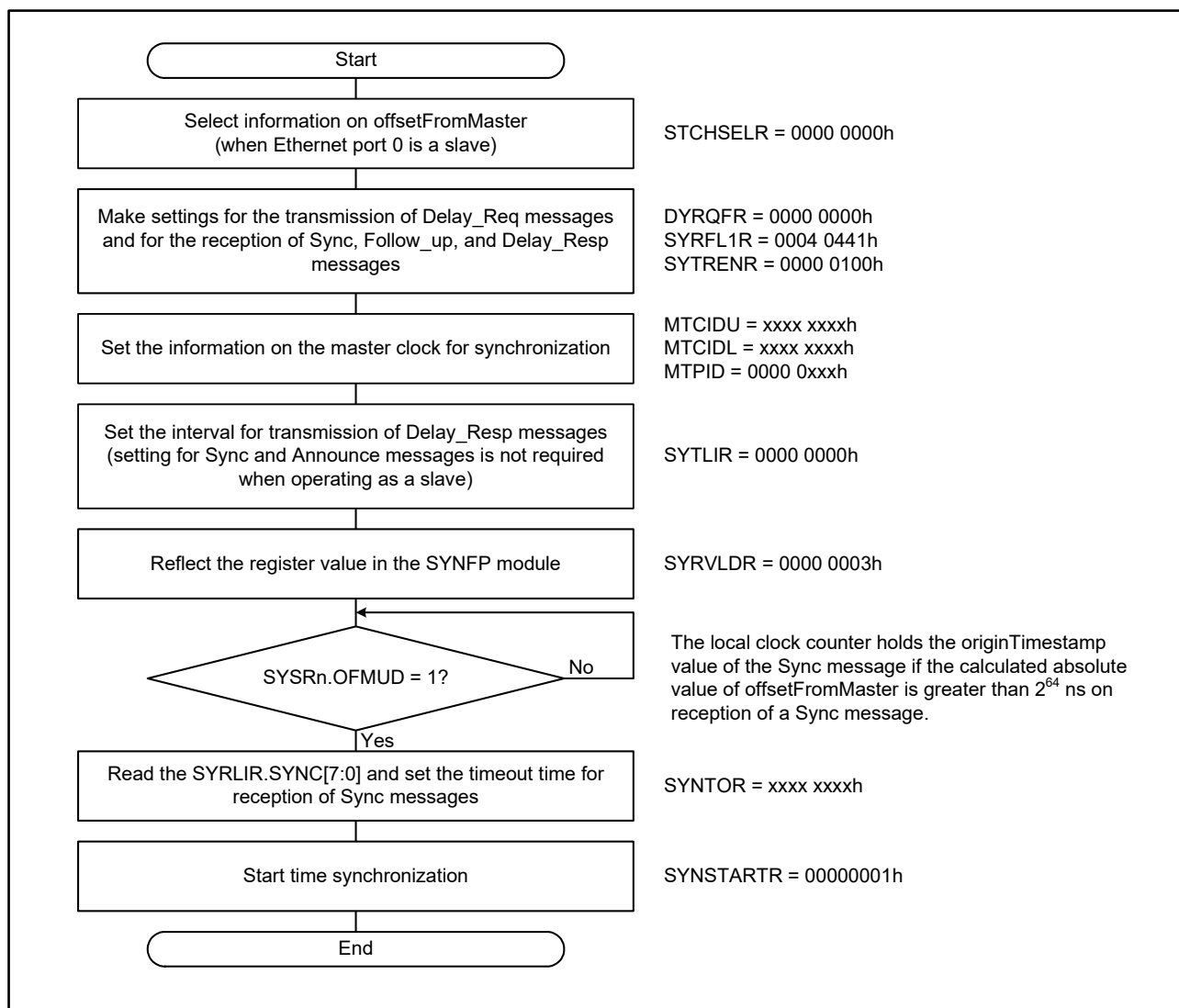


Figure 37.13 Procedure for Starting Operation as an E2E Slave

37.3.8.3 Procedure for Changing the Settings

IEEE 1588 stipulates that the average interval for the transmission of Delay_Req messages must be adjusted in response to changes in the value of the logMessageInterval field of received Delay_Resp messages. The EPTPC sets the SYSR.INTCHG flag to 1 if the logMessageInterval value of a received message differs from that of the previous message.

When this happens, set the SYTLIR.DREQ[7:0] bits to the value of the SYRLIR.DRESP[7:0] bits.

The SYTLIR.DREQ[7:0] bits specify a value in the range from -7 to +6. Set the SYTLIR.DREQ[7:0] bits to -7 if the value indicated by the SYRLIR.DRESP[7:0] flags is less than or equal to -8 and to 6 if the value indicated by the SYRLIR.DRESP[7:0] bits is greater than or equal to 7.

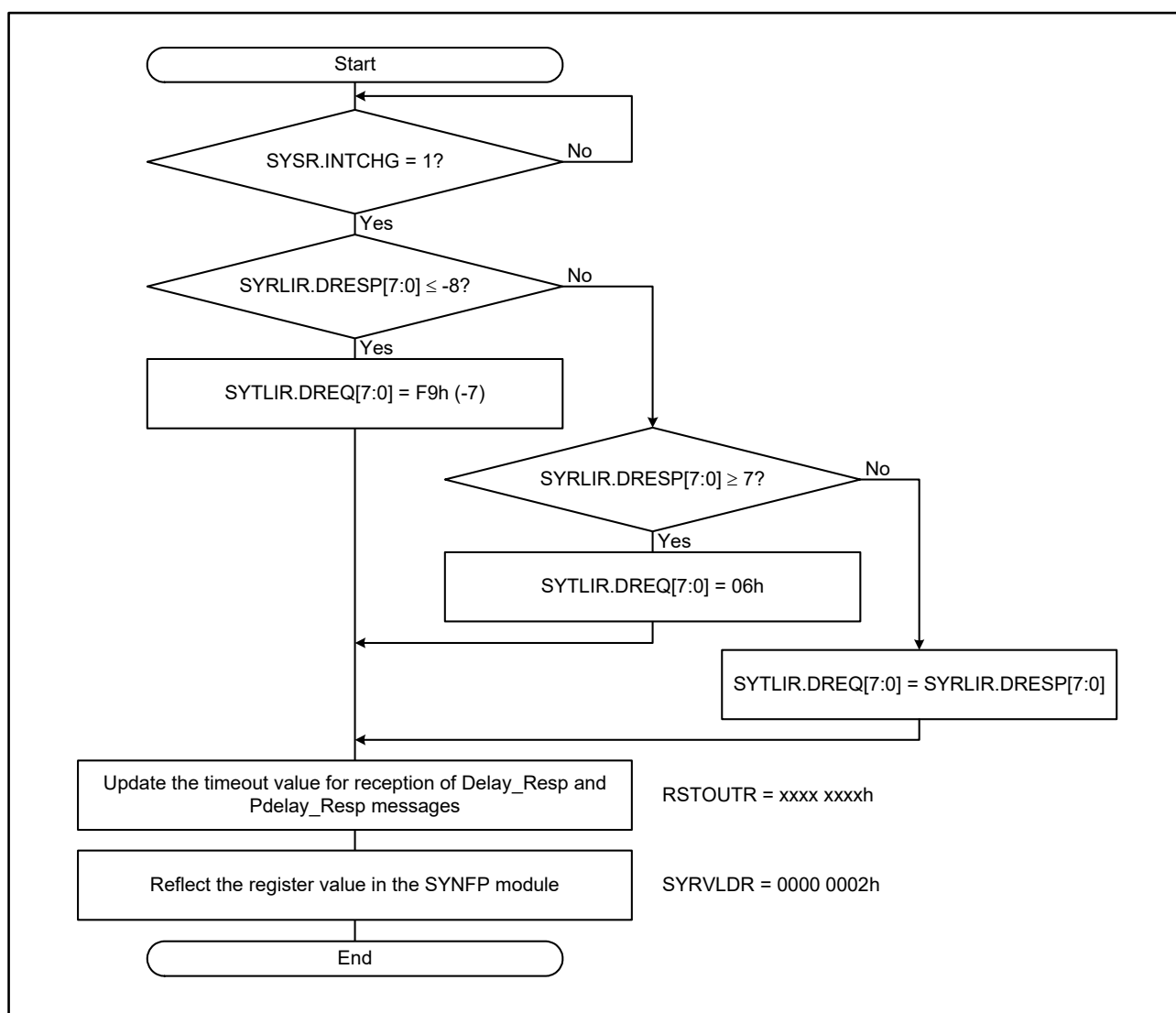


Figure 37.14 Procedure for Changing the Interval for Transmission of Delay_Req Messages

37.3.8.4 Procedure for Stopping Operations

Figure 37.15 shows the procedure for stopping operation as an E2E slave. To confirm that operation as an E2E slave is completely stopped, read the SYSR.GENDN flag to check that generation is completely stopped.

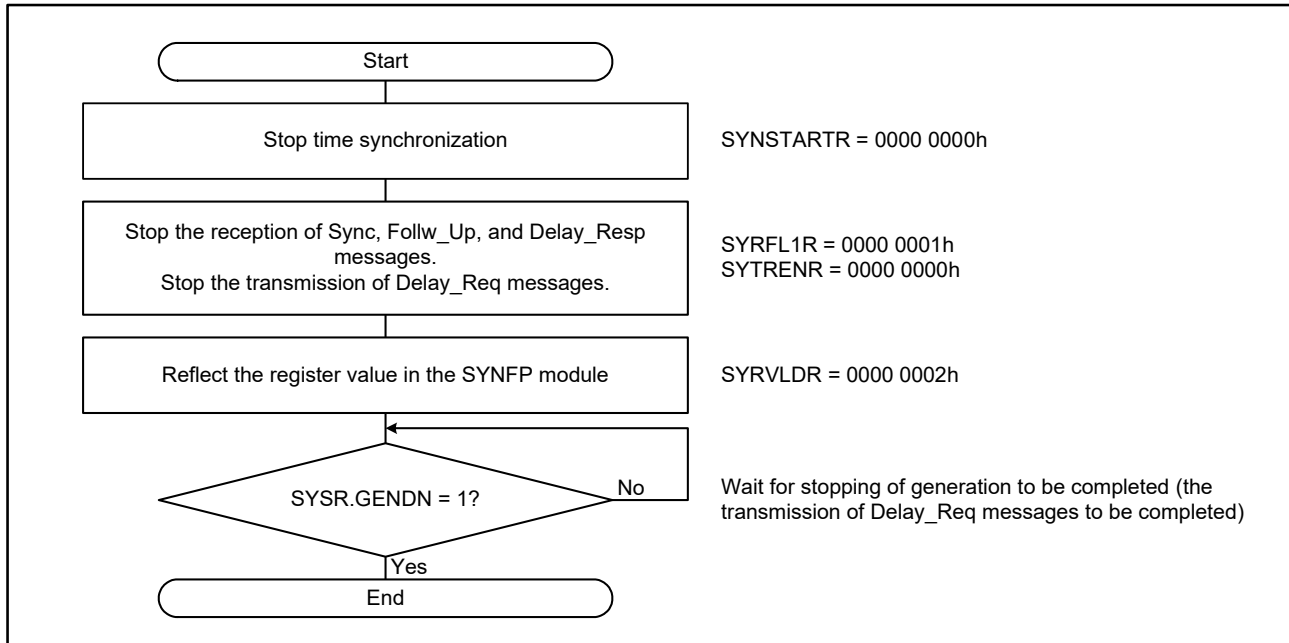


Figure 37.15 Procedure for Stopping Operation as an E2E Slave

37.3.9 P2P Operation (Common to Master and Slave)

If the EPTPC is to be operated as a P2P, the SYNFP module handles the processing of PTP-pdelay messages regardless of whether operation is to be as a master or slave. The interval for Pdelay_Req transmission and the parameters for monitoring of Pdelay_Resp messages must be set at the same time. Registers for use in operation as a P2P are listed in Table 37.23.

Table 37.23 Registers for Use in Operation as a P2P

Register Name	SYRVLDR Register Bits Used for Loading Direction	Settings	Description
MTCID	BMUP	As desired	This is the clockIdentity value of the synchronized master clock.
MTPID	BMUP	As desired	This is the portNumber value of the synchronized master clock.
SYTLIR	ANUP STUP	0x00000000	Announce: — Sync: — Pdelay_Req: 1 s
RSTOUTR	STUP	As desired	
SYRFL1R	STUP	0x44400001	This enables the reception of Pdelay_Req, Pdelay_Resp, and Pdelay_Resp_Follow_Up messages and the transfer of Announce messages to the PTPEDMAC.
SYRFL2R	STUP	0x00000011	This enables the transfer of Signaling and Management messages to the PTPEDMAC.
SYTRENR	STUP	0x00001000	This enables the generation of Pdelay_Req messages.

37.3.9.1 Procedure for Starting Operations

Figure 37.16 shows the procedure for starting operation as a P2P (sending and receiving PTP-pdelay messages).

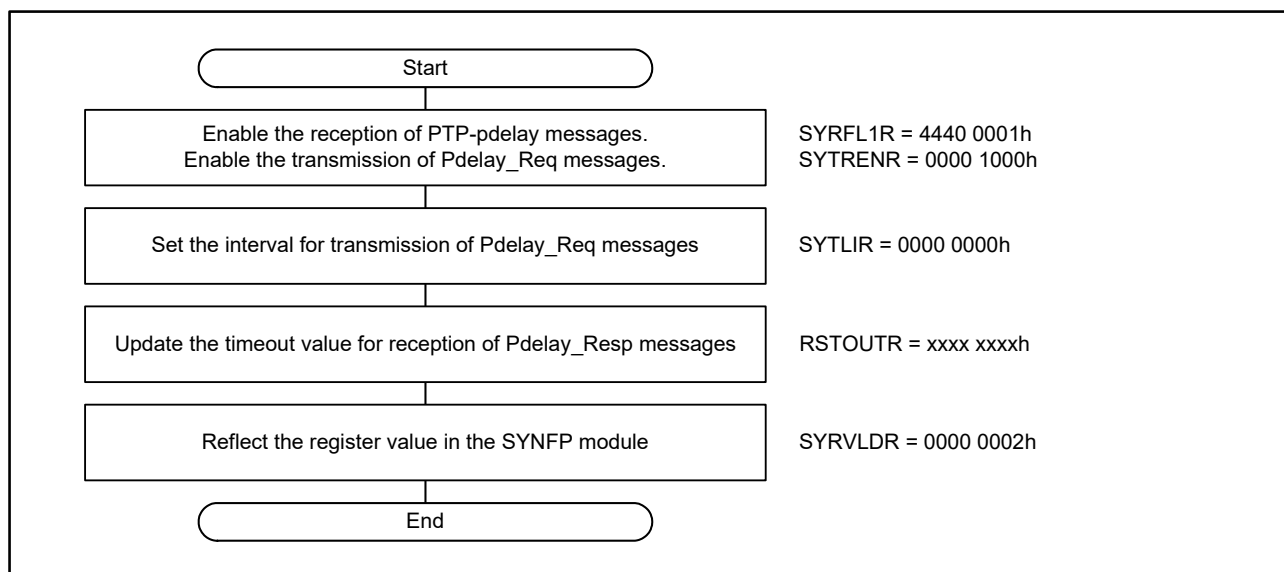


Figure 37.16 Procedure for Starting Operation as a P2P

37.3.9.2 Procedure for Stopping Operations

Figure 37.17 shows the procedure for stopping operation as a P2P (sending and receiving PTP-pdelay messages).

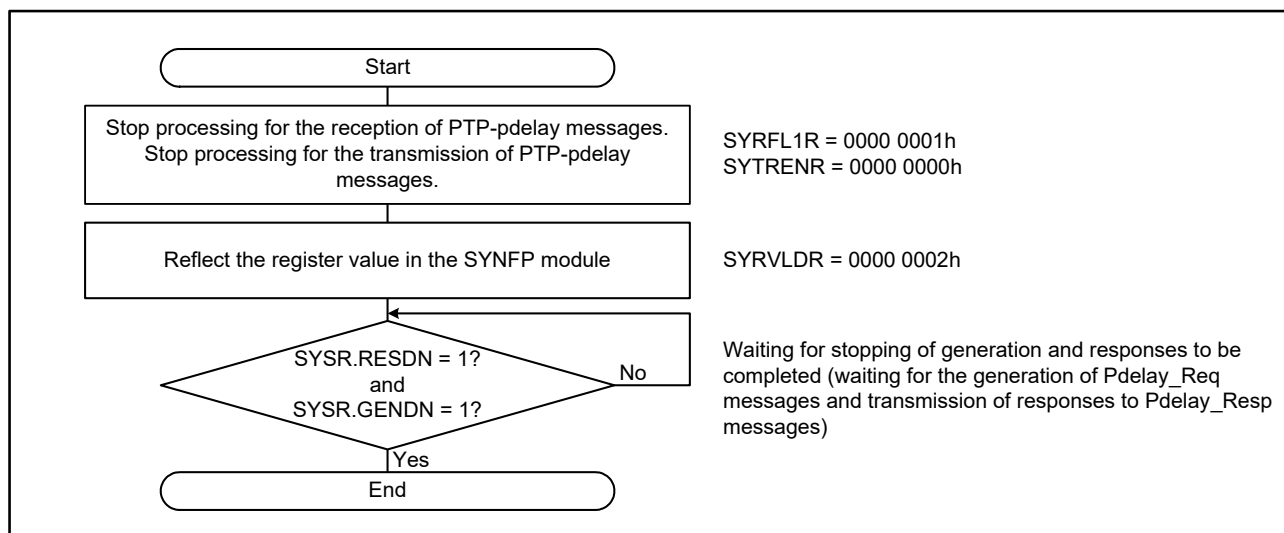


Figure 37.17 Procedure for Stopping Operation as a P2P

37.3.10 Operation as a P2P Master

When the EPTPC operates as OC or BC that uses both ports as the master, set the initial value of time information in advance as required. Refer to section 37.2.18, Local Clock Counter Initial Value Registers (LCIVRU, LCIVRM, LCIVRL) for the initial value of time information. Registers for use in operation as a P2P master are listed in Table 37.24.

Table 37.24 Registers for Use in Operation as a P2P Master

Register Name	SYRVLDR Register Bits Used for Loading Direction	Settings	Description
SYCONFR	—	0x00000028	
ANFR	ANUP	0x00000000	flagField for Announce messages
SYNFR	STUP	0x00000000	flagField for Sync messages
SYTLIR	ANUP STUP	0x00000001	Announce: 2 s Sync: 1 s Pdelay_Req: 1 s
GMPR	ANUP	As desired	Grandmaster Priority1 and Priority2
GMCQR	ANUP	As desired	Grandmaster Quality
GMIDR	ANUP	As desired	Grandmaster Identity
CUOTSR	ANUP	As desired	currentUtcOffset, timeSource
SRR	ANUP	As desired	StepsRemoved
RSTOUTR	STUP	As desired	
SYRFL1R	STUP	0x44400000	This enables the reception of Pdelay_Req, Pdelay_Resp, and Pdelay_Resp_Follow_Up messages.
SYRFL2R	STUP	0x00000011	This enables the transfer of Signaling and Management messages to the PTPEDMAC.
SYTRENr	STUP	0x00001011	This enables the transmission of Pdelay_Req, Sync, and Announce messages.

37.3.10.1 Procedure for Starting Operations

When transmission of Sync and Announce messages is started during operation as a P2P (sending and receiving PTP-pdelay messages), the EPTPC operates as a P2P master. Figure 37.18 shows the procedure for starting operation as a P2P master.

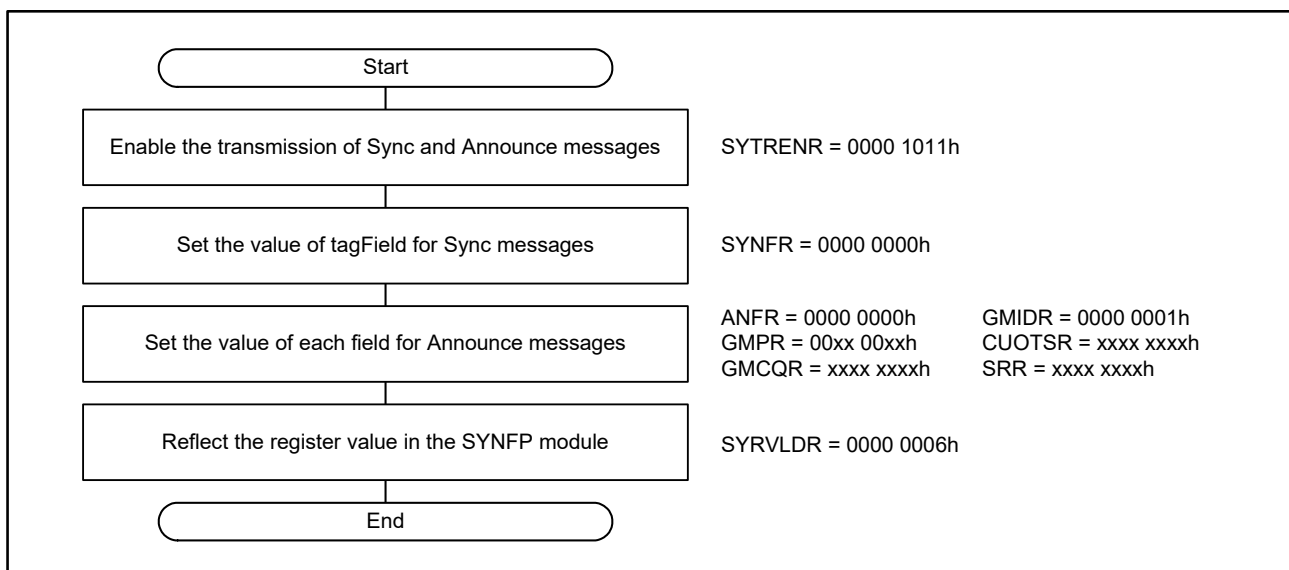


Figure 37.18 Procedure for Starting Operation as a P2P Master

37.3.10.2 Procedure for Stopping Operations

Figure 37.19 shows the procedure for stopping the transmission of Sync and Announce messages to stop operation as a P2P master.

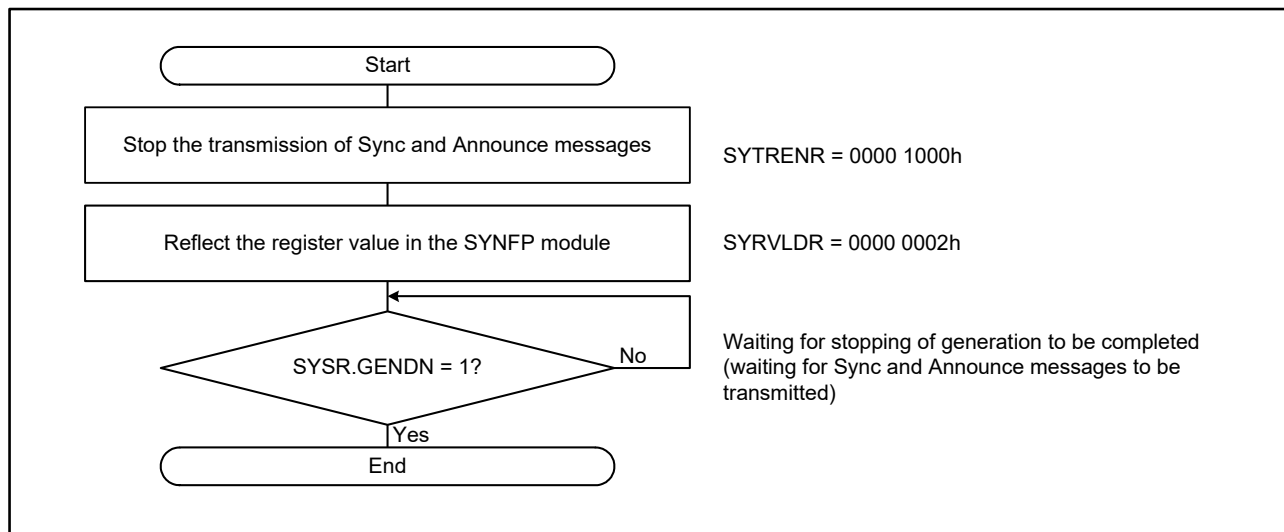


Figure 37.19 Procedure for Stopping Operation as a P2P Master

37.3.11 Operation as a P2P Slave

Setting a SYNFP module to receive Sync messages and Follow_Up messages during P2P operation obtains operation as a P2P slave. Information on the master clock for synchronization must be specified.

Table 37.25 lists the registers for use in operation as a P2P slave.

Table 37.25 Registers for Use in Operation as a P2P Slave

Register Name	SYRVLDR Register Bits Used for Loading Direction	Settings	Description
MTCID	BMUP	As desired	This is the clockIdentity value of the synchronized master clock.
MTPID	BMUP	As desired	This is the portNumber value of the synchronized master clock.
RSTOUTR	STUP	As desired	
SYRFL1R	STUP	0x44400441	This enables the reception of Pdelay_Req, Pdelay_Resp, Pdelay_Resp_Follow_Up, Follow_Up, and Sync messages and the transfer of Announce messages to the PTPEDMAC.

37.3.11.1 Procedure for Starting Operations

Figure 37.20 shows the procedure for making the additional settings for shifting to operation as a slave during operation as a P2P (sending and receiving PTP-pdelay messages).

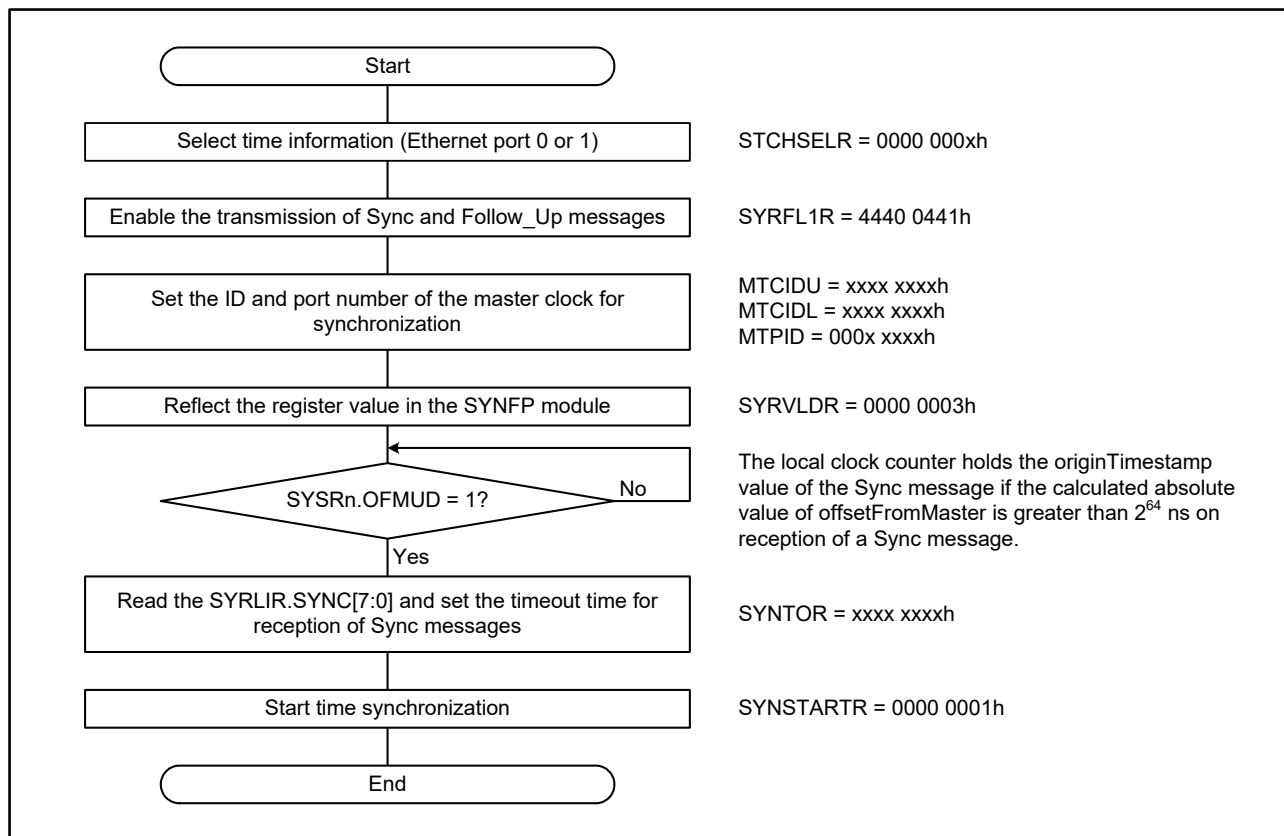


Figure 37.20 Procedure for Starting Operation as a P2P Slave

37.3.11.2 Procedure for Stopping Operations

Figure 37.21 shows the procedure for stopping the reception of Sync and Follow_Up messages to stop operation as a P2P slave.

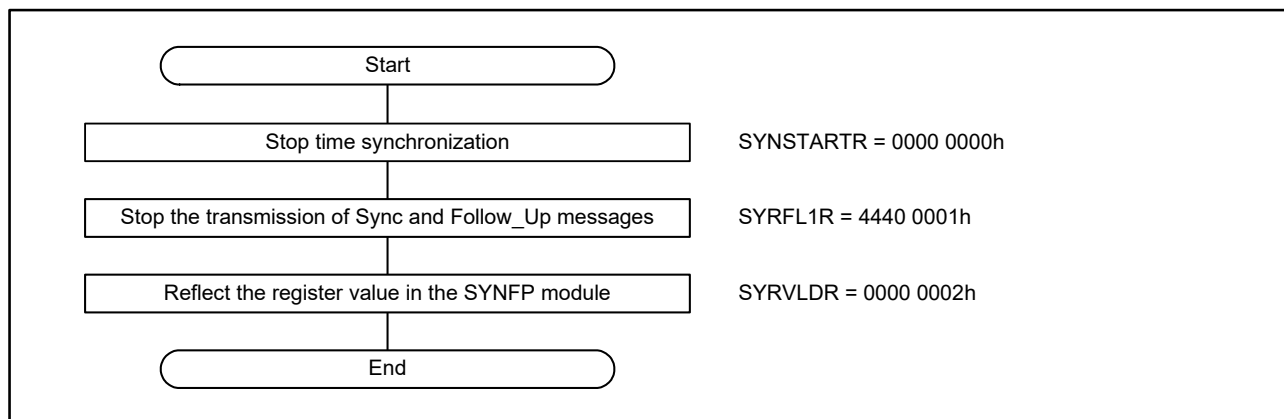


Figure 37.21 Procedure for Stopping Operation as a P2P Slave

37.3.12 Operation as an E2E TC

37.3.12.1 Preparatory Setting

Table 37.26 lists the registers for use in operation as an E2E TC.

Make the settings for both the SYNFP0 and SYNFP1 modules.

Table 37.26 Registers for Use in Operation as an E2E TC

Register Name	SYRVLDR Register Bits Used for Loading Direction	Settings	Description
SYCONF	—	Example: 0x00000028	Set the TCMOD bit to 0 as a preparatory setting for the PTP device.
SYRFL1R	STUP	0x22222222	This enables the relaying of messages by the PRC-TC.
SYRFL2R	STUP	0x20000033	This enables the transfer of Signaling and Management messages to the PTPEDMAC and the relaying of these messages by the PRC-TC.
SYTRENR	STUP	0x00000000	No generation of messages

37.3.12.2 Procedure for Starting Operations

Figure 37.22 shows the procedure for starting operation as an E2E TC.

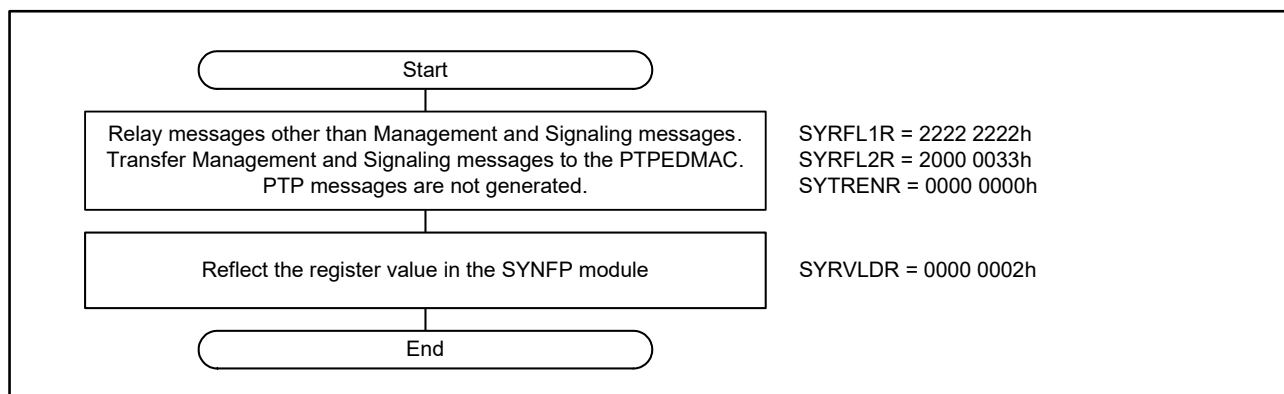


Figure 37.22 Procedure for Starting Operation as an E2E TC

37.3.13 Operation as a P2P TC

Table 37.27 lists the registers for use in operation a P2P TC.
Make the settings for both the SYNFP0 and SYNFP1 modules.

Table 37.27 Registers for Use in Operation as a P2P TC

Register Name	SYRVLDR Register Bits Used for Loading Direction	Settings	Description
SYCONFR	—	Example: 0x00100028	Set the TCMOD bit to 1 as a preparatory setting for the PTP device.
RSTOUTR	STUP	As desired	
SYTLIR	STUP	Example: 0x00000000	Pdelay_Req: 1 s
SYRFL1R	STUP	0x44400222	This enables the discarding of Delay_Req and Delay_Resp messages and the relaying of Sync, Follow_Up, and Announce messages by the PRC-TC.
SYRFL2R	STUP	0x20000033	This enables the transfer of Signaling and Management messages to the PTPEDMAC and the relaying of these messages by the PRC-TC
SYTRENR	STUP	0x00001000	This enables the generation of Pdelay_Req messages.

37.3.13.1 Procedure for Starting Operations

Figure 37.23 shows the procedure for starting operation as a P2P TC.

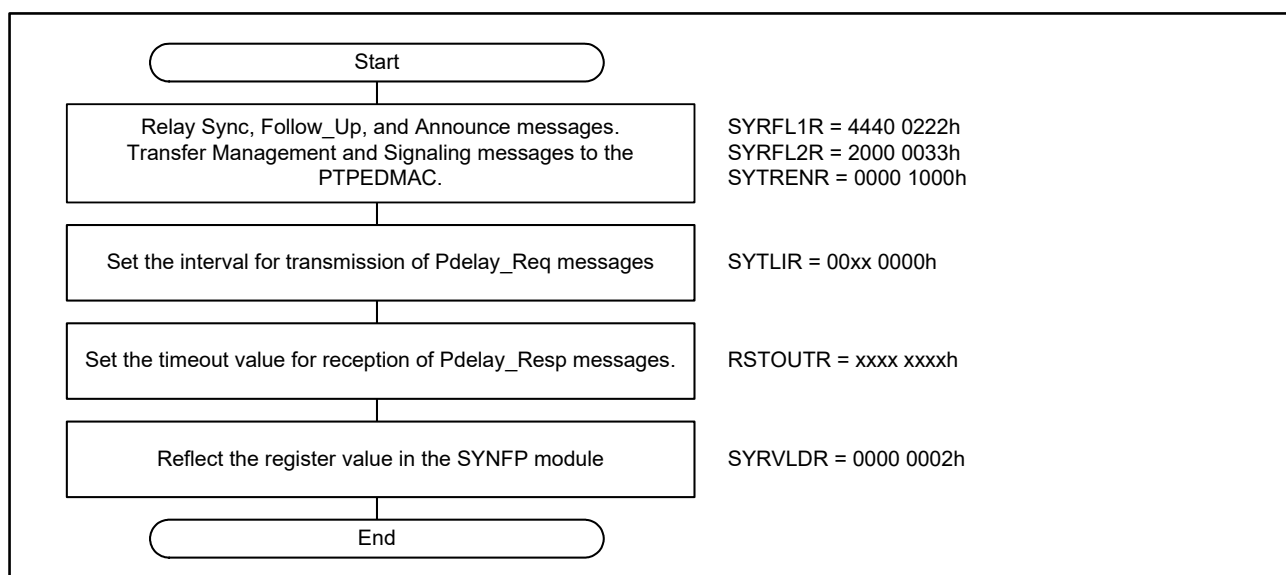


Figure 37.23 Procedure for Starting Operation as a P2P TC

37.3.14 Monitoring of Received Messages

37.3.14.1 Reception of Announce Messages

The EPTPC does not detect timeouts in the reception of Announce messages. To detect timeouts in the reception of these messages, monitor the reception of Announce messages by software.

37.3.14.2 Reception of Sync Messages

The STSR.SYNTOUT flag becomes 1 when a timeout in the reception of a Sync message occurs during the correction of time synchronization.

The SYSR.OFMUD flag becomes 1 when a Sync message is received, regardless of whether time synchronization is being corrected. Accordingly, the reception of Sync messages is detectable by reference to the SYSR.OFMUD flag even if the correction of time synchronization stops due to a timeout in the reception of a Sync message.

37.3.14.3 Reception of Delay_Resp and Pdelay_Resp Messages

The SYSR.DRPTO flag becomes 1 when a timeout in the reception of a Delay_Resp message occurs after the transmission of a Delay_Req message during operation as an E2E slave, or when a timeout in the reception of a Pdelay_Resp message occurs after the transmission of a Pdelay_Req message during operation as a P2P.

The SYSR.MPDUD flag becomes 1 when a Delay_Resp or Pdelay_Resp message is received, so the reception of these messages is still detectable in the case of a timeout in reception.

37.3.15 Correcting Time Synchronization

A slave detects differences in the clock gradient relative to the master clock. The `offsetFromMaster` values calculated by using the standard IEEE 1588 algorithm are used to calculate the clock gradient, so the result includes elements of network fluctuation besides frequency differences. The EPTPC has a “worst 10” function to eliminate fluctuations due to network load, etc. With these functions, the time is corrected from the calculated gradient difference values and results of correction are obtained as shown in Figure 37.25.

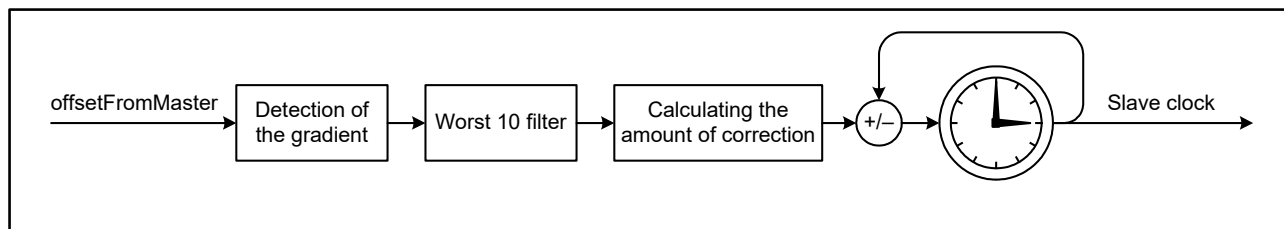


Figure 37.24 Configuration of the Time Correction Circuit

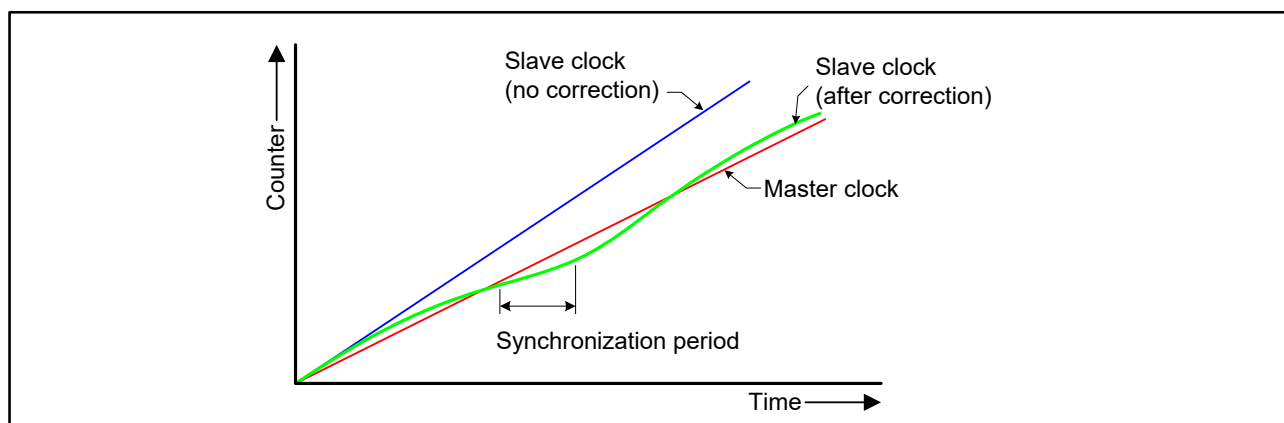


Figure 37.25 Overview of Time Correction

37.3.15.1 Judging Synchronization and Loss of Synchronization

The absolute value of `offsetFromMaster` reaching or exceeding the value specified in the `SYNTDARU` or `SYNTDARL` register is considered a loss of synchronization. The absolute value of `offsetFromMaster` being less than the absolute values of the synchronization detection threshold registers (`SYNTDBRU` and `SYNTDBRL`) is considered synchronization.

The values of the `STSR.SYNCOUT` and `SYNC` flags become 1 when synchronization is lost and obtained, respectively. Hysteresis can be obtained by setting the respective threshold registers to appropriate different values. Furthermore, the `STMR.DVTH[3:0]` and `SYTH[3:0]` bits can be used to set the consecutive number of times detection must occur for the judgment of loss of synchronization and of synchronization.

Set registers `SYNTDARU` and `SYNTDARL` to low values and the number of times detection is required to judge loss of synchronization to one in systems where control must be aborted if synchronization is lost due to fluctuations in network conditions. Set registers `SYNTDARU` and `SYNTDARL` and the number of times detection is required to large values in systems where the above condition does not apply.

Figure 37.26 shows an example of a situation where synchronization is being lost and regained. In this example, the number of consecutive times detection is required is three for both synchronization and loss of synchronization.

Note: The setting of the `STSR.SYNCOUT` flag is 1 when time synchronization starts, even though the condition for judging loss of synchronization has not been satisfied at this stage. For this reason, detection of loss of synchronization must be ignored immediately after time synchronization starts.

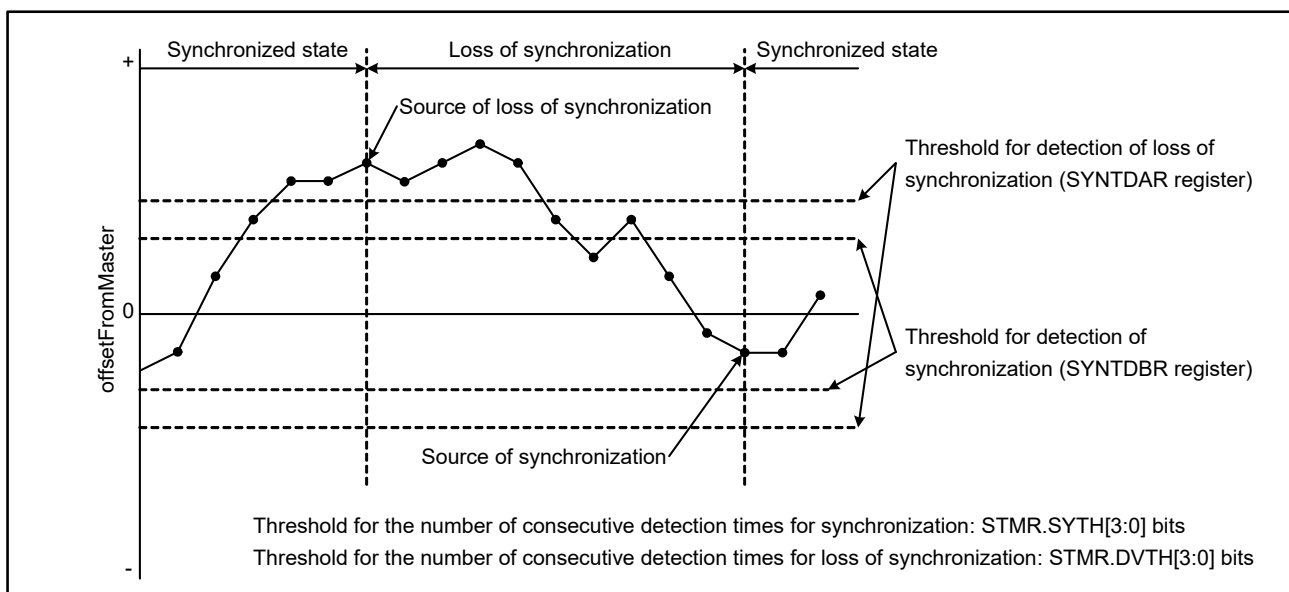


Figure 37.26 Example of a Situation where Synchronization is being Lost and Regained (the number of consecutive detection times is specified as three by the `STMR.DVTH[3:0]` and `SYTH[3:0]` bits)

37.3.15.2 Worst 10 Function

The worst 10 function is for imposing limits with regard to exceedingly large and small values among the calculated values for differences in clock gradient.

Values for difference in clock gradient are collected by observing the state of transfer over a specified interval and threshold values to impose limits are extracted from the observed values. With regard to differences in gradient, not only errors in the clock but also fluctuations in network conditions must be considered, and differences in both the positive and negative directions as shown in Figure 37.27 are collected.

From among the collected values for positive and negative gradient difference, the largest values for gradient, from the first to the tenth place (worst to tenth worst), are acquired and the tenth worst is used as a threshold value. Fluctuations in the time kept by a slave clock can be suppressed by continually overwriting the tenth worst value for gradient difference with new values large enough to exceed the threshold. Furthermore, periodic collections of gradient values can also be made for updating the threshold values during operations or for the method of setting threshold values from previously measured results

However, note that while fluctuations in the time kept by a slave clock can be suppressed by the effective filtering of values for gradient difference (collecting the worst 10 values and using the tenth worst), following of the time kept by a master clock also becomes slower.

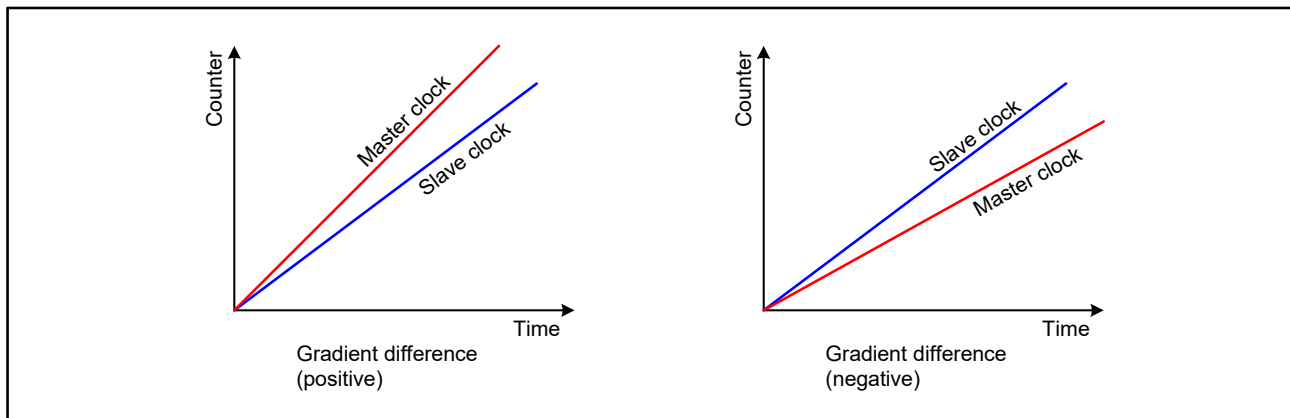


Figure 37.27 Overview of Gradient Differences

37.3.15.3 Collecting Differences in Clock Gradient and Extracting the Worst Ten Values

During slave operation, the EPTPC is capable of calculating the offsetFromMaster values from received messages and then calculating gradient differences between the local clock (acting as a slave clock) and master clock from those values. Specifically, the worst ten values are extracted from the sets of collected values for gradient difference. Either automatic filtering by the hardware or software-triggered filtering can be designated for acquisition of the sets of worst-10 values.

Figure 37.28 gives an overview of the collection of values for gradient difference.

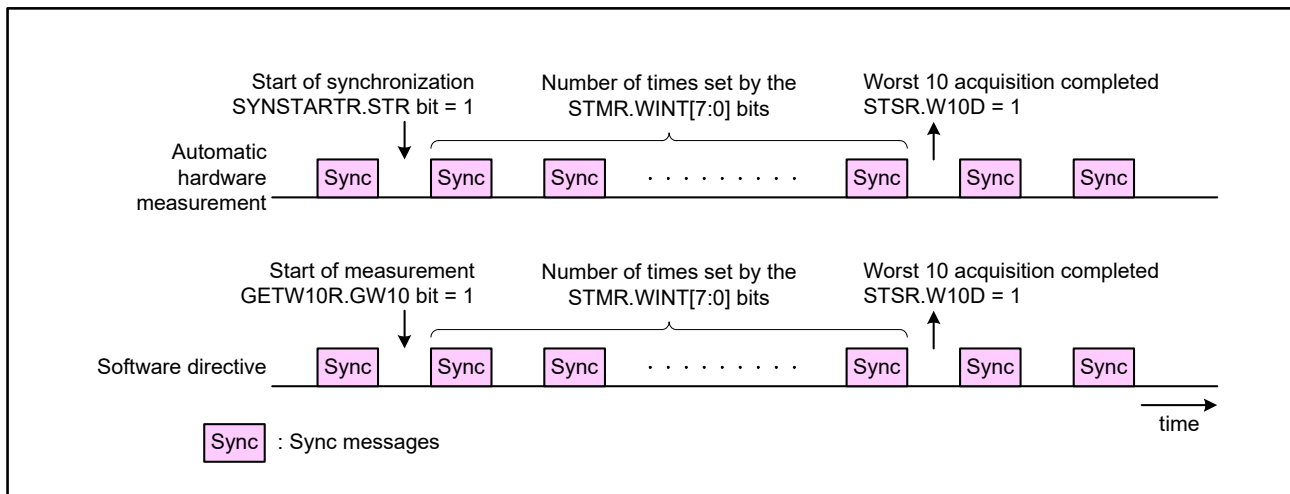


Figure 37.28 Overview of the Collection of Values for Gradient Differences

(1) Collecting Gradient Differences and Extracting the Worst Ten Values by Hardware

The EPTPC automatically collects the values of gradient difference by hardware if the STMR.W10S bit is 0. When the SYNSTARTR.STR bit is set to 1 (starting slave time synchronization), the EPTPC collects gradient difference values for the number of times set by the STMR.WINT[7:0] bits. When the collection of gradient difference values is finished, the tenth largest values on the positive and negative gradient difference sides are stored as the tenth worst values in registers PW10VRU, PW10VRM, and PW10VRL and registers MW10RU, MW10RM, and MW10RL. When acquisition of the worst 10 values is complete, the STSR.W10D flag becomes 1.

Filtering of gradient difference values by using the stored tenth worst values then proceeds automatically.

Note that if the number of times set by STMR.WINT[7:0] bits are fewer than ten times, the double of the best of the collected values on the positive side is stored in registers PW10VRU, PW10VRM, and PW10VRL, and the half of the best of the collected values on the negative side is stored in registers MW10RU, MW10RM, and MW10RL.

(2) Collecting Gradient Differences and Extracting the Worst Ten Values by Software

The EPTPC collects values of gradient differences by software if the STMR.W10S bit is 1.

When the GETW10R.GW10 bit is set to 1 after time synchronization has started, the EPTPC collects gradient difference values for the number of times set by the STMR.WINT[7:0] bits. When the collection of gradient difference values is finished, the tenth largest values on the positive and negative gradient difference sides are stored as the tenth worst values in registers PW10VRU, PW10VRM, and PW10VRL and registers MW10RU, MW10RM, and MW10RL. When acquisition of the worst 10 values is complete, the STSR.W10D flag becomes 1.

Since filtering of gradient difference values proceeds with the values set in registers PLIMITRU, PLIMITRM, and PLIMITRL and registers MLIMITRU, MLIMITRM, and MLIMITRL as the upper and lower limits for filtering, the values stored in registers PW10VRU, PW10VRM, and PW10VRL and registers MW10RU, MW10RM, and MW10RL must be written to the PLIMITRU, PLIMITRM, and PLIMITRL and registers MLIMITRU, MLIMITRM, and MLIMITRL, respectively.

Note that if the number of times set by the STMR.WINT[7:0] bits is fewer than ten or times, the double of the best of the collected values on the positive is stored in registers PW10VRU, PW10VRM, and PW10VRL, and the half of the best of the collected values on the negative side is stored in registers MW10RU, MW10RM, and MW10RL.

The flowchart in Figure 37.29 shows an example of the procedure for software-triggered acquisition of the worst 10 values.

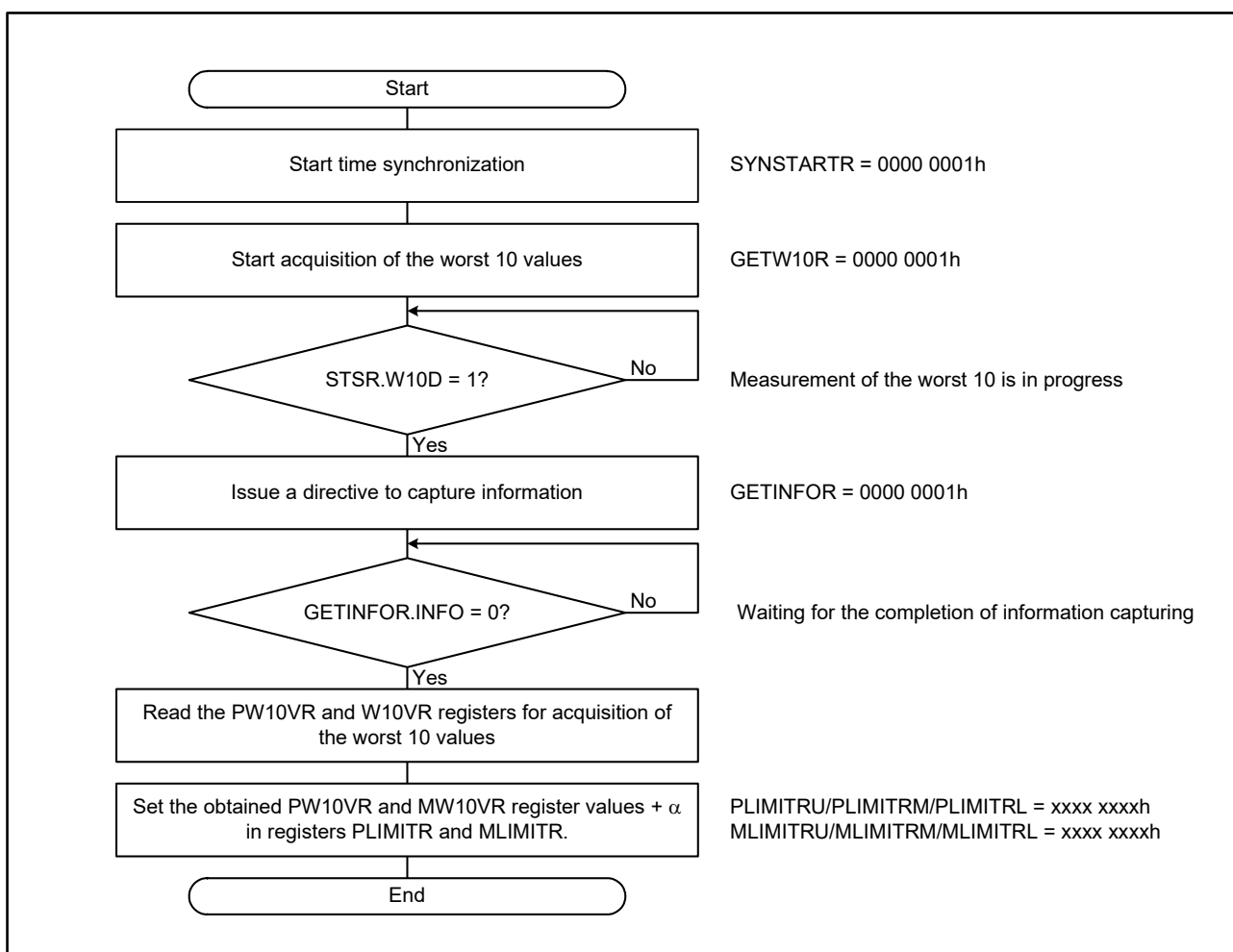


Figure 37.29 Example of the Procedure for Software-Triggered Acquisition of the Worst 10 Values

37.3.16 Local Clock Counter

The local clock counter retains the synchronized time information. The counter starts counting from 0 after the ETHERC is released from the module stop state or the EPTPC is released from the software reset state. The local clock counter can then be set to any desired value. The procedure for setting the initial value in the local clock counter is shown in Figure 37.30.

The time information kept by the local clock counter is also readable. Figure 37.31 shows the procedure for reading the time information kept by the local clock counter.

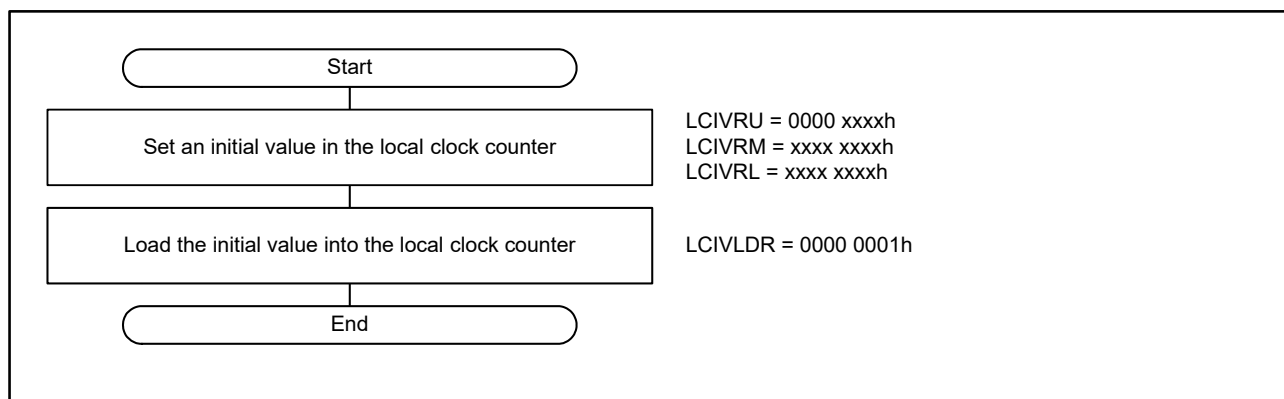


Figure 37.30 Procedure for Setting a New Initial Value in the Local Clock Counter

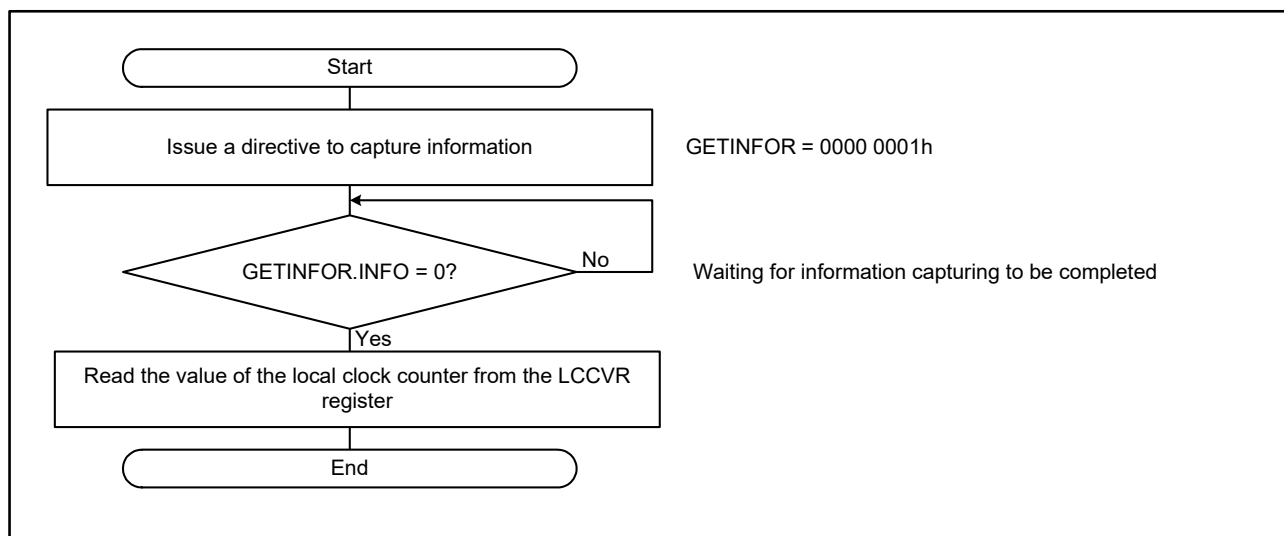


Figure 37.31 Procedure for Reading the Time Kept by the Local Clock Counter

37.3.17 Pulse Output Timer

The STCA module of the EPTPC incorporates six timers (pulse output timers 0 to 5) which operate independently of each other.

The pulse output timers produce periodic pulses, and the rising or falling edges of these pulses can be used as interrupt requests or output to the ELC as event signals. The pulses generated by the pulse output timers 0 and 1 can be output from the EPLSOUT0 and EPLSOUT1 pins, respectively.

The time at which a pulse output timer starts operating (t_{start}), and the cycle time (t_c) and pulse width (t_w) of the output pulses, can be specified for each channel by register settings.

The timing of pulse output timer operation is shown in Figure 37.32, and limits on the settings are listed in Table 37.28.

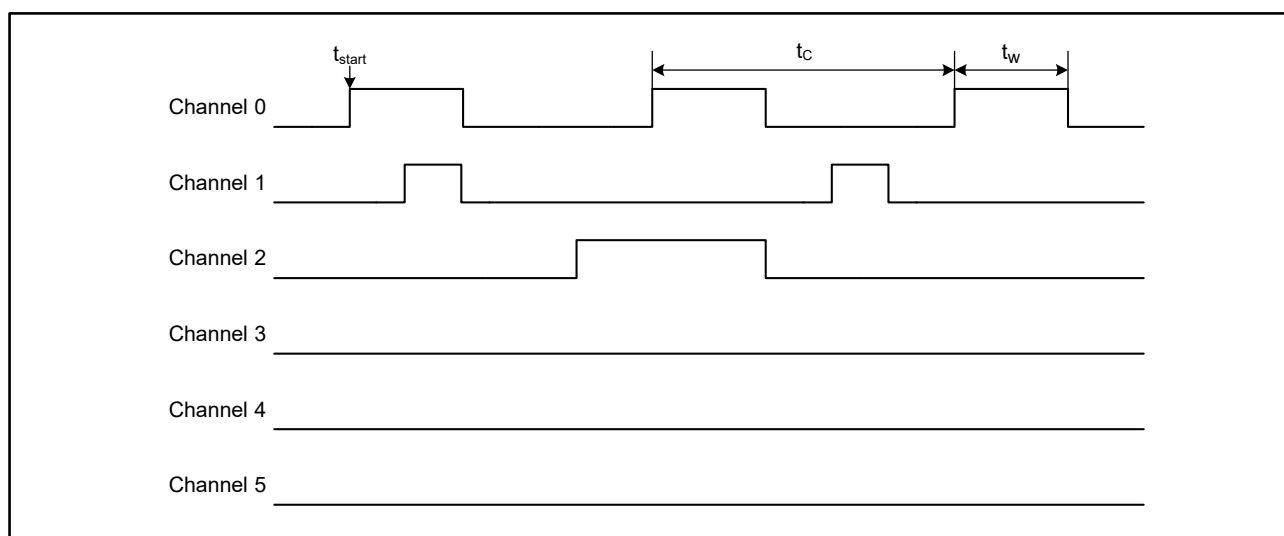


Figure 37.32 Time at which a Pulse Output Timer Starts Operating

Table 37.28 Limits on the Values that can be Specified for a Pulse Output Timer

Item	Description
Cycle time (t_c)	From four cycles of the STCA clock to 1 s
Resolution of the cycle	Set in units of nanoseconds. However, the timing of rising edges is rounded by the period of the system clock (50 ns, 40 ns, 20 ns, or 10 ns).
Pulse width (t_w)	From two cycles of the STCA clock to 500 ms
Resolution of the pulse width	Set in units of nanoseconds. However, the timing of falling edges is rounded by the period of the system clock (50 ns, 40 ns, 20 ns, or 10 ns).

37.3.17.1 Procedure for Setting a Pulse Output Timer

Figure 37.33 shows the procedure for setting a pulse output timer.

Note that a timer does not produce periodic pulses if the time set in registers TMSTTRUm and TMSTTRLm (m = 0 to 5) has already passed. Set the time for a pulse output timer to start to a later time than the time when the timer is being set.

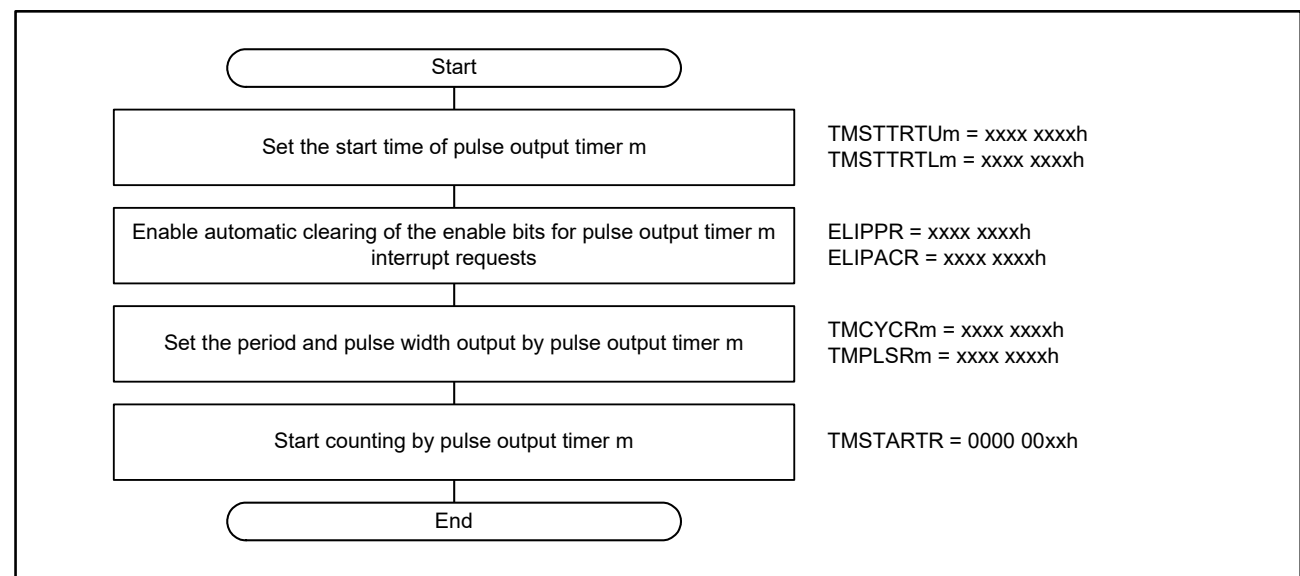


Figure 37.33 Procedure for Setting a Pulse Output Timer

37.3.17.2 Output of periodic pulses as interrupt requests or event signals

MINT interrupt requests, IPLS interrupt requests, or event output signals for the ELC can be generated by detecting rising or falling edges of the periodic pulses from the pulse output timer. The edge for detection and pulse output timer to use can be selected, and automatic clearing of enable bits for the IPLS interrupt or event output can be set. Make the required settings before setting the TMSTARTR.ENm bit to 1 (starting pulse output timer m).

(1) MINT Interrupt Request

Rising edges of the periodic pulses from the pulse output timers can be detected to generate MINT interrupt requests. Falling edges cannot be used in this way. The pulse output timers for use in generating MINT interrupt requests are selected by the MITSELR.MINTENm bits.

Automatic clearing of the enable bits for MINT interrupt requests is not available.

(2) IPLS Interrupt Request

Rising or falling edges of the periodic pulses from the pulse output timers can be detected to generate IPLS interrupt requests. The pulse output timers for use in generating IPLS interrupt requests are selected by the IPTSELR.IPTSELM bits.

Setting the ELIPACR.PLSP or PLSN bit enables automatic clearing of the enable bits for IPLS interrupt requests.

(3) Output of Event Signals to the ELC

Rising or falling edges of the periodic pulses from the pulse output timers can be detected to output event signals to the ELC. The pulse output timers for use in outputting event signals and the edge are selected by the ELIPPR.CYCPm or CYCNm bits.

Setting the ELIPACR.CYCPm or CYCNm bits enables automatic clearing of the event output enable bits.

37.3.18 Priority Control in Transmission

37.3.18.1 Arbitration

In cases of contention between a message for transmission from the PTPEDMAC and a request for relaying from another channel, the PRC-TC module gives priority to the request for relaying from the other channel.

Cases of contention between multiple requests for the transmission of messages by the SYNFP module are also arbitrated in the order of priority shown in Table 37.29.

Table 37.29 Priority in Arbitration for Messages for Transmission

Messages for Transmission	Priority Order	Remark
Sync	1 Highest priority	
Delay_Req, Pdelay_Req	2	There is no device type that simultaneously transmits Delay_Req and Pdelay_Req messages.
Delay_Resp, Pdelay_Resp	3	There is no device type that simultaneously transmits Delay_Resp and Pdelay_Resp messages.
Announce	4	
Relay messages from another channel	5	The PRC-TC module determines the order of priority for these two items.
Messages for transmission from the PTPEDMAC	5	
Messages for transmission from the EDMAC	6	

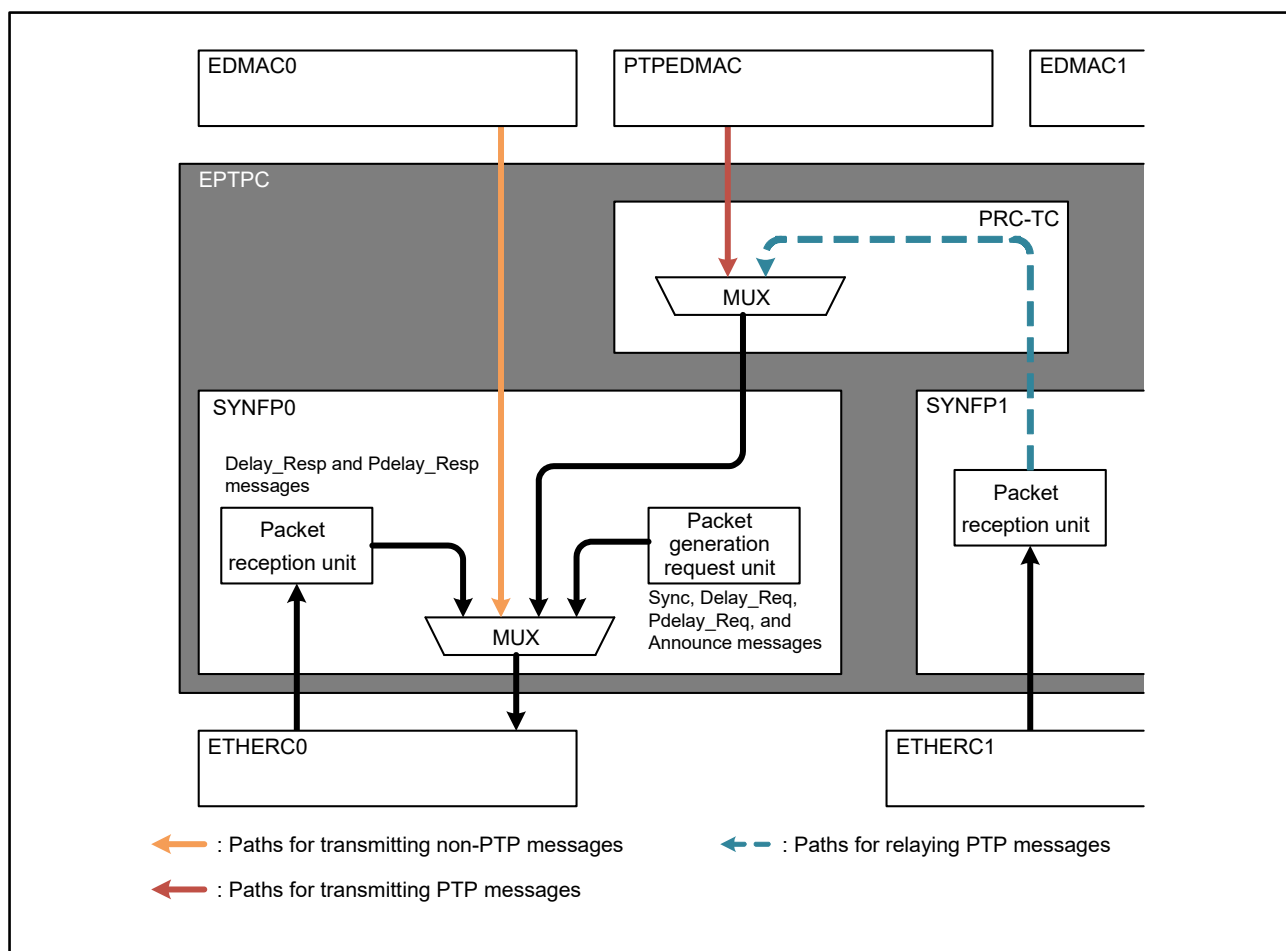


Figure 37.34 Arbitration in Message Transmission

37.3.18.2 Securing of Bandwidth for the Transmission of Sync Messages

The EPTPC secures bandwidth for the transmission of Sync messages, and is capable of handling transmission at very precise intervals.

If the transmission of a Sync message at a fixed interval proceeds at the same time as transmission by the PTPEDMAC or the relaying of messages by another channel is in progress, because transmission of the Sync message proceeds when the other processing is complete, the interval for transmission will no longer be fixed.

By securing bandwidth for the transmission of Sync messages, the transmission of messages from EDMAC0, EDMAC1, and the PTPEDMAC is limited, and the transmission of Sync messages can be handled without fluctuations. Securing of bandwidth for the transmission of Sync messages can be disabled by setting the SYCONFR.SBDIS bit to 1. Figure 37.35 gives a schematic view of securing bandwidth for the transmission of Sync messages.

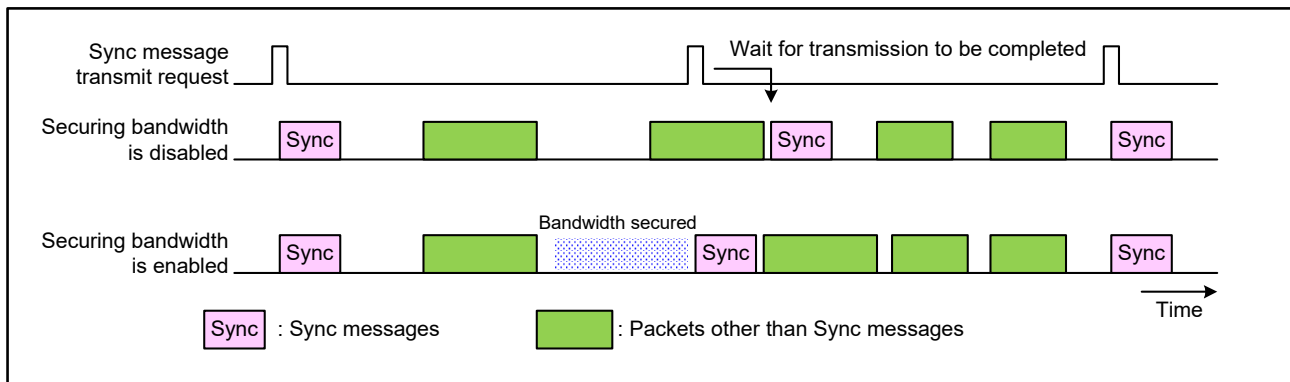


Figure 37.35 Securing of Bandwidth for the Transmission of Sync Messages

37.3.18.3 Securing of Transmission Interval

In the transmission of messages by the ETHERC, if there is a fixed delay from the time of a request for transmission to the time of transmission on the MII of Ethernet port 0 or 1, time stamp values of PTP messages can be used for accurately obtaining the size of the delay during operation as a slave. However, in the case of continuous transfer and so on, in situations where processing of messages to wait for interpacket gap times is required, delay times may fluctuate. Specifying an interval for frame transmission in SYCONFR.TCYC[7:0] bits to control the interval between the completion of transmission and the next request for transmission makes the ETHERC able to secure the reliability of time stamp values, thus avoiding the effects of interpacket gap times and obtaining a fixed delay for transmission.

37.4 Interrupts

The EPTPC has the MINT interrupt request and IPLS interrupt request.

Figure 37.36 shows the relation between the MINT interrupt request and IPLS interrupt request. Figure 37.37 shows the details on interrupt requests of the pulse output timer.

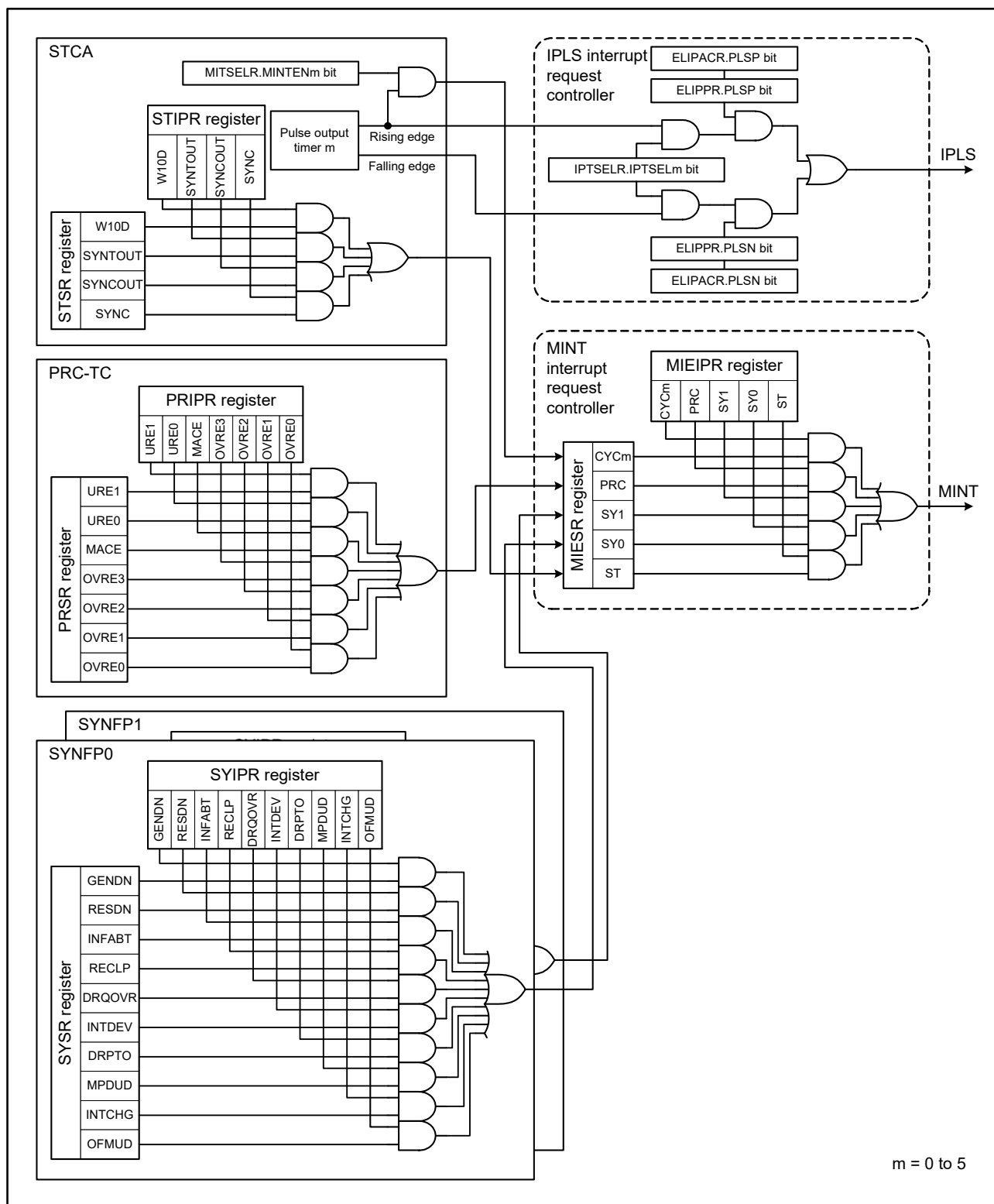


Figure 37.36 MINT Interrupt Request and IPLS Interrupt Request

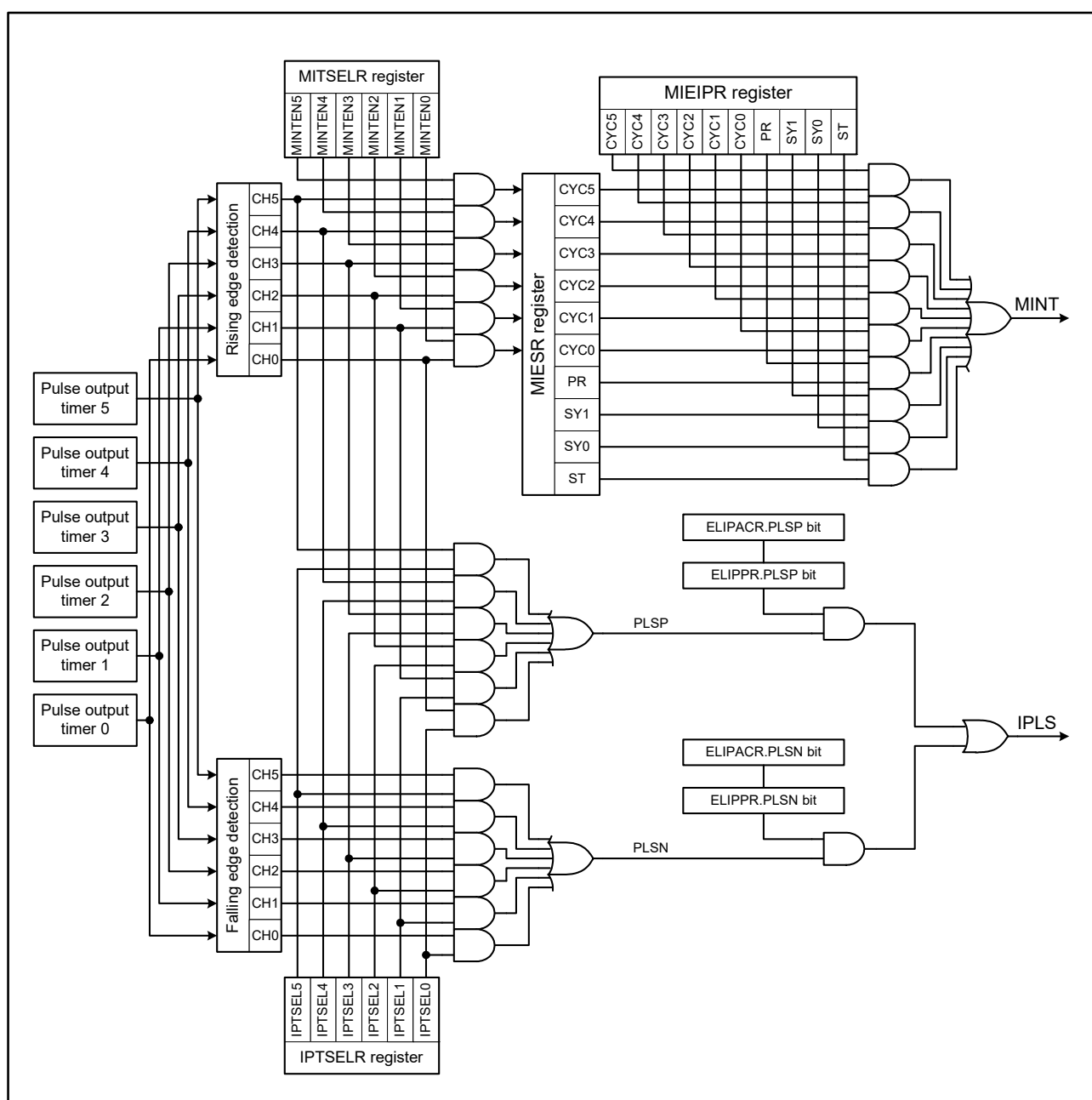


Figure 37.37 Details on Interrupt Requests of the Pulse Output Timer

37.5 Event Link (Output)

The EPTPC can output an event to the ELC by detecting the rising edge or falling edge of the pulse that the pulse output timer outputs.

Figure 37.37 shows the relation between the pulse output timer and the ELC.

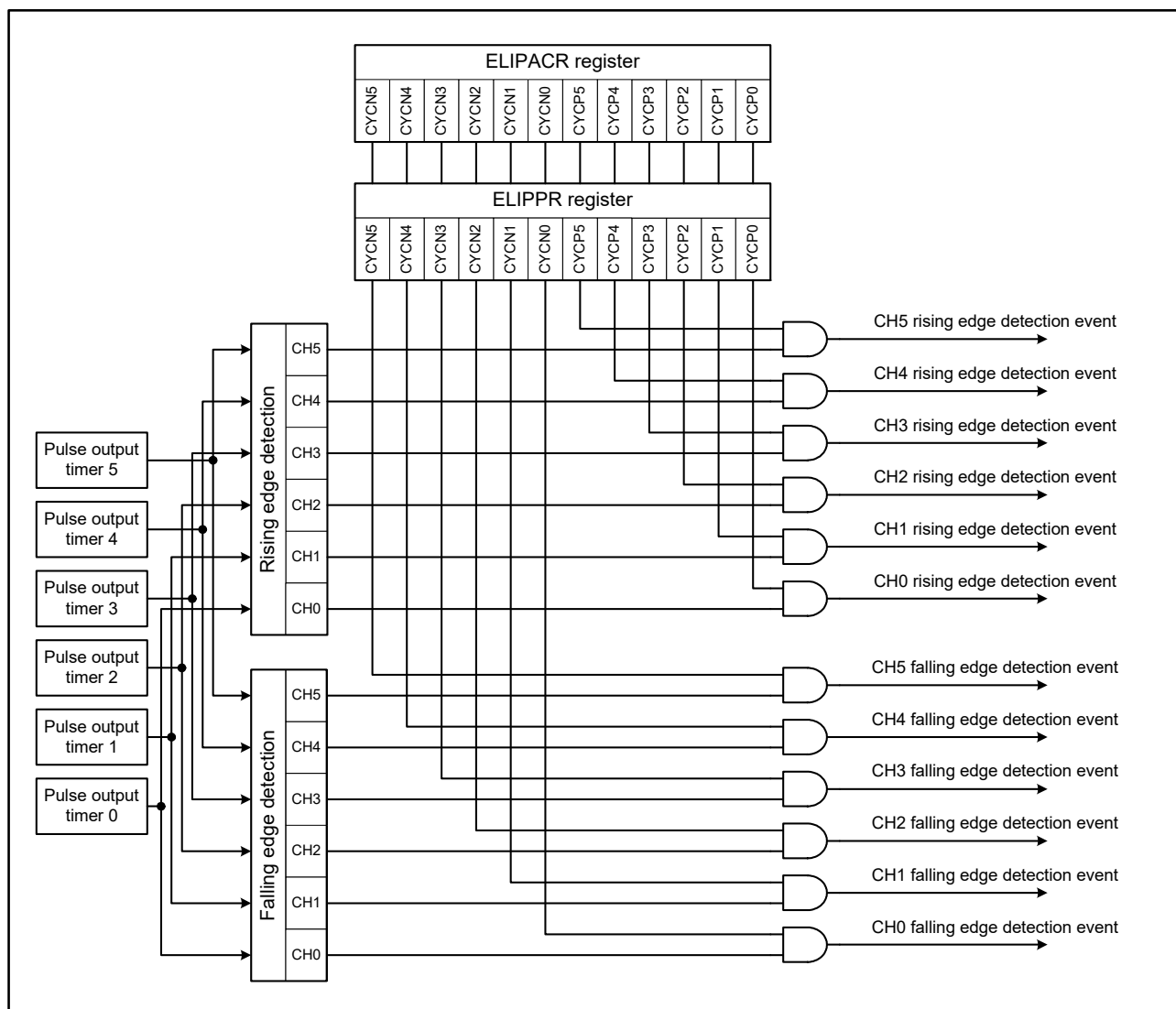


Figure 37.38 Relation between the Pulse Output Timer and ELC

37.6 Usage Notes

37.6.1 Setting the Module-Stop Function

Module stop control register B (MSTPCRB) is used to enable/disable the EPTPC operation. After a reset, the EPTPC is stopped. After exiting the module-stop state and 3 μ s is elapsed after setting the SYBYPSSR.BYPASSn bit to 0, registers can be accessed. Refer to section 11, Low Power Consumption for details on the module-stop function.

Table 37.30 lists the relations between bit settings and accessible registers.

STCA clock and MII clock are required to be supplied to access the registers in the STCA module and SYNFP modules, respectively.

Table 37.30 Register Settings and Accessible Registers

MSTPCRB Register			SYBYPSSR Register		Allocation of Register Addresses for Access					
MSTPB15 (ETHERC0, EDMAC0)	MSTPB14 (ETHERC1, EDMAC1)	MSTPB13 (EPTPC, PTPEDMAC)	BYPASS0	BYPASS1	000C 0500h to 000C 05FFh	000C 4000h to 000C 403Fh	000C 4040h to 000C 43FFh (STCA)	000C 4400h to 000C 47FFh (PRC-TC)	000C 4800h to 000C 4BFFh (SYNFP0)	000C 4C00h to 000C 4FFFh (SYNFP1)
0	0	0	0	0	Accessible	Accessible	Accessible	Accessible	Accessible	Accessible
0	1	0	0	1	Accessible	Accessible	Accessible	Accessible	Accessible	Access prohibited
1	0	0	1	0	Accessible	Accessible	Accessible	Accessible	Access prohibited	Accessible
0	0	1	1	1	Accessible	Access prohibited	Access prohibited	Access prohibited	Access prohibited	Access prohibited
0	1	1	1	1	Accessible	Access prohibited	Access prohibited	Access prohibited	Access prohibited	Access prohibited
1	0	1	1	1	Accessible	Access prohibited	Access prohibited	Access prohibited	Access prohibited	Access prohibited
1	1	1	1	1	Access prohibited	Access prohibited	Access prohibited	Access prohibited	Access prohibited	Access prohibited

Note: The combinations other than those listed above are prohibited.

Access to an access-prohibited register or access to a register while the MII clock or STCA clock is not supplied may cause a bus freeze. To avoid a bus freeze, set the BSC.BEREN.TOEN bit to 1 to enable the bus timeout detection. If a bus timeout error occurs, use the PTRSTR.RESET bit to reset the EPTPC by software. Refer to section 16.7.1.2, Timeout for details on the bus timeout.

37.6.1.1 Exiting the Module-Stop State

When the EPTPC is to be used, follow the procedure below for exit from the module-stop state. Do not execute other processes while executing this procedure.

- (1) Set the MSTPB13, MSTPB14, and MSTPB15 bits in the MSTPCRB register to 0.
- (2) Set the BYPASS0 and BYPASS1 bits in the SYBYPSSR register to 0.
- (3) Wait for 3 μ s.

Note: The above procedure applies when enabling all modules. Set the corresponding bits as required.

37.6.1.2 Transitions to the Module-Stop State

To place the EPTPC, EPTPC, and EDMAC in the module-stop state, follow the procedure below. Do not execute other processes while executing this procedure.

- (1) Reset the EPTPC, EPTPC, and EDMAC.*¹
- (2) Set the BYPASS0 and BYPASS1 bits in the SYBYP SR register to 1.
- (3) Set the MSTPB13, MSTPB14, and MSTPB15 bits in the MSTPCR B register to 1.
- (4) Wait for 3 μ s.

Note 1. Refer to section 36.5.4, Procedure to Reset Ethernet Controller.

37.6.2 Note on Transitions to the Software Standby Mode

When the MCU is to enter the software standby mode, place the EPTPC, EPTPC, and EDMAC in the module-stop state before executing a WAIT instruction.

37.6.3 Wait Cycles for Register Access

Access to registers in the EPTPC involves the arbitration of different clock signals, specifically the peripheral module clock signal (PCLKA), the STCA clock signal, and the MII clock signals such as TX_CLK. Accordingly, numbers of wait cycles for register access differ with the combination of the frequency settings of the peripheral module clock signal, of the STCA clock signal, and of the MII clock signals.

Table 37.31 gives examples of numbers of wait cycles for different combinations of clock frequency. Add 1 to 2 cycles to these values to obtain the number of access cycles.

Table 37.31 Wait Cycles for Register Access (when the frequency of the STCA clock is 20 MHz)

Address Range	STCA Clock = 20 MHz							
	Peripheral Module Clock PCLKA = 120 MHz				Peripheral Module Clock PCLKA = 20 MHz			
	MII Clock 25 MHz (100 Mbps)		MII Clock 2.5 MHz (10 Mbps)		MII Clock 25 MHz (100 Mbps)		MII Clock 2.5 MHz (10 Mbps)	
	Read	Write	Read	Write	Read	Write	Read	Write
000C 0500h to 000C 05FFh	2	2	2	2	2	2	2	2
000C 4000h to 000C 403Fh	4	4	4	4	4	4	4	4
000C 4040h to 000C 43FFh (STCA)	7	27 to 41*1	7	27 to 41*1	7	15 to 17*1	7	15 to 17*1
000C 4400h to 000C 47FFh (PRC-TC)	8	8	8	8	8	8	8	8
000C 4800h to 000C 4BFFh (SYNFP0)	8	23 to 33*2	8	111 to 209*2	8	15 to 17*2	8	31 to 49*2
000C 4C00h to 000C 4FFFh (SYNFP1)	8	23 to 33*2	8	111 to 209*2	8	15 to 17*2	8	31 to 49*2

Note 1. The number of wait cycles in access to the STCA-related registers (Wstca) can be calculated to be in the following range from the periods of the peripheral module clock ($t_{c(PCLKA)}$) and STCA clock ($t_{c(STCA)}$).

$$\begin{aligned} \text{The minimum value of Wstca} &= \text{Int} (t_{c(STCA)} / t_{c(PCLKA)}) \times 2 + 15 \quad (t_{c(PCLKA)} \leq t_{c(STCA)}) \\ &= 15 \quad (t_{c(PCLKA)} > t_{c(STCA)}) \end{aligned}$$

$$\begin{aligned} \text{The maximum value of Wstca} &= \text{Int} (t_{c(STCA)} / t_{c(PCLKA)}) \times 4 + 17 \quad (t_{c(PCLKA)} \leq t_{c(STCA)}) \\ &= 17 \quad (t_{c(PCLKA)} > t_{c(STCA)}) \end{aligned}$$

- Int(A) is the calculation of the largest integer not greater than A.
- This calculation presumes that the CPU clock and peripheral module clock have the same periods.

For example, if the frequency of the peripheral module clock is 120 MHz and that of the STCA clock is 1/6 that of the peripheral module clock (= 20 MHz),

$$\text{the minimum value of Wstca} = \text{Int} (50 \text{ (ns)} / 8.3 \text{ (ns)}) \times 2 + 15 = 27, \text{ and}$$

$$\text{the maximum value of Wstca} = \text{Int} (50 \text{ (ns)} / 8.3 \text{ (ns)}) \times 4 + 17 = 41.$$

If REF50CK0 or REF50CK1 is used as the STCA clock, the frequency of the STCA clock is 25 MHz.

Note 2. The number of wait cycles in access to the SYNFP-related registers (Wsynf) can be calculated to be in the following range from the periods of the peripheral module clock ($t_{c(PCLKA)}$) and MII clock ($t_{c(MII)}$).

$$\begin{aligned} \text{The minimum value of Wsynf} &= \text{Int} (t_{c(MII)} / t_{c(PCLKA)}) \times 2 + 15 \quad (t_{c(PCLKA)} \leq t_{c(MII)}) \\ &= 15 \quad (t_{c(PCLKA)} > t_{c(MII)}) \end{aligned}$$

$$\begin{aligned} \text{The maximum value of Wsynf} &= \text{Int} (t_{c(MII)} / t_{c(PCLKA)}) \times 4 + 17 \quad (t_{c(PCLKA)} \leq t_{c(MII)}) \\ &= 17 \quad (t_{c(PCLKA)} > t_{c(MII)}) \end{aligned}$$

- Int(A) is the calculation of the largest integer not greater than A.
- This calculation presumes that the CPU clock and peripheral module clock have the same periods.

For example, if the frequency of the peripheral module clock is 120 MHz and the transmission rate is 10 Mbps (so the MII clock is running at 2.5 MHz),

$$\text{the minimum value of Wsynf} = \text{Int} (400 \text{ (ns)} / 8.3 \text{ (ns)}) \times 2 + 15 = 111, \text{ and}$$

$$\text{the maximum value of Wsynf} = \text{Int} (400 \text{ (ns)} / 8.3 \text{ (ns)}) \times 4 + 17 = 209.$$

37.6.4 Note on Message Transmission in Operation as a Transparent Clock (TC)

When a message is transmitted by using the PTPEDMAC after a PTP event message (Sync, Delay_Resp, Pdelay_Req, Pdelay_Resp) has been relayed by the PRC-TC module and before a non-PTP message or PTP general message is relayed, the residence time is erroneously added to the section of the message transmitted. The section corresponds to the location of the correctionField of the message relayed immediately before the message is transmitted.

When any PTP event message is set to be relayed (when at least one of the SYNC[1], DRP[1], PDRQ[1], and PDRP[1] bits in the SYRFL1R register is set to 1), use EDMAC0 or EDMAC1 instead of the PTPEDMAC to transmit messages.